SQANTI3 report

Unique Genes: 10277

Unique Isoforms: 18735

Transcript Classification

Category	Isoforms, count
FSM	18054
ISM	194
NIC	272
NNC	90
Genic Genomic	27
Antisense	81
Fusion	17
Intergenic	0
Genic Intron	0

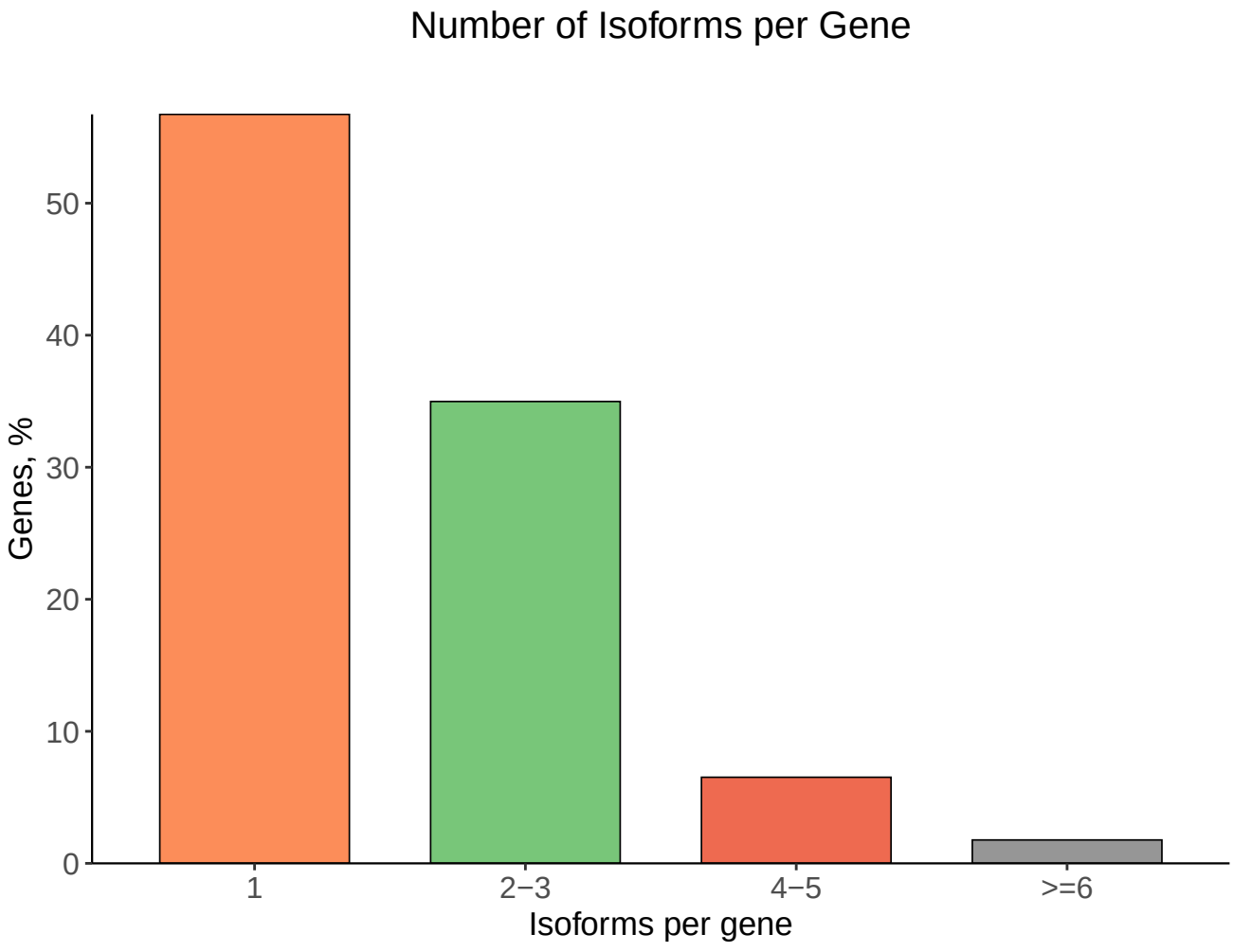
Gene Classification

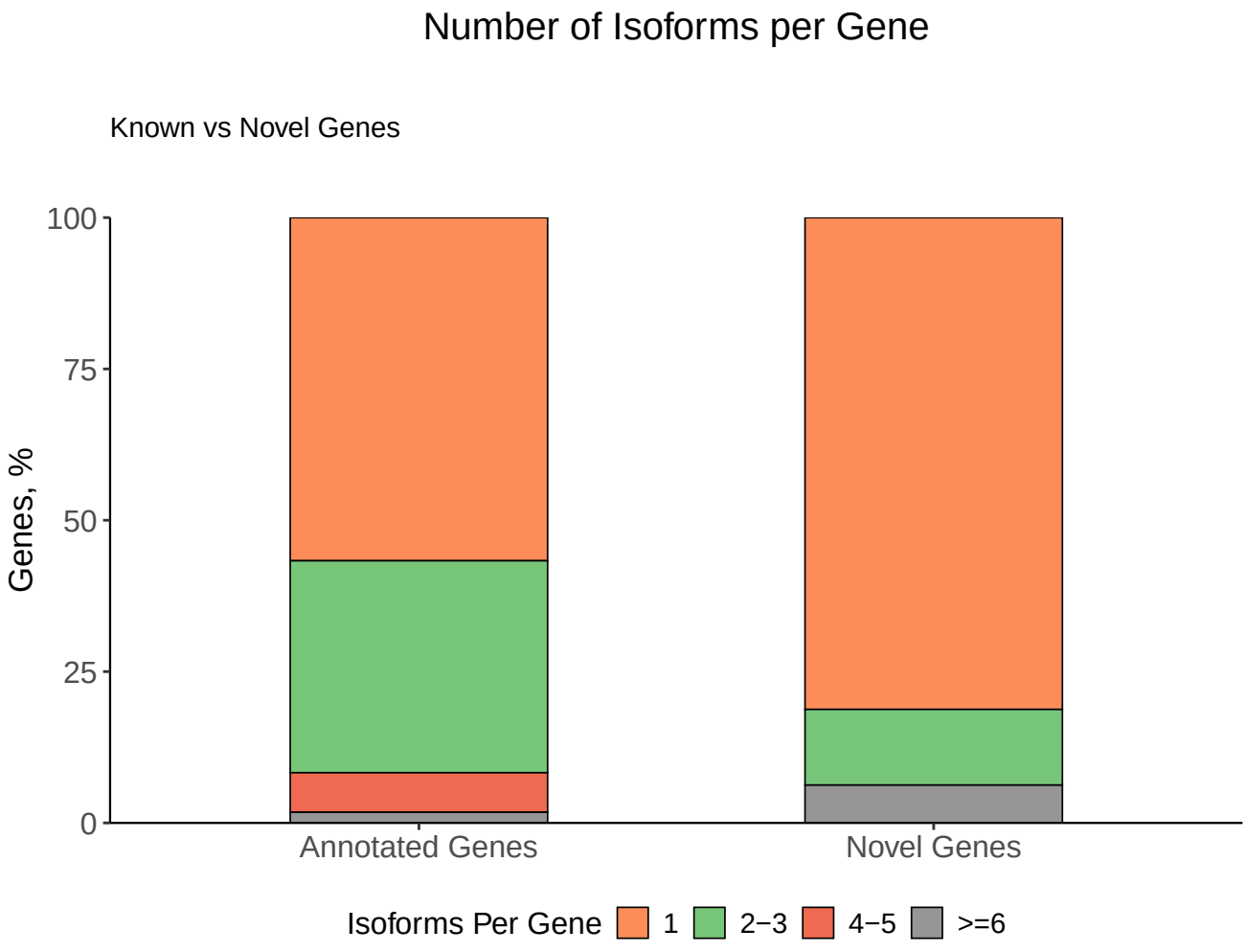
Category	Genes, count
Annotated Genes	10261
Novel Genes	16

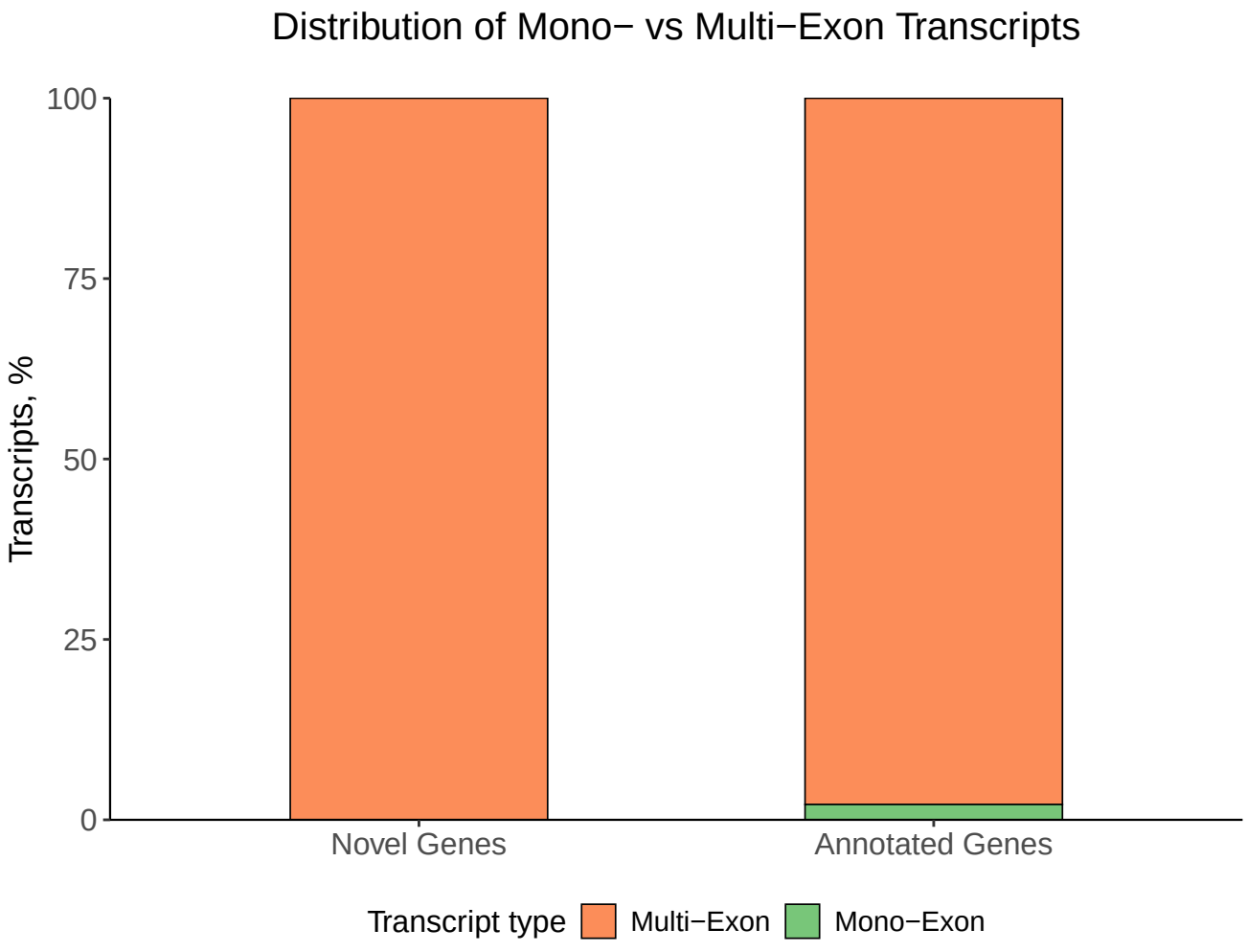
Splice Junction Classification

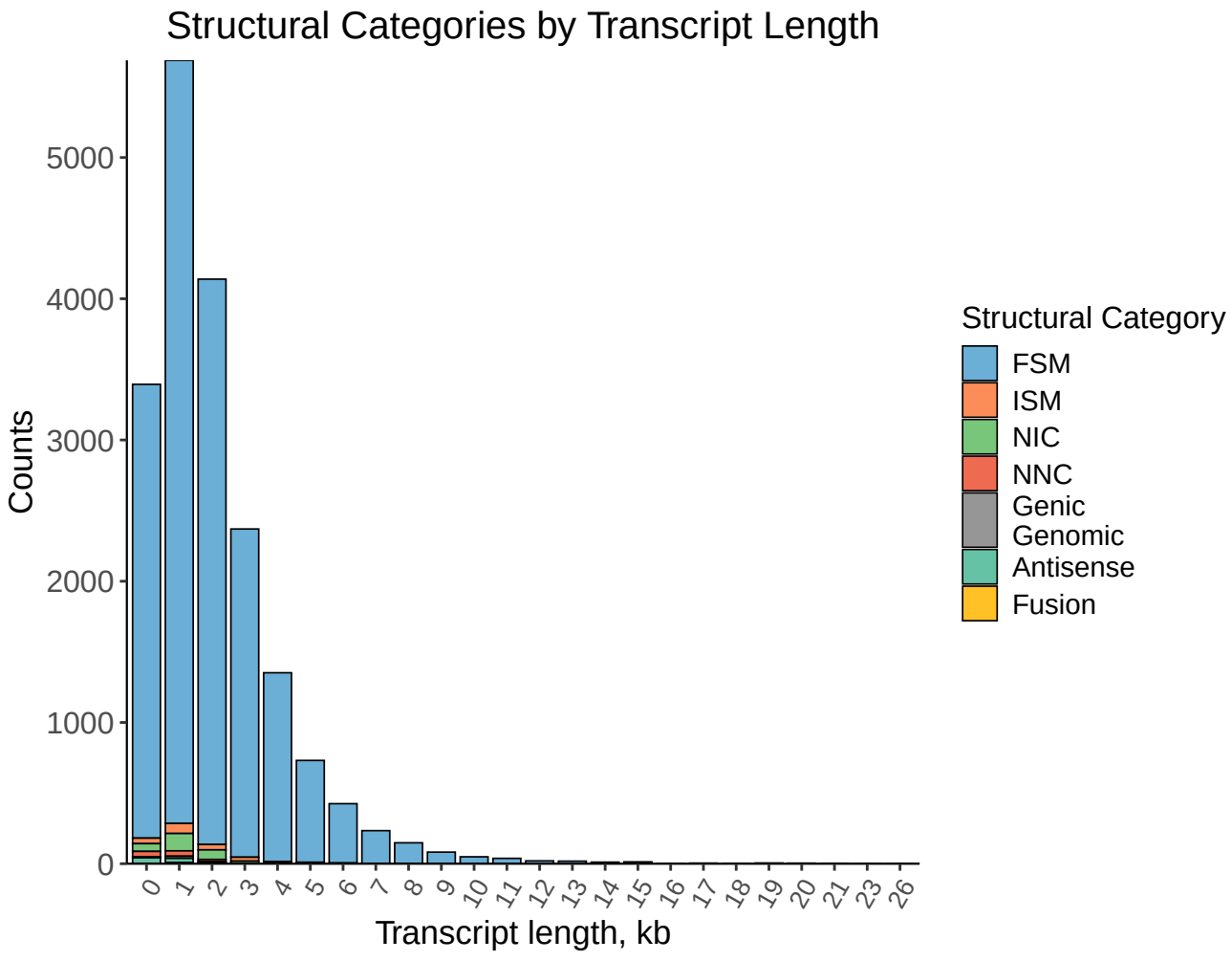
Category	SJs, count	Percent
Known canonical	96857	99.75
Known Non−canonical	50	0.05
Novel canonical	180	0.19
Novel Non−canonical	9	0.01

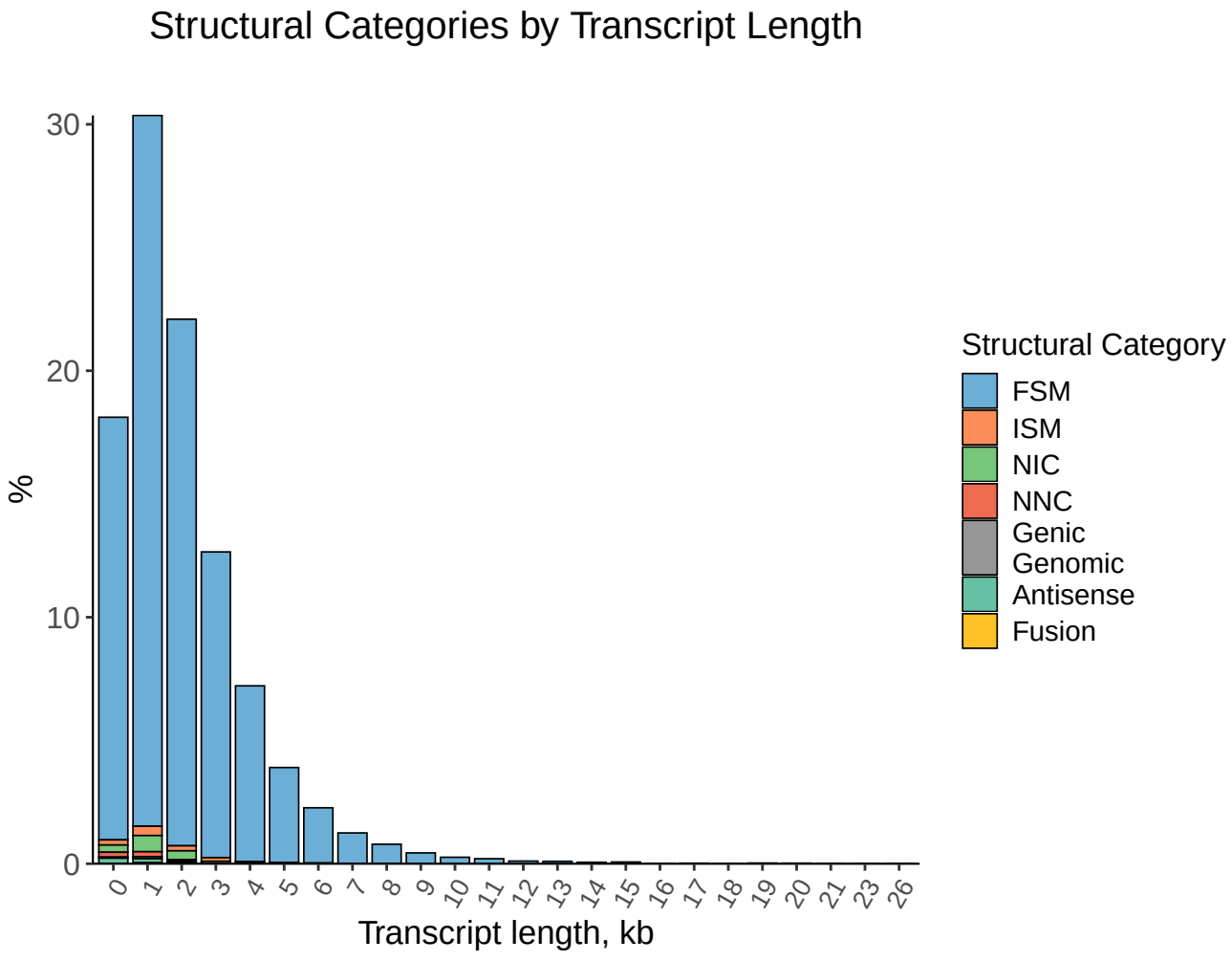
Gene Characterization

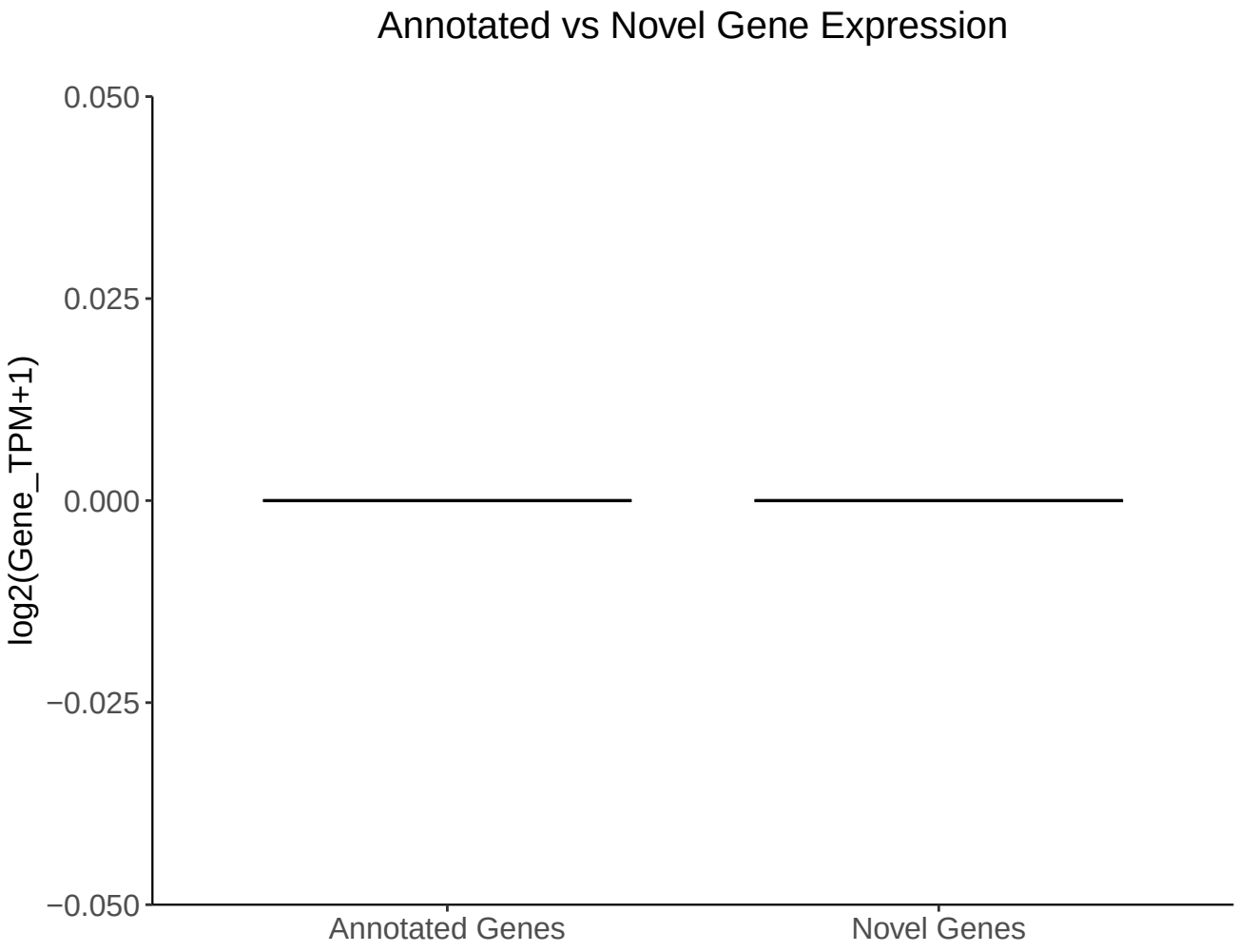

Number of Isoforms per Gene
Genes, %
0
10
20
30
40
50
1
2−3
4−5
>=6
Isoforms per gene

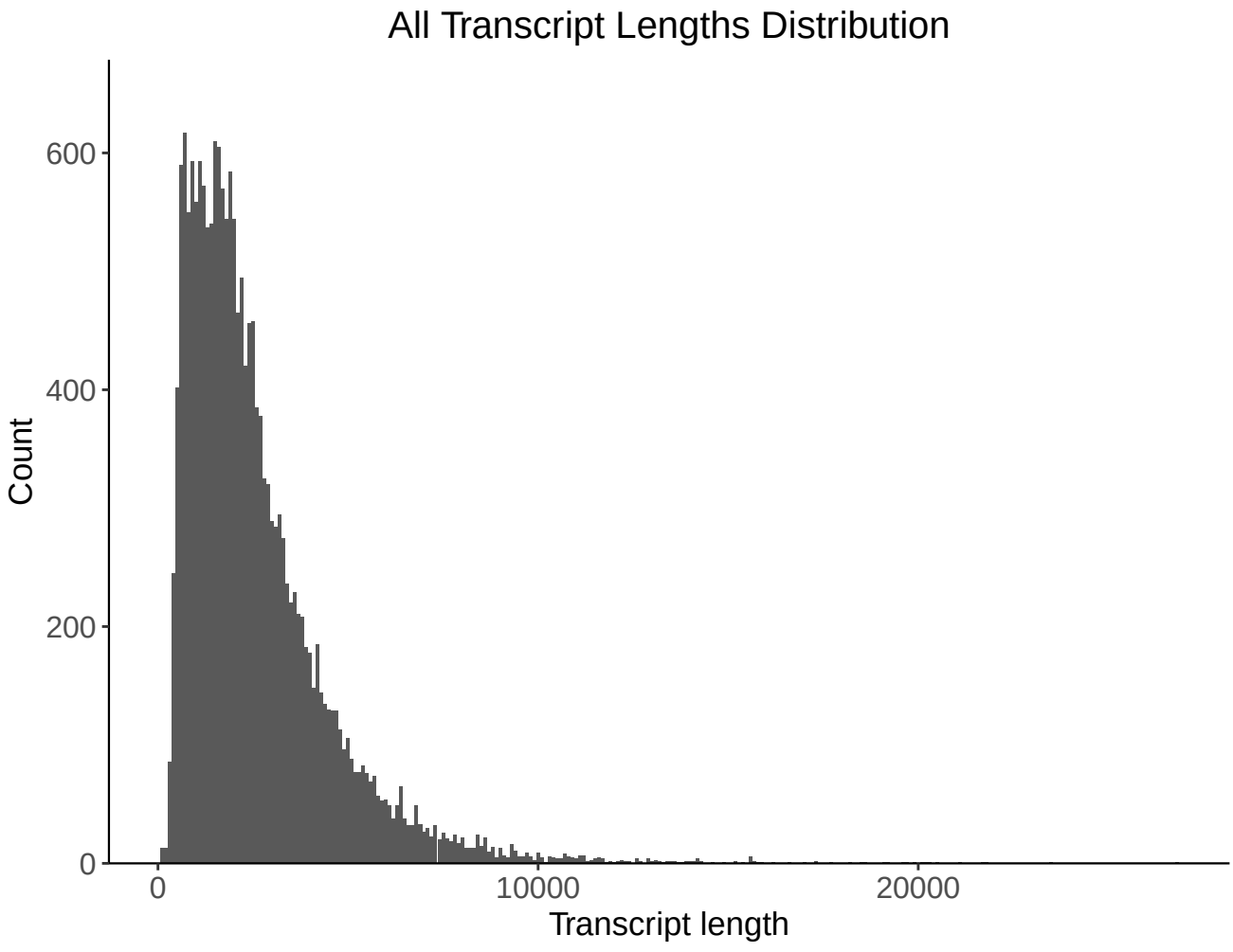

Number of Isoforms per Gene
Known vs Novel Genes
Genes, %
100
75
50
25
0
Annotated Genes
Novel Genes
Isoforms Per Gene
1
2−3
4−5
>=6

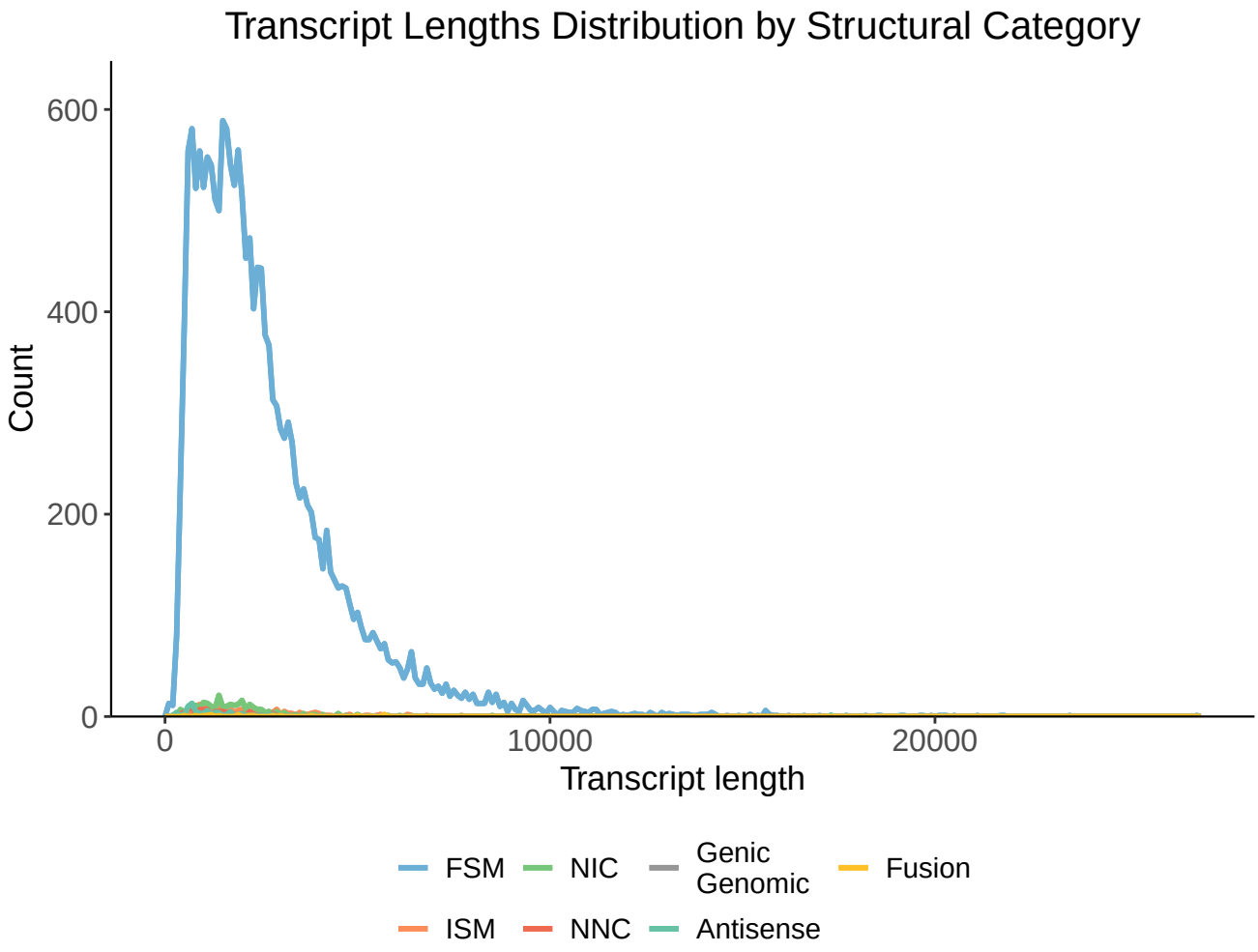

Distribution of Mono− vs Multi−Exon Transcripts
Transcripts, %
100
75
50
25
0
Novel Genes
Annotated Genes
Transcript type
Multi−Exon
Mono−Exon

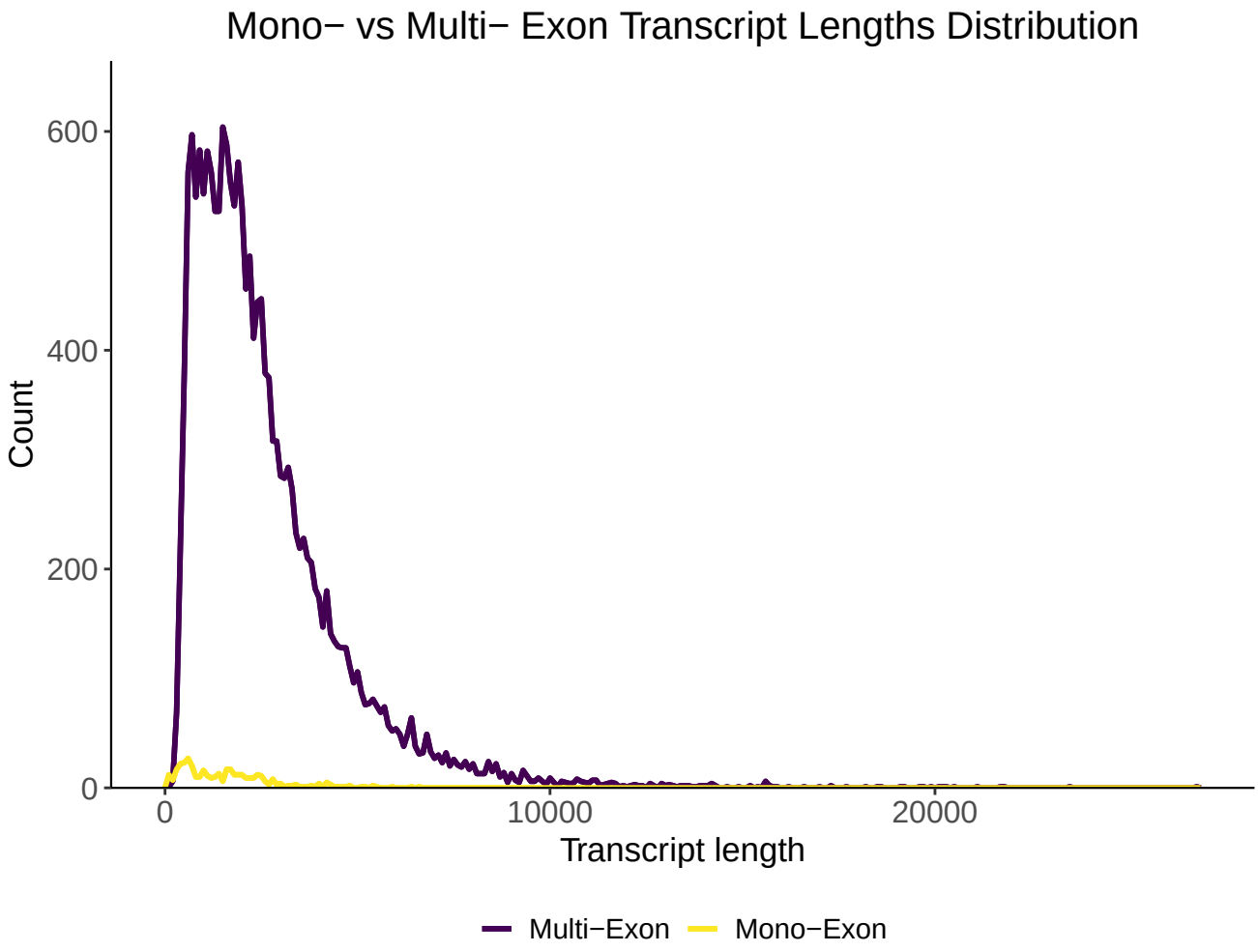

Structural Categories by Transcript Length
Counts
5000
4000
3000
2000
1000
0
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
23
26
Transcript length, kb
Structural Category
FSM
ISM
NIC
NNC
Genic Genomic
Antisense
Fusion


Structural Categories by Transcript Length
30
20
10
0
%
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
23
26
Transcript length, kb
Structural Category
FSM
ISM
NIC
NNC
Genic
Genomic
Antisense
Fusion

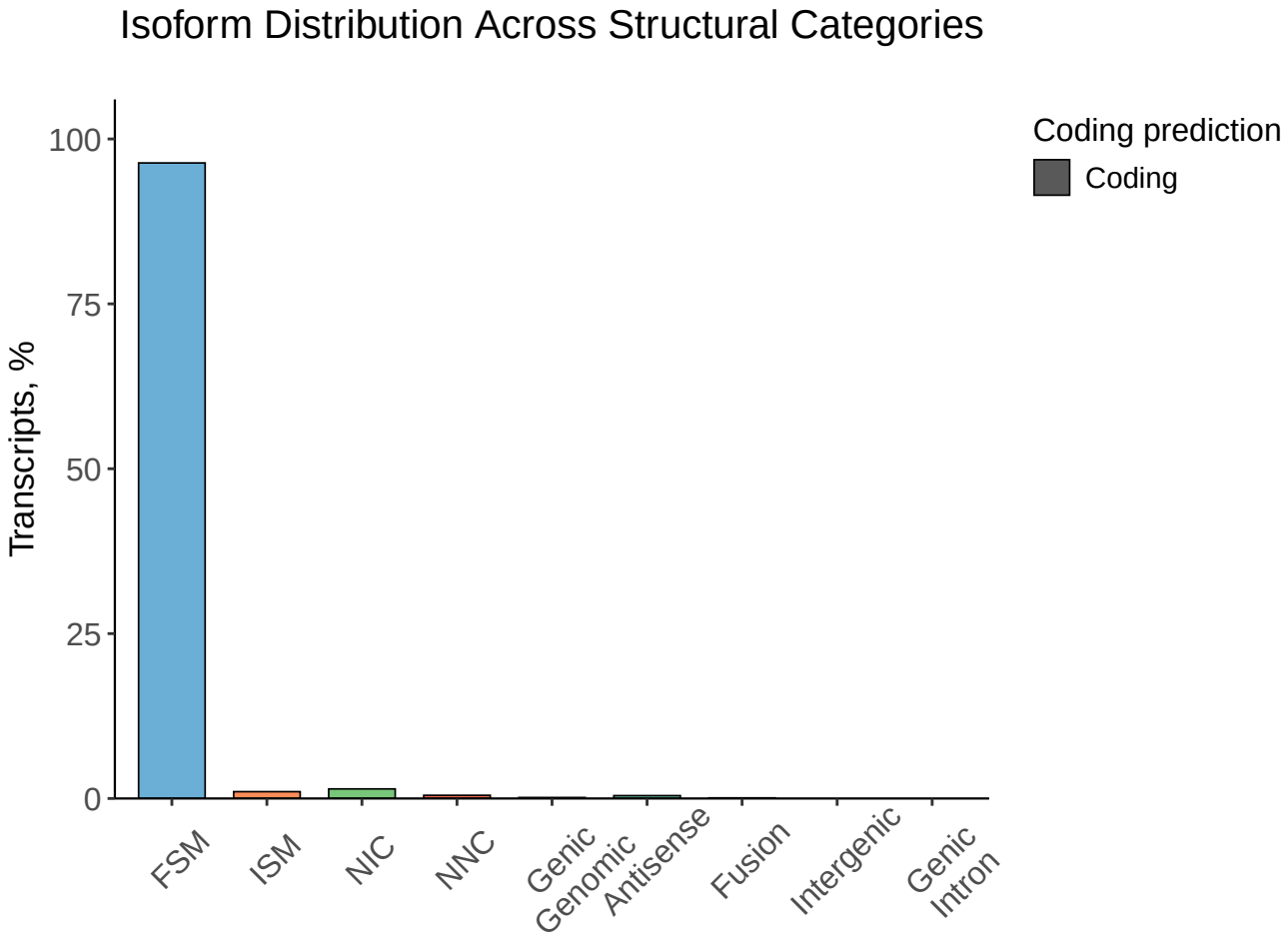

Annotated vs Novel Gene Expression
0.050
0.025
0.000
−0.025
−0.050
log2(Gene_TPM+1)
Annotated Genes
Novel Genes

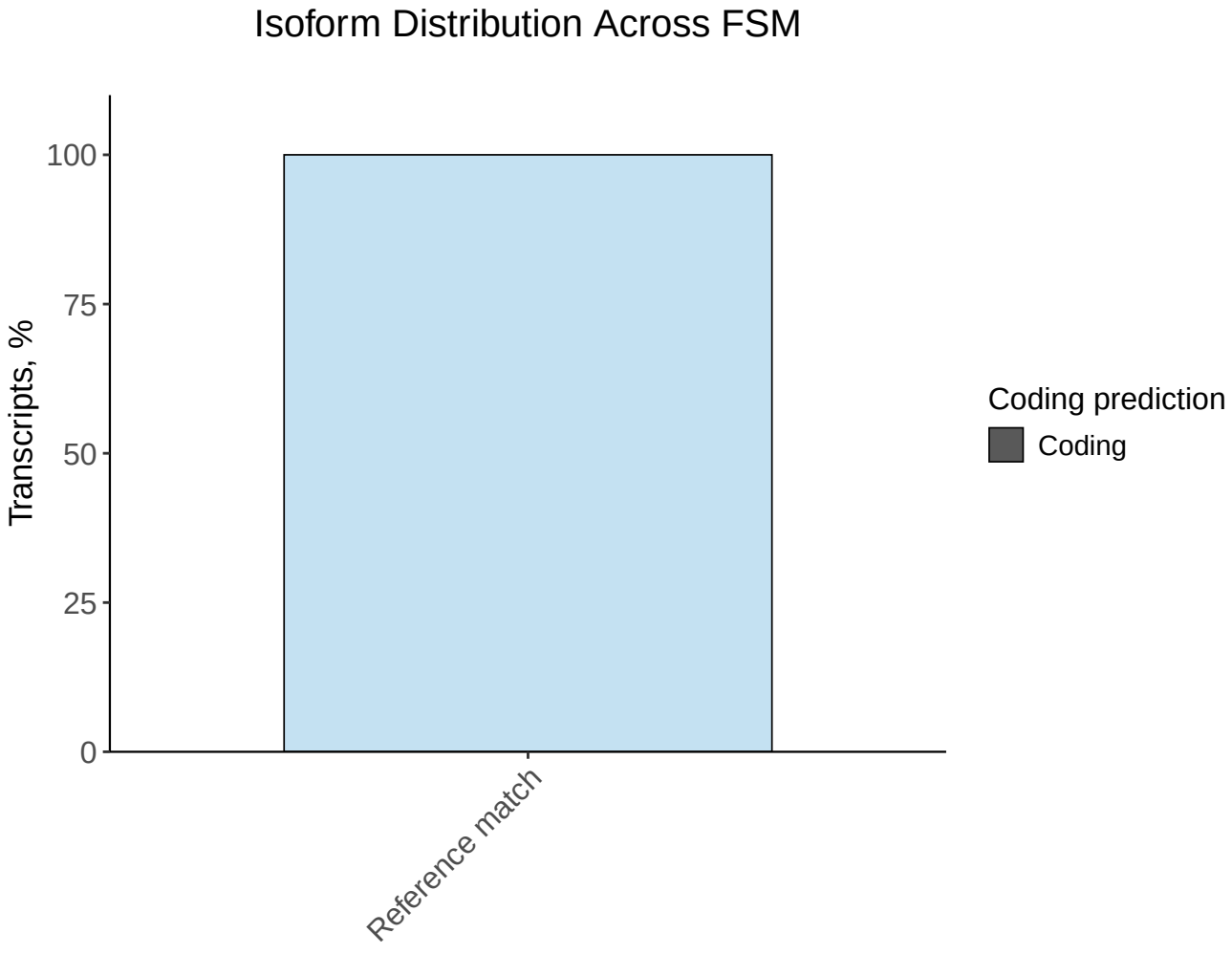

All Transcript Lengths Distribution
Count
600
400
200
0
0
10000
20000
Transcript length

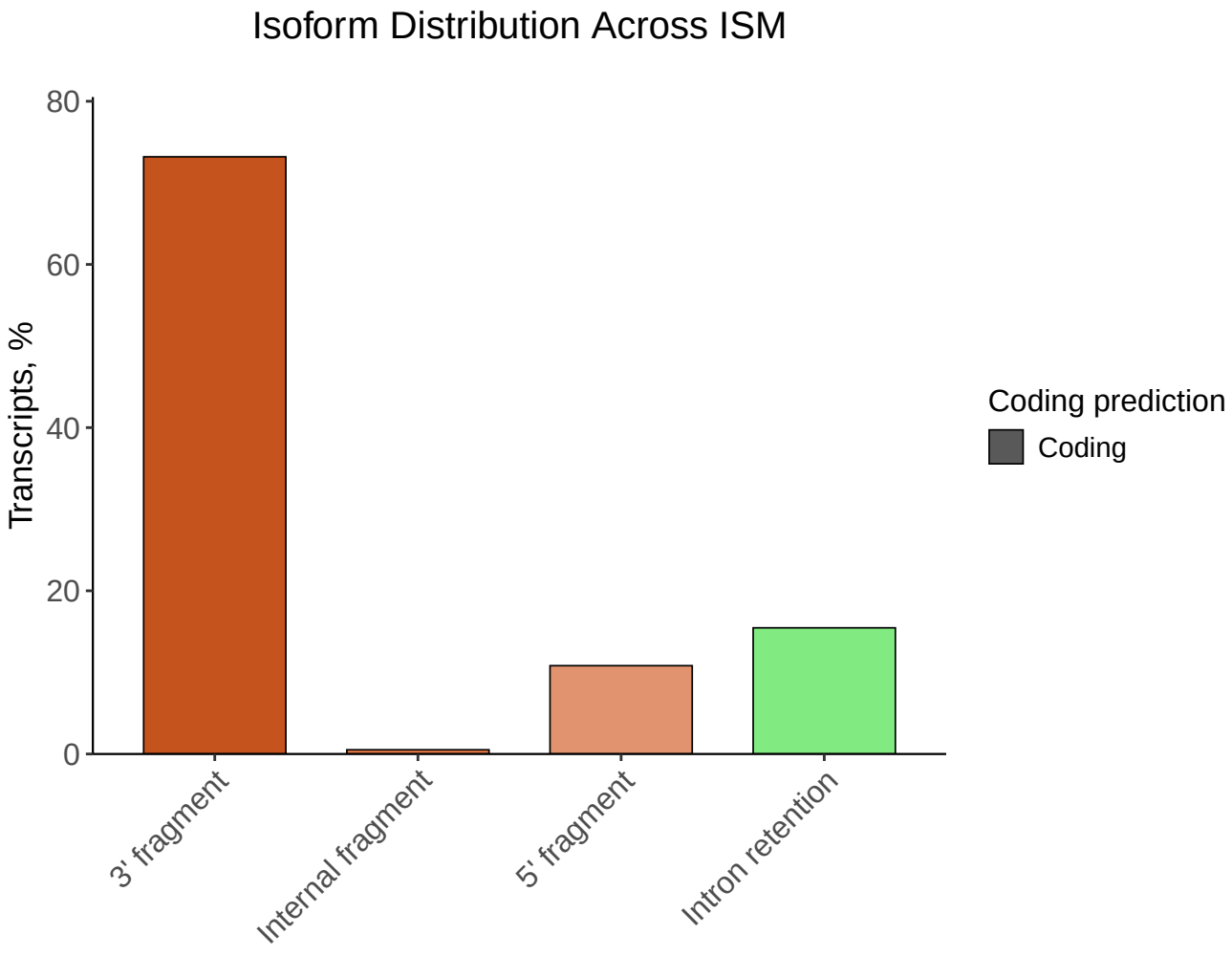

Transcript Lengths Distribution by Structural Category
Count
600
400
200
0
0
10000
20000
Transcript length
FSM
NIC
Genic Genomic
Fusion
ISM
NNC
Antisense

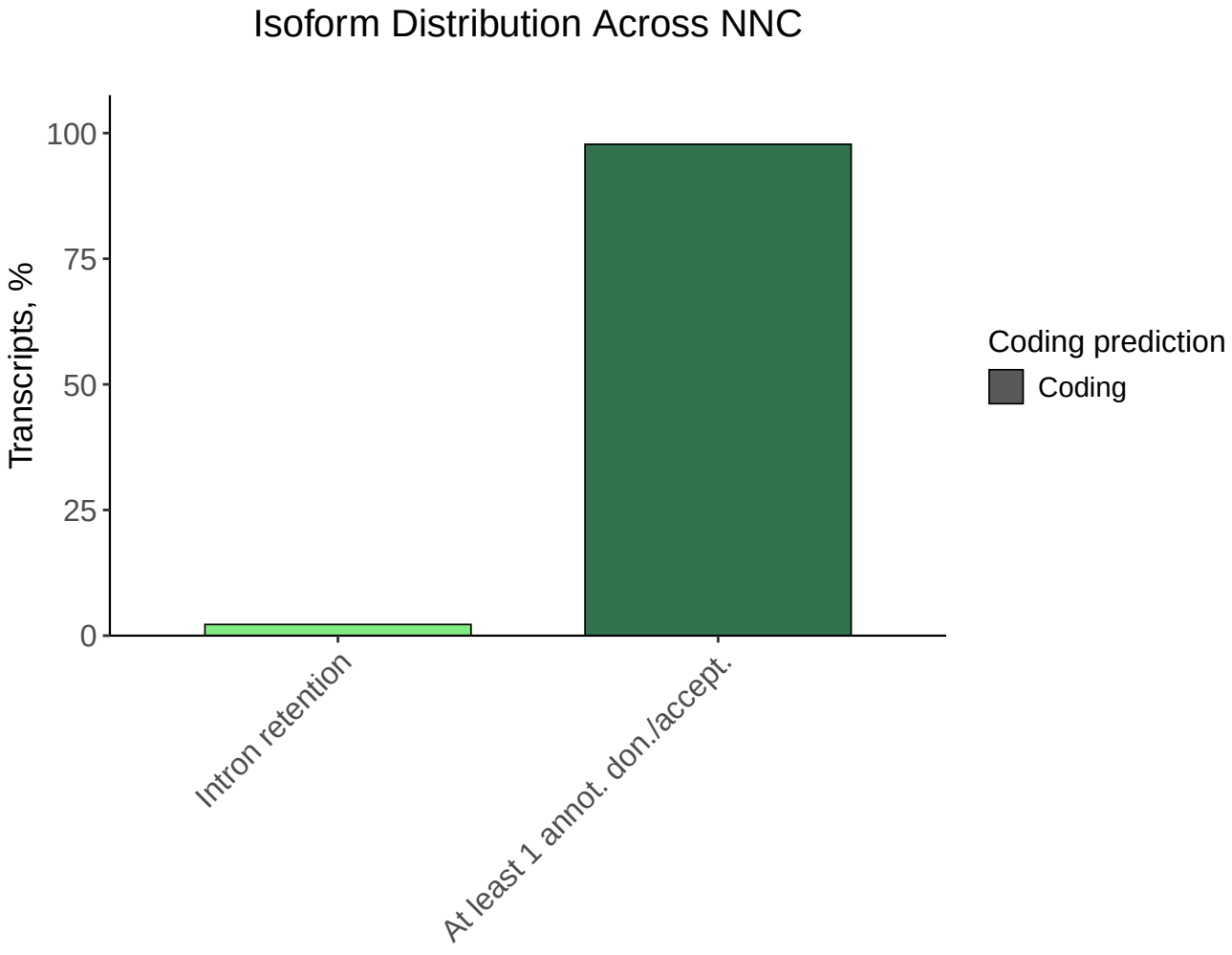

Mono− vs Multi− Exon Transcript Lengths Distribution
Count
600
400
200
0
0
10000
20000
Transcript length
Multi−Exon
Mono−Exon

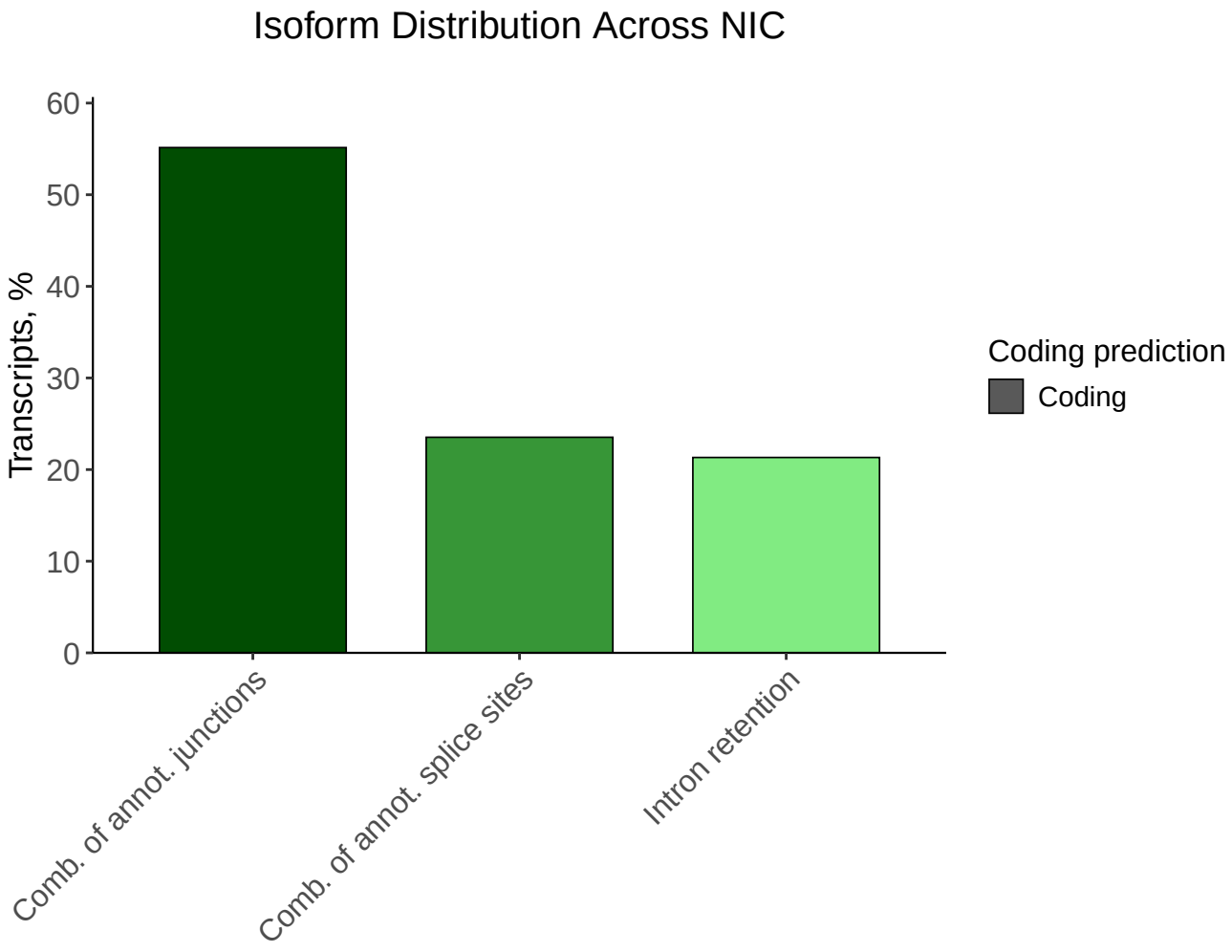
Structural Isoform Characterization

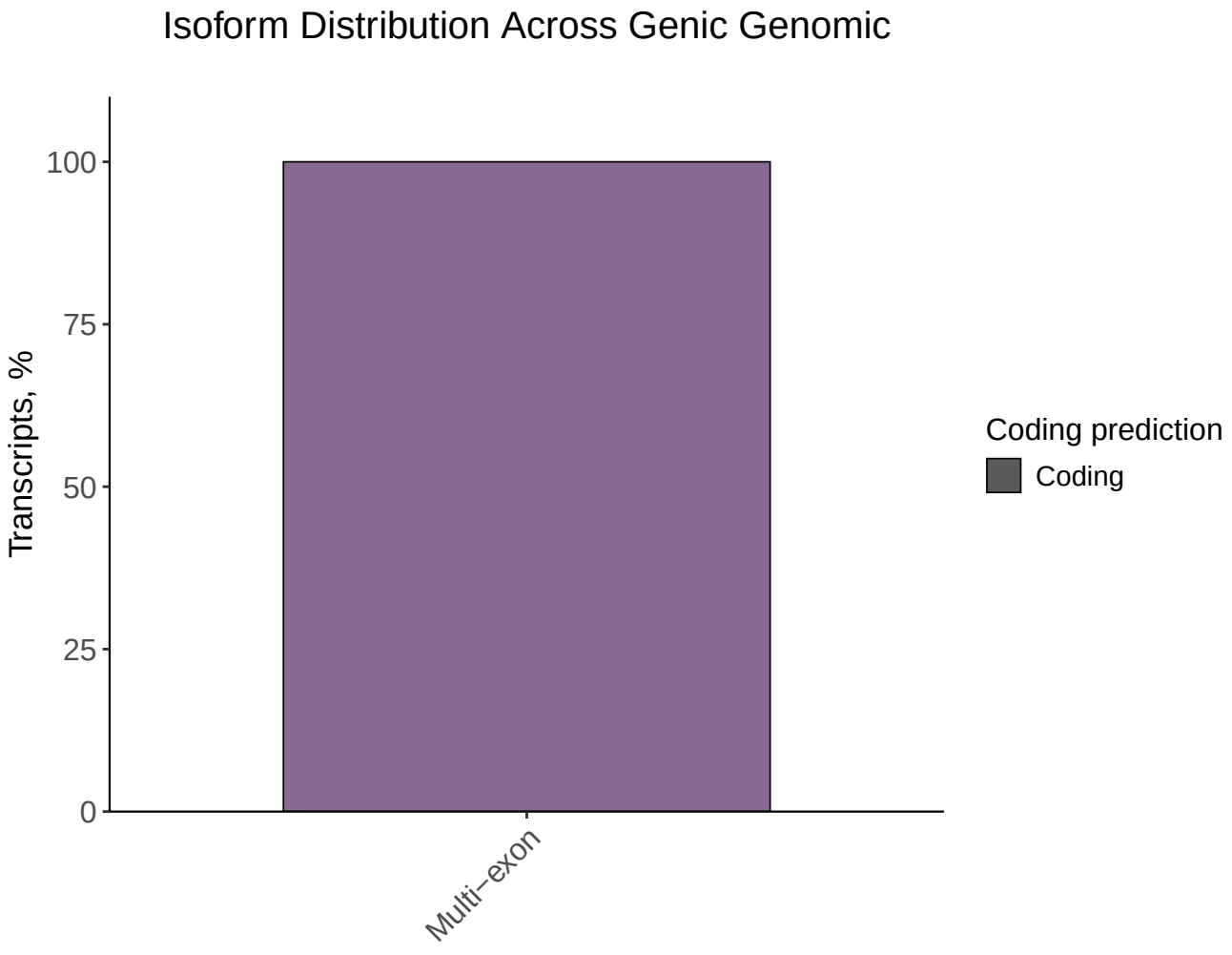

Isoform Distribution Across Structural Categories
Transcripts, %
100
75
50
25
0
FSM
ISM
NIC
NNC
Genic Genomic
Antisense
Fusion
Intergenic
Genic Intron
Coding prediction
Coding

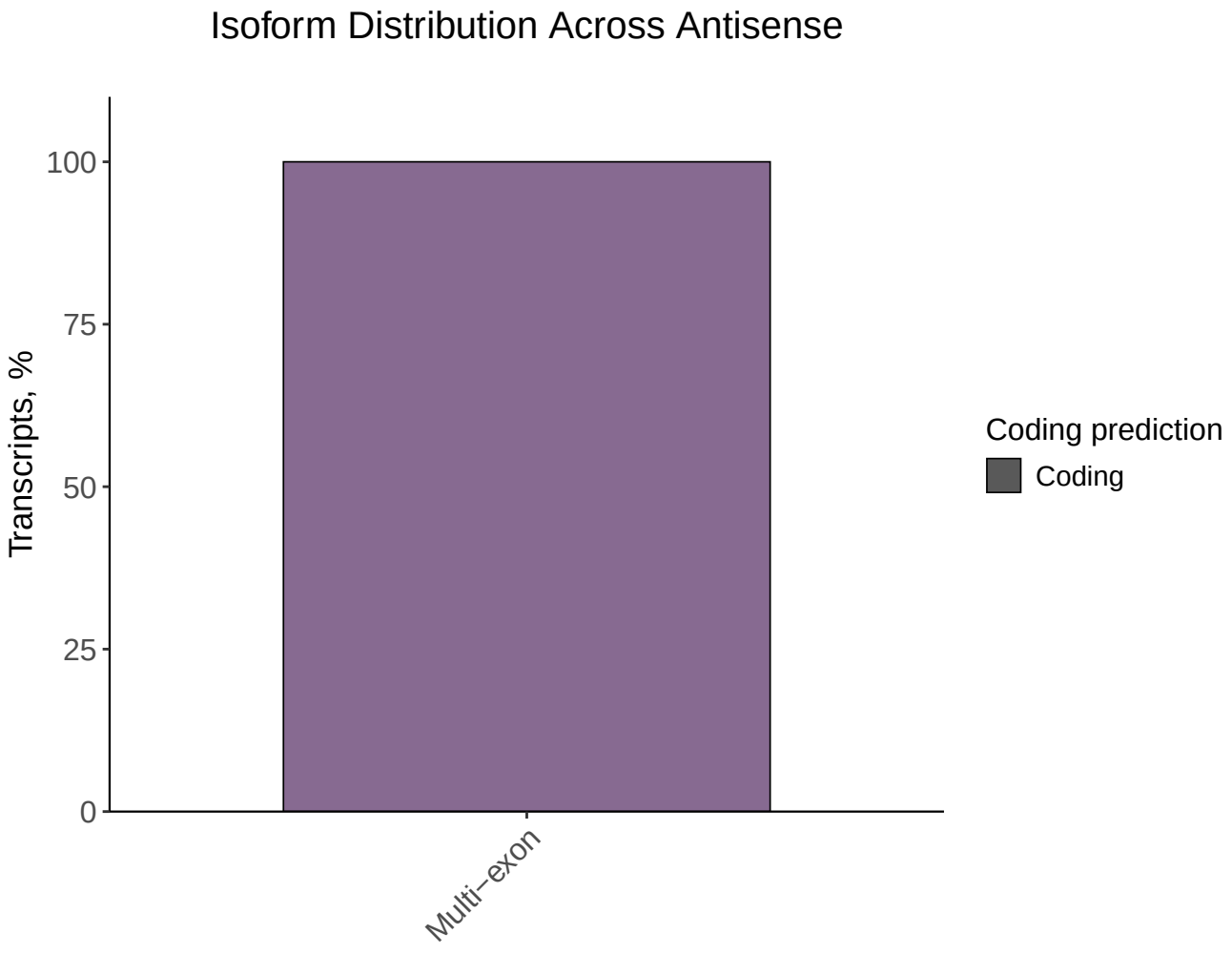

Isoform Distribution Across FSM
Transcripts, %
100
75
50
25
0
Reference match
Coding prediction
Coding

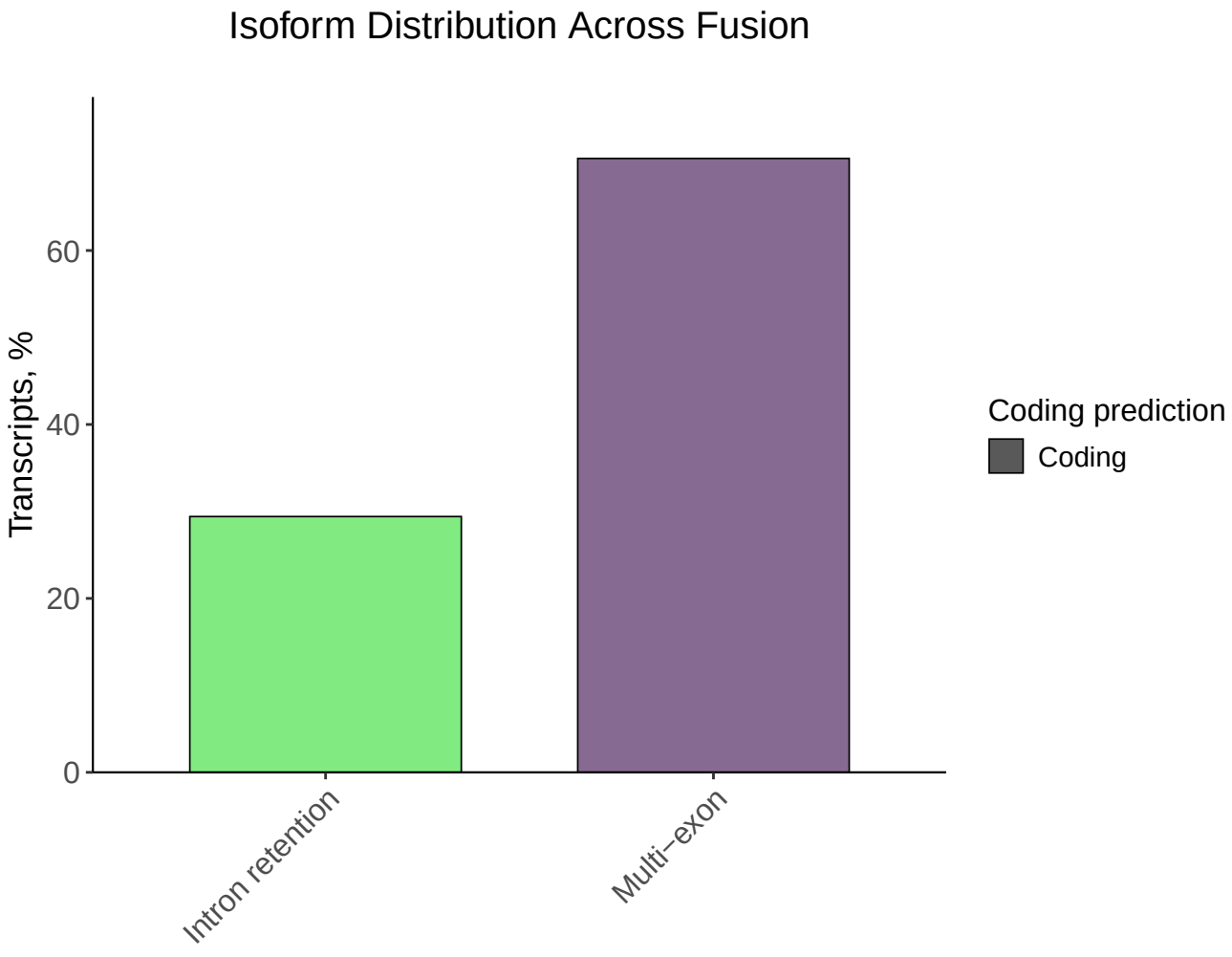

Isoform Distribution Across ISM
Transcripts, %
80
60
40
20
0
3' fragment
Internal fragment
5' fragment
Intron retention
Coding prediction
Coding

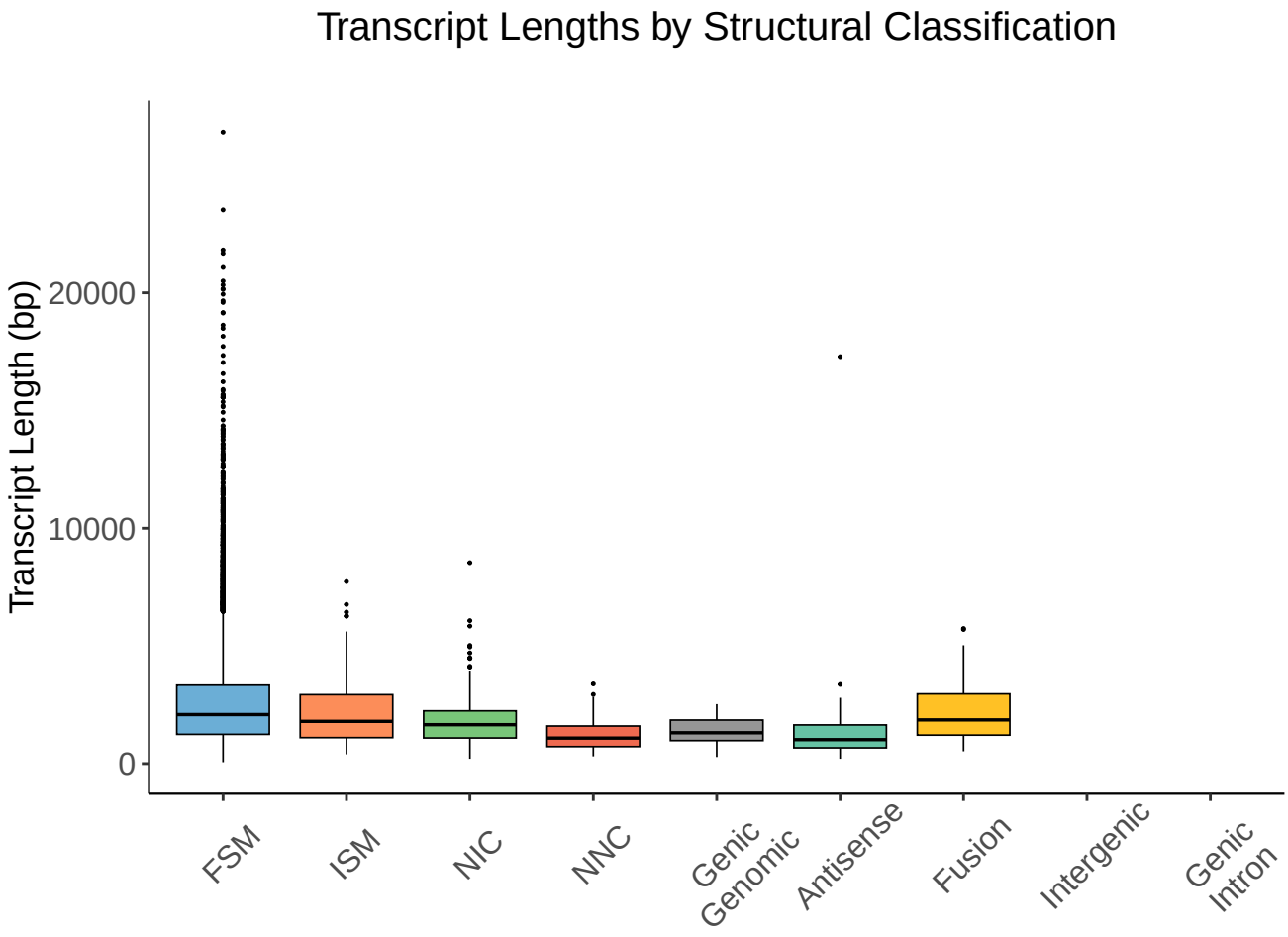

Isoform Distribution Across NNC
Transcripts, %
100
75
50
25
0
Intron retention
At least 1 annot. don./accept.
Coding prediction
Coding

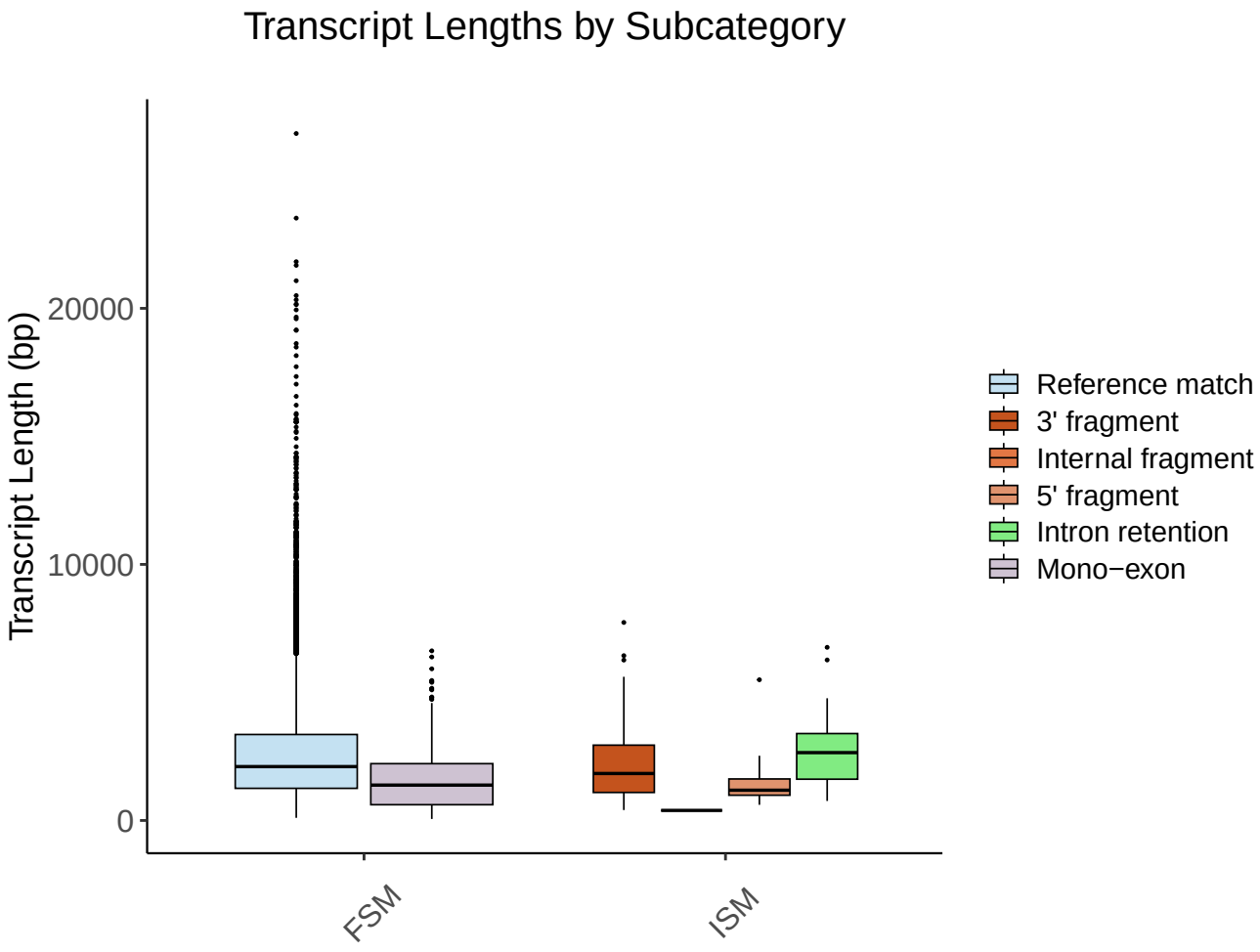

Isoform Distribution Across NIC
Transcripts, %
60
50
40
30
20
10
0
Comb. of annot. junctions
Comb. of annot. splice sites
Intron retention
Coding prediction
Coding

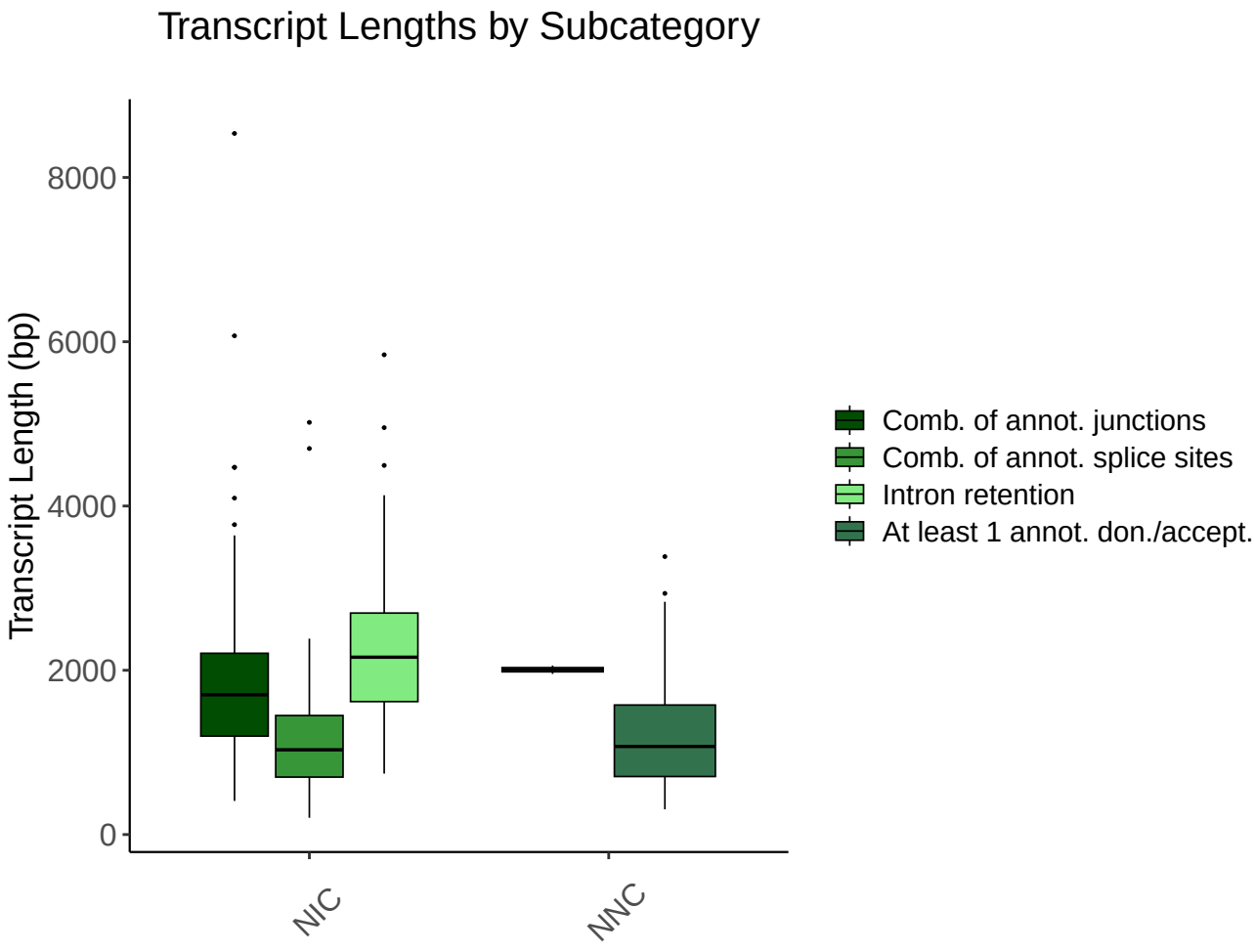

Isoform Distribution Across Genic Genomic
Transcripts, %
100
75
50
25
0
Multi−exon
Coding prediction
Coding

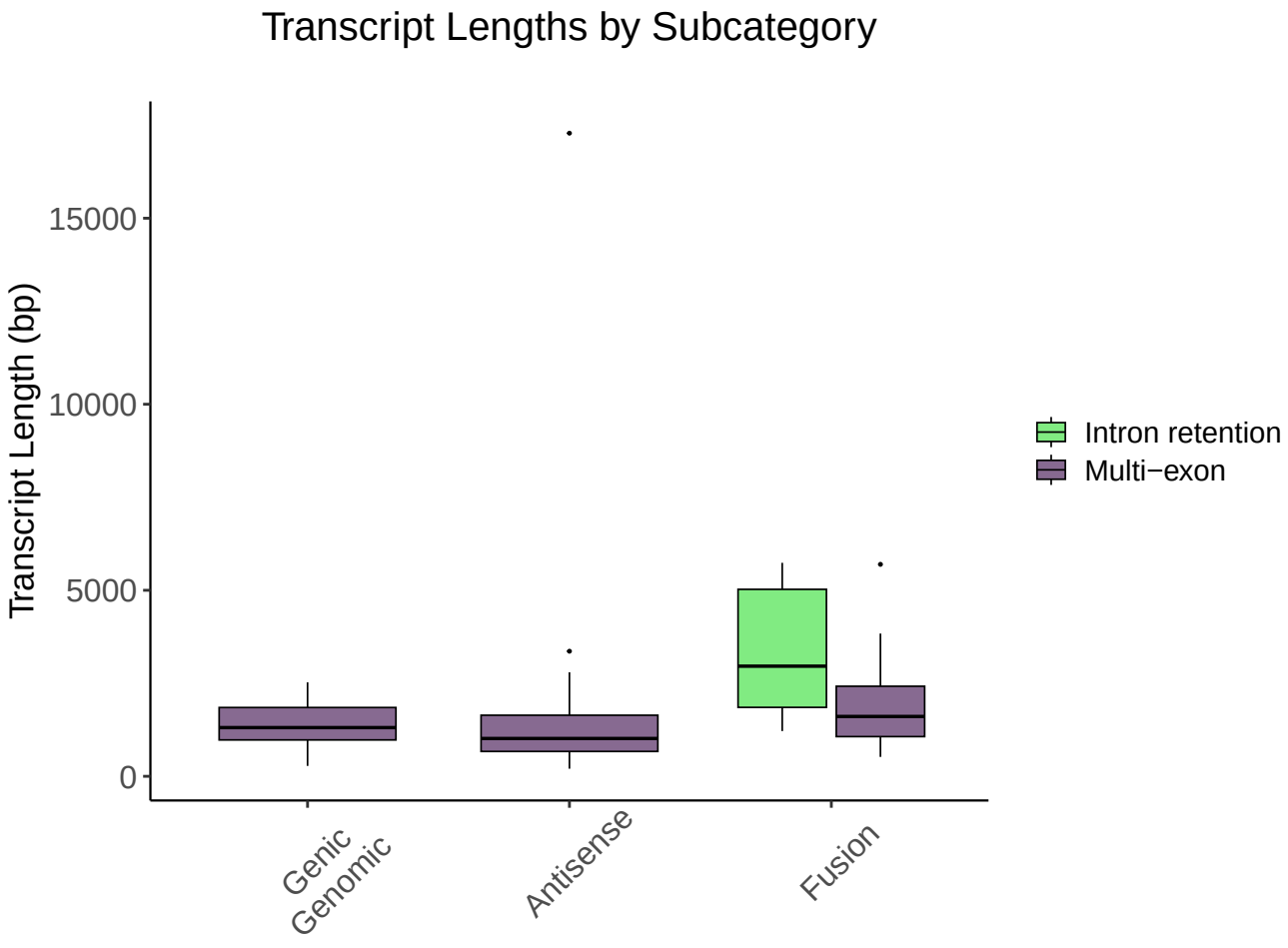

Isoform Distribution Across Antisense
Transcripts, %
100
75
50
25
0
Multi−exon
Coding prediction
Coding

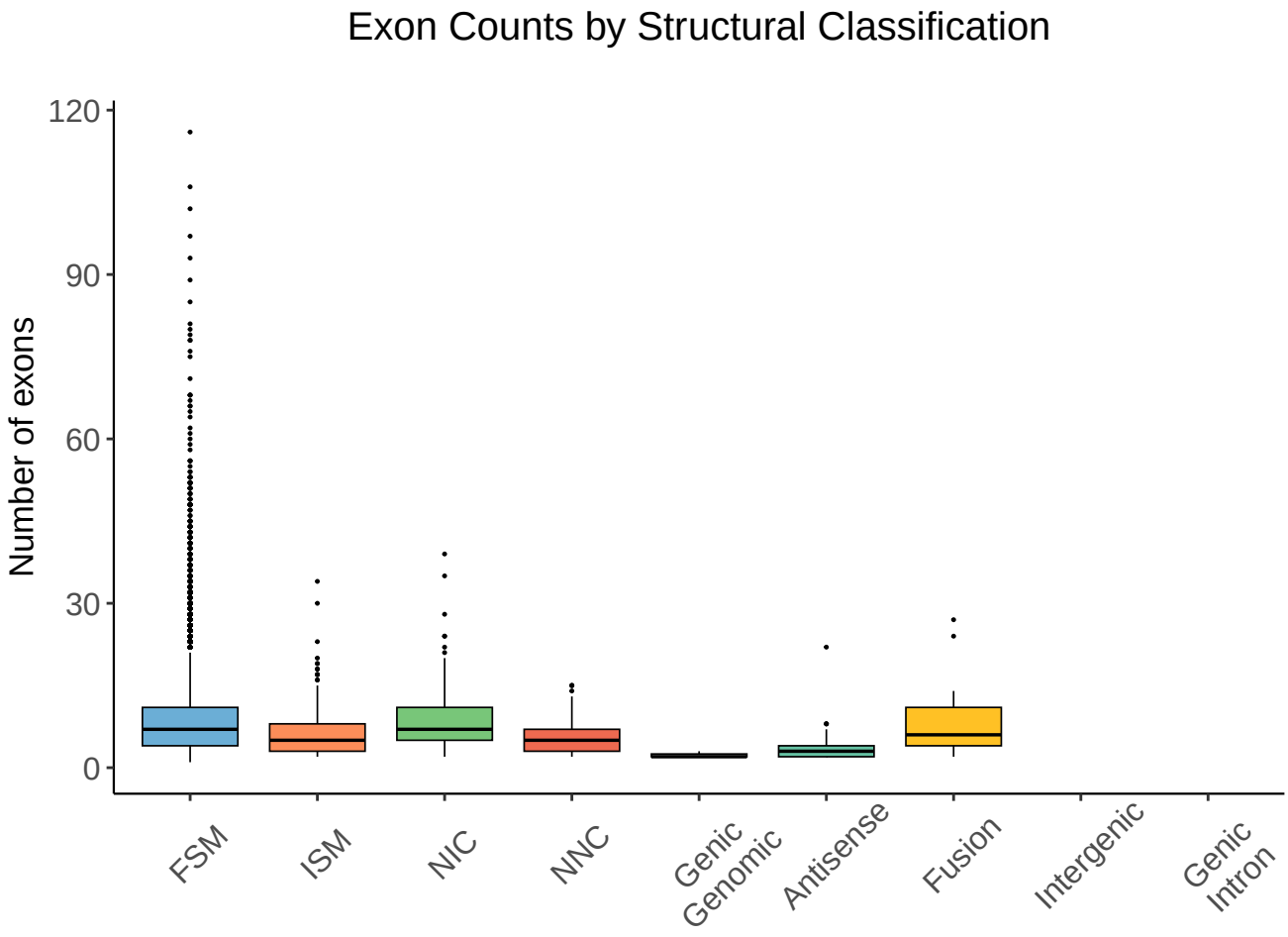

Isoform Distribution Across Fusion
Transcripts, %
0
20
40
60
Intron retention
Multi−exon
Coding prediction
Coding

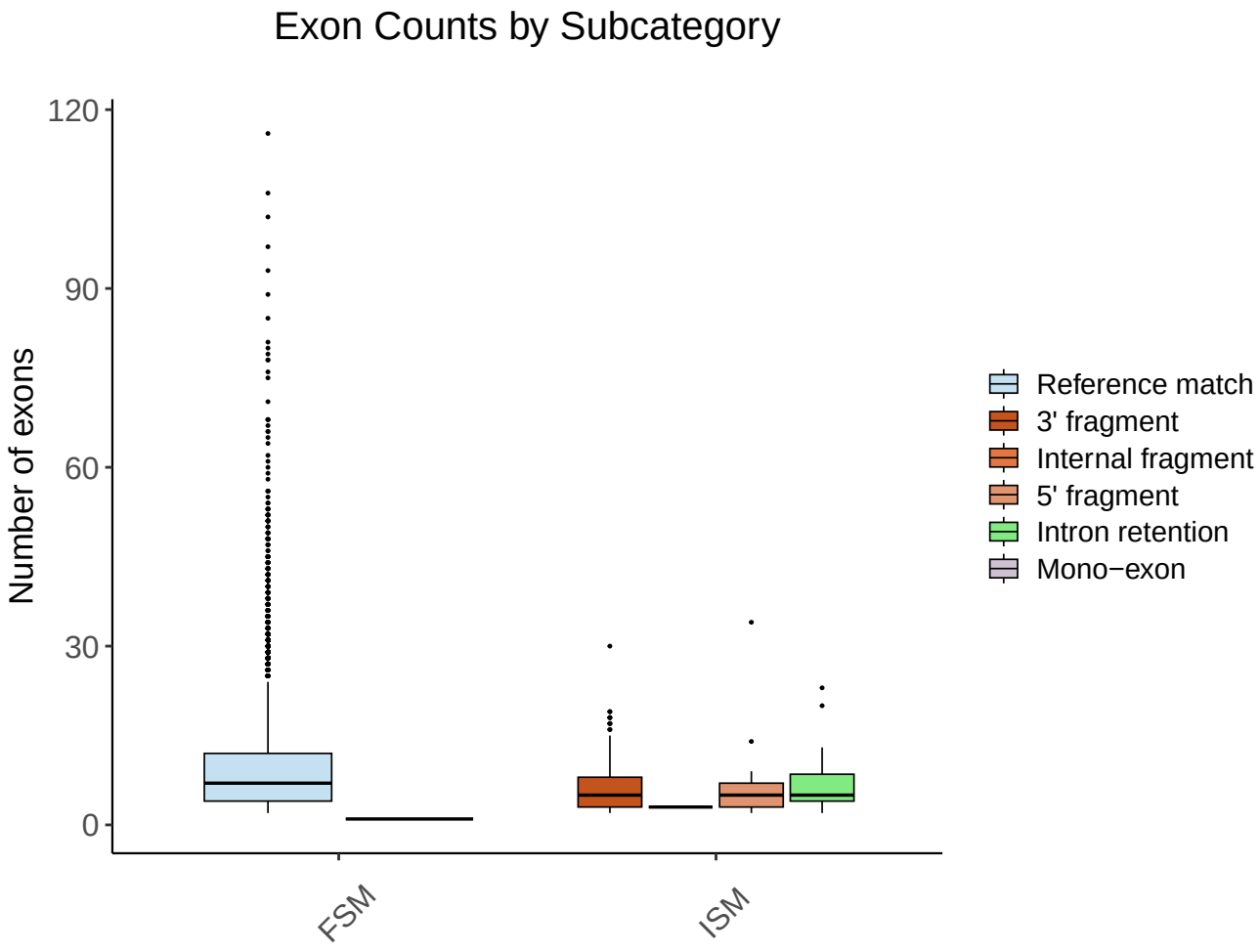

Transcript Lengths by Structural Classification
Transcript Length (bp)
20000
10000
0
FSM
ISM
NIC
NNC
Genic Genomic
Antisense
Fusion
Intergenic
Genic Intron

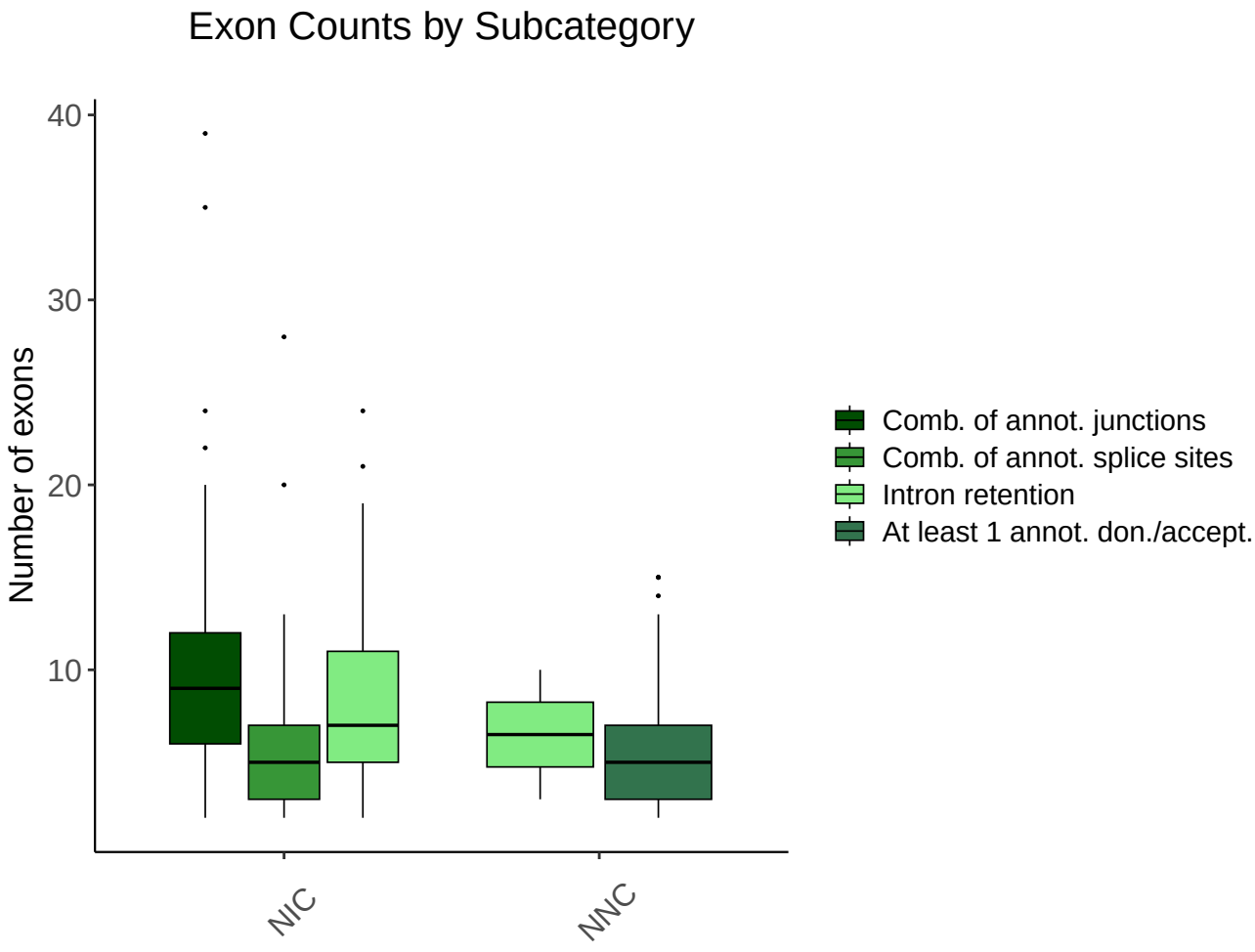

Transcript Lengths by Subcategory
Transcript Length (bp)
20000
10000
0
FSM
ISM
Reference match
3' fragment
Internal fragment
5' fragment
Intron retention
Mono−exon

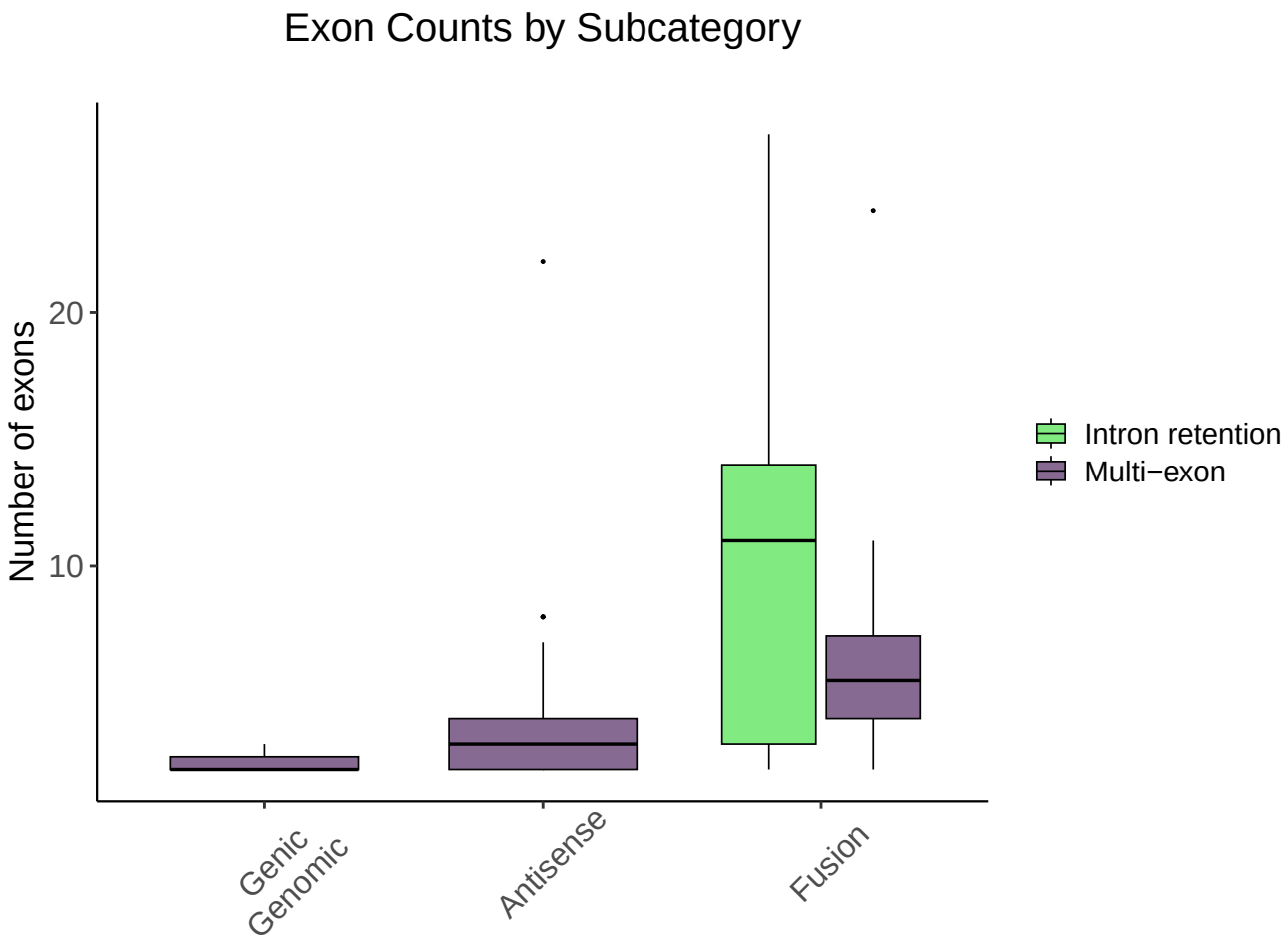

Transcript Lengths by Subcategory
Transcript Length (bp)
8000
6000
4000
2000
0
NIC
NNC
Comb. of annot. junctions
Comb. of annot. splice sites
Intron retention
At least 1 annot. don./accept.

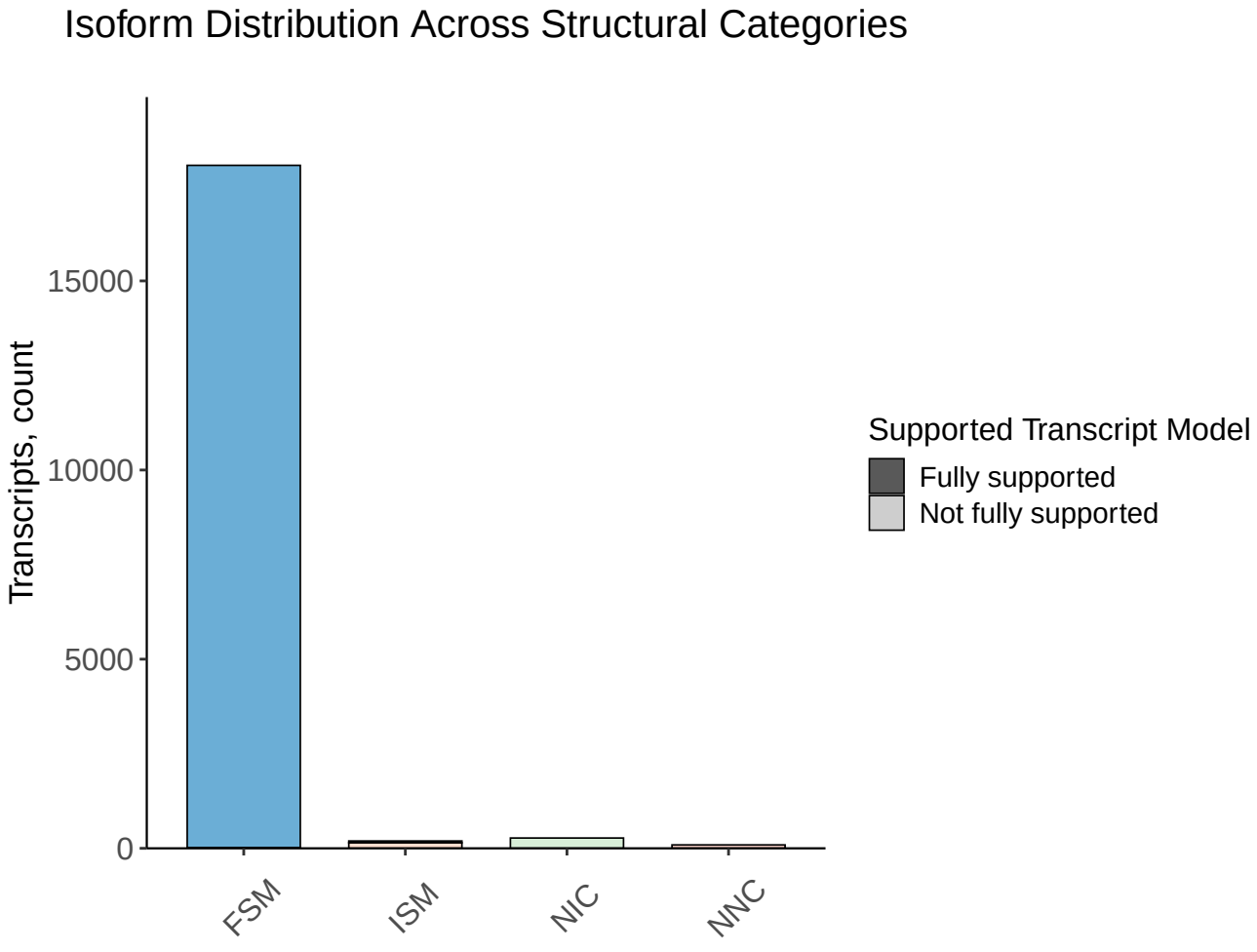

Transcript Lengths by Subcategory
Transcript Length (bp)
15000
10000
5000
0
Genic Genomic
Antisense
Fusion
Intron retention
Multi−exon

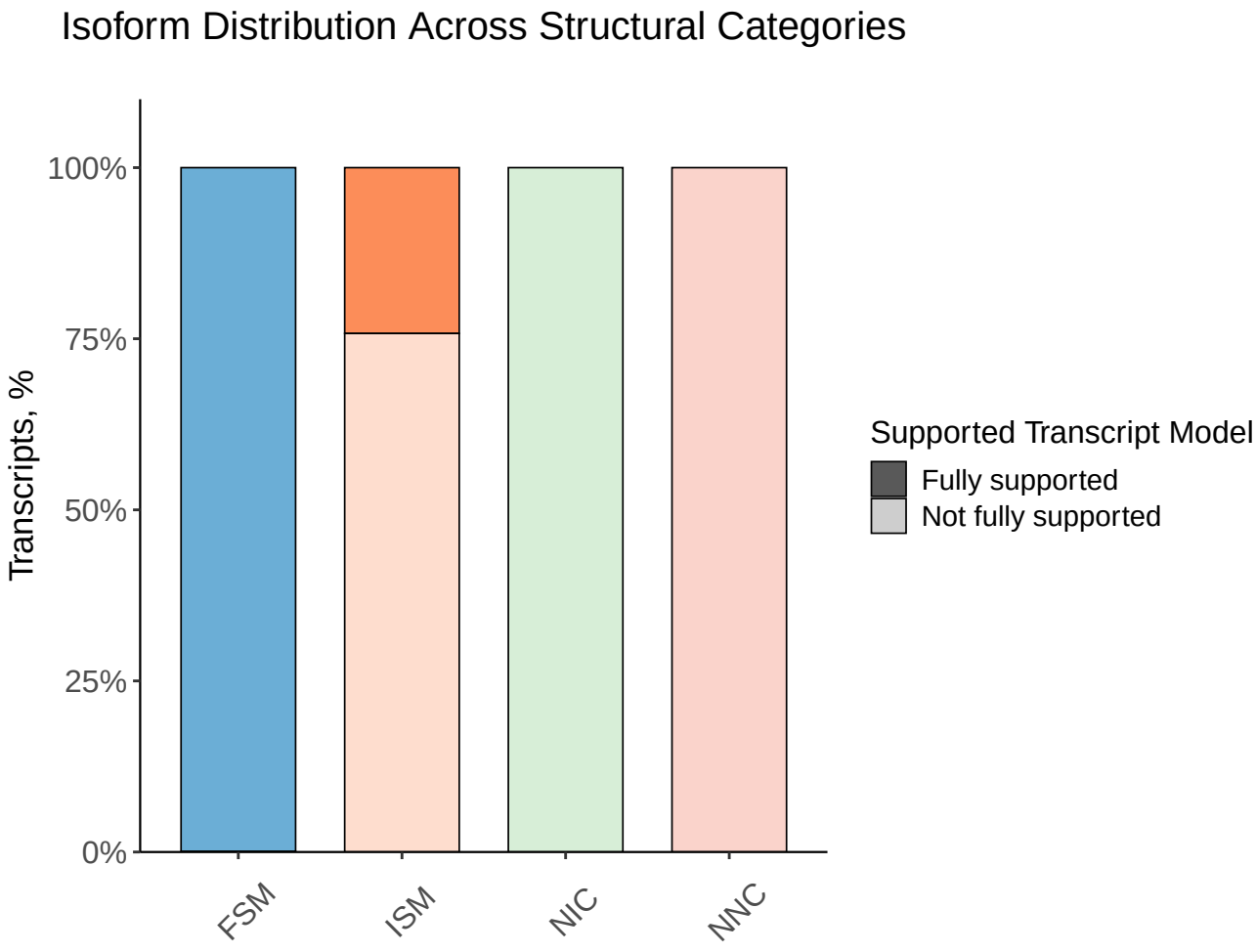

Exon Counts by Structural Classification
Number of exons
120
90
60
30
0
FSM
ISM
NIC
NNC
Genic Genomic
Antisense
Fusion
Intergenic
Genic Intron

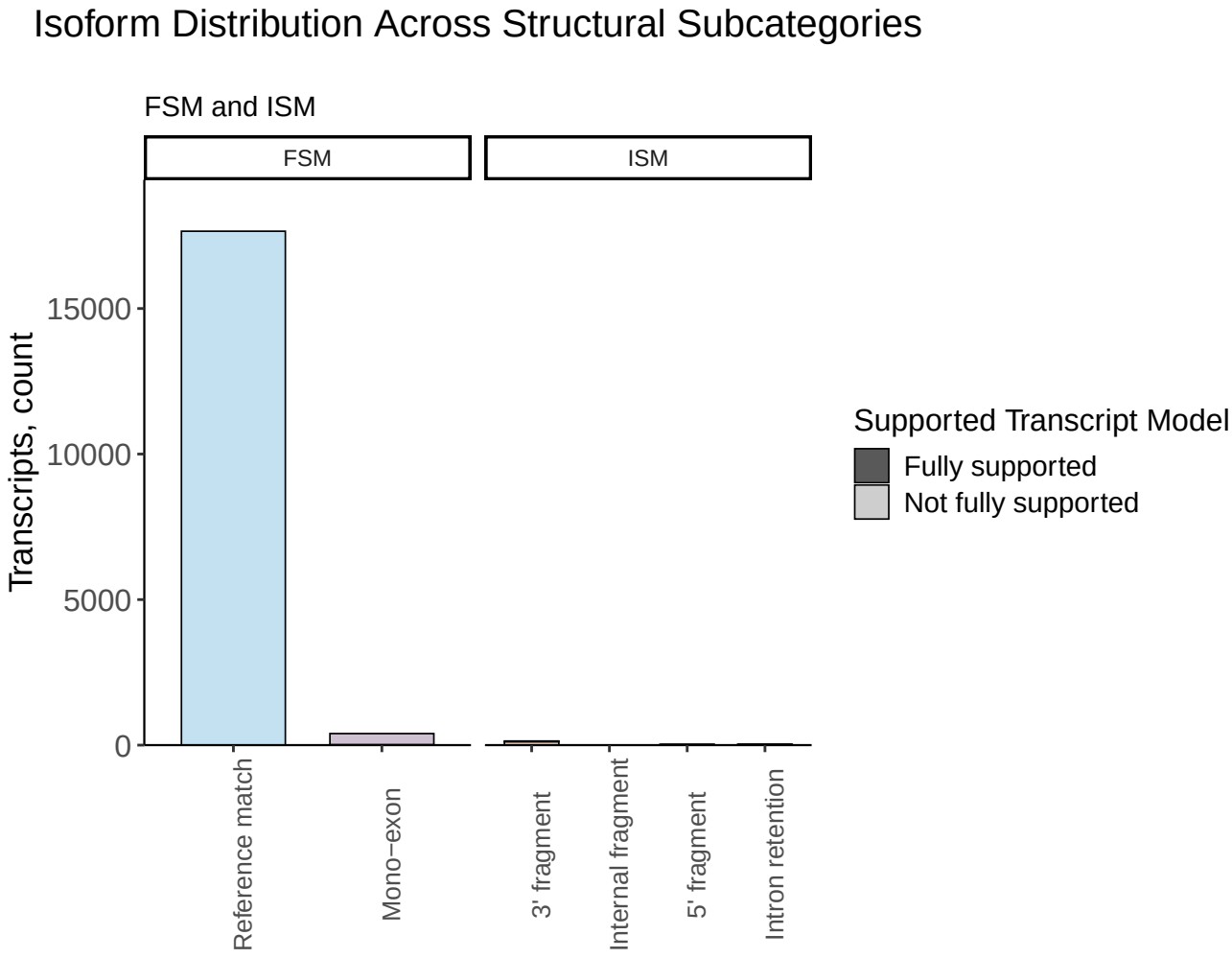

Exon Counts by Subcategory
Number of exons
120
90
60
30
0
FSM
ISM
Reference match
3' fragment
Internal fragment
5' fragment
Intron retention
Mono−exon

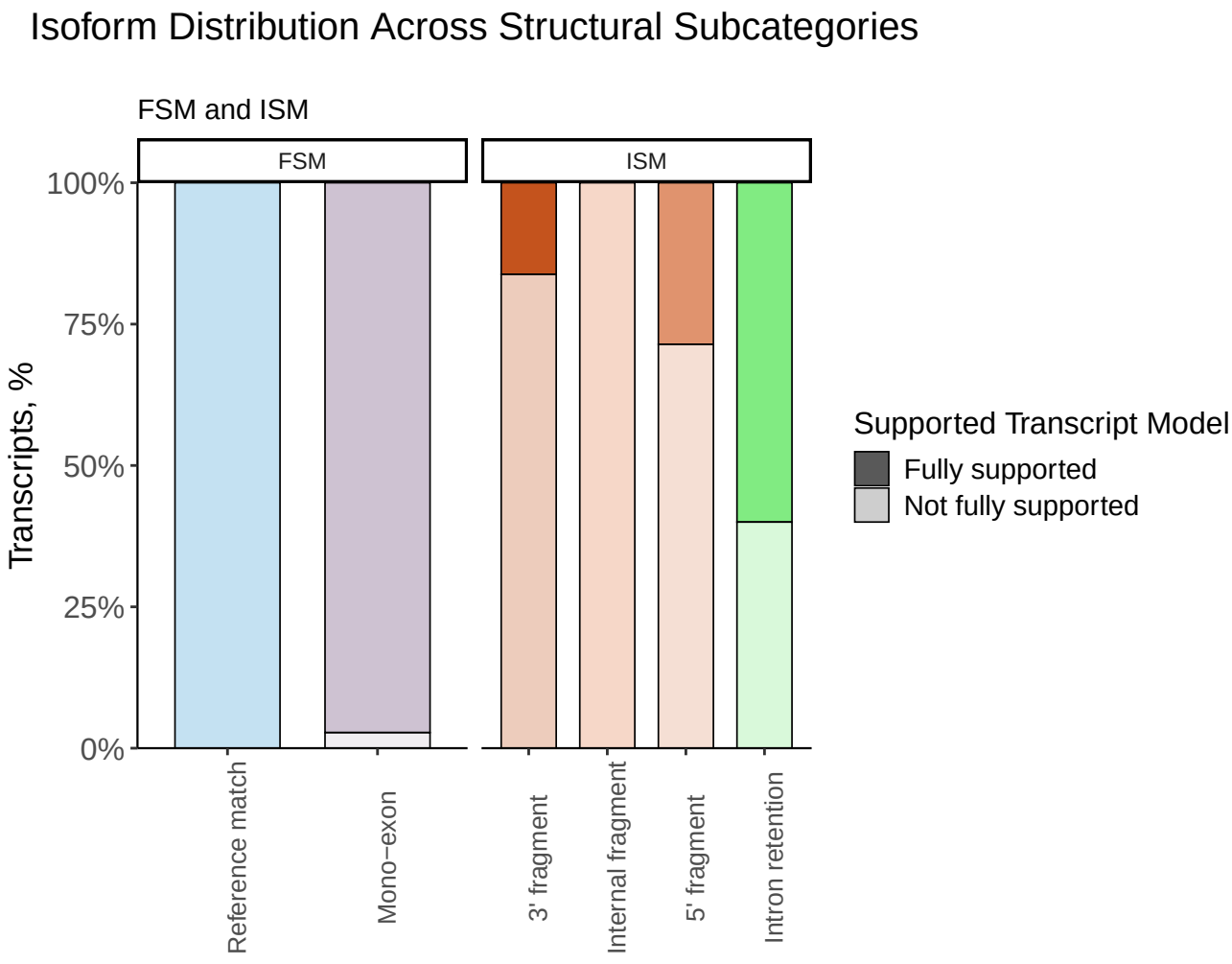

Exon Counts by Subcategory
Number of exons
40
30
20
10
NIC
NNC
Comb. of annot. junctions
Comb. of annot. splice sites
Intron retention
At least 1 annot. don./accept.

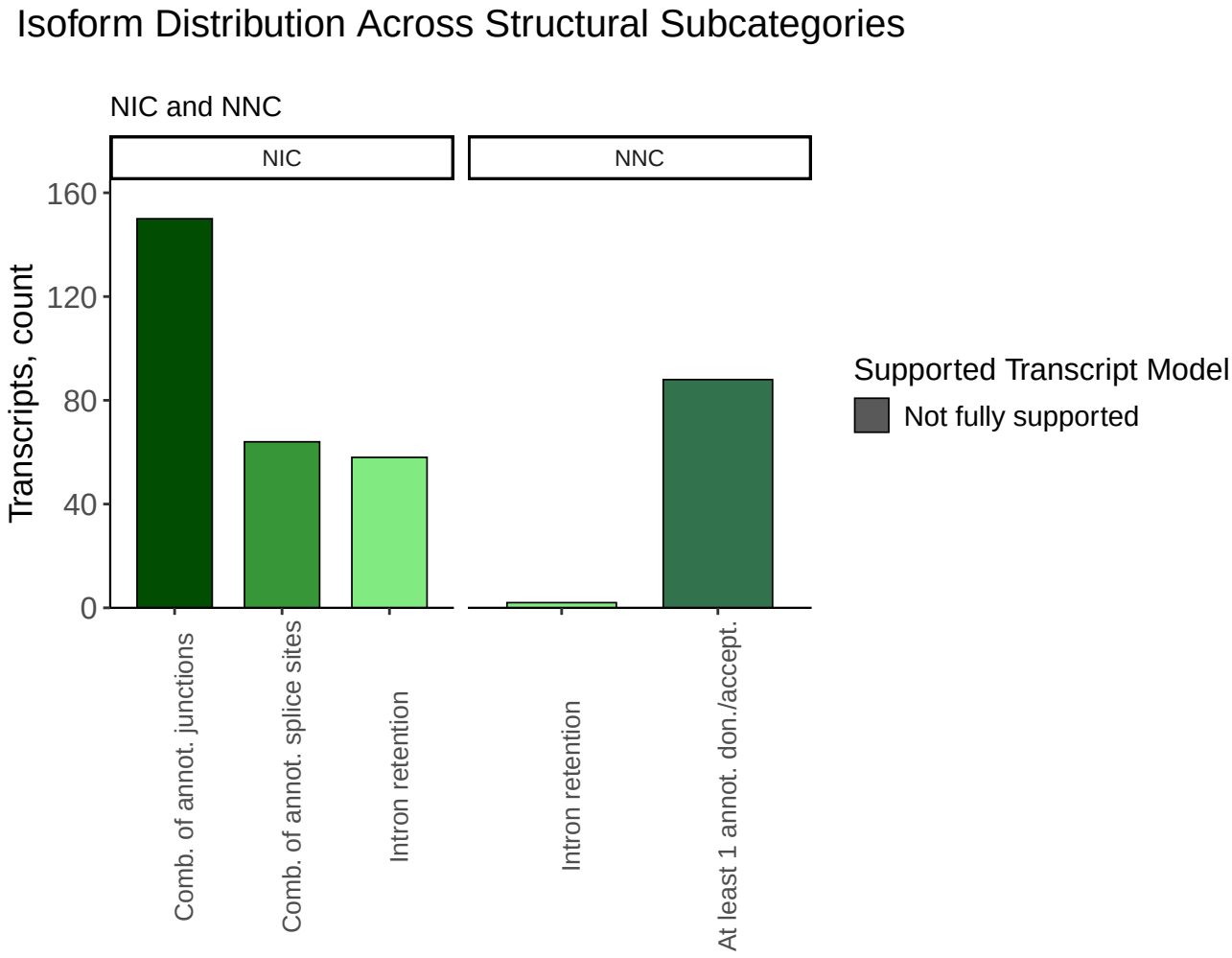

Exon Counts by Subcategory
Number of exons
20
10
Genic Genomic
Antisense
Fusion
Intron retention
Multi−exon

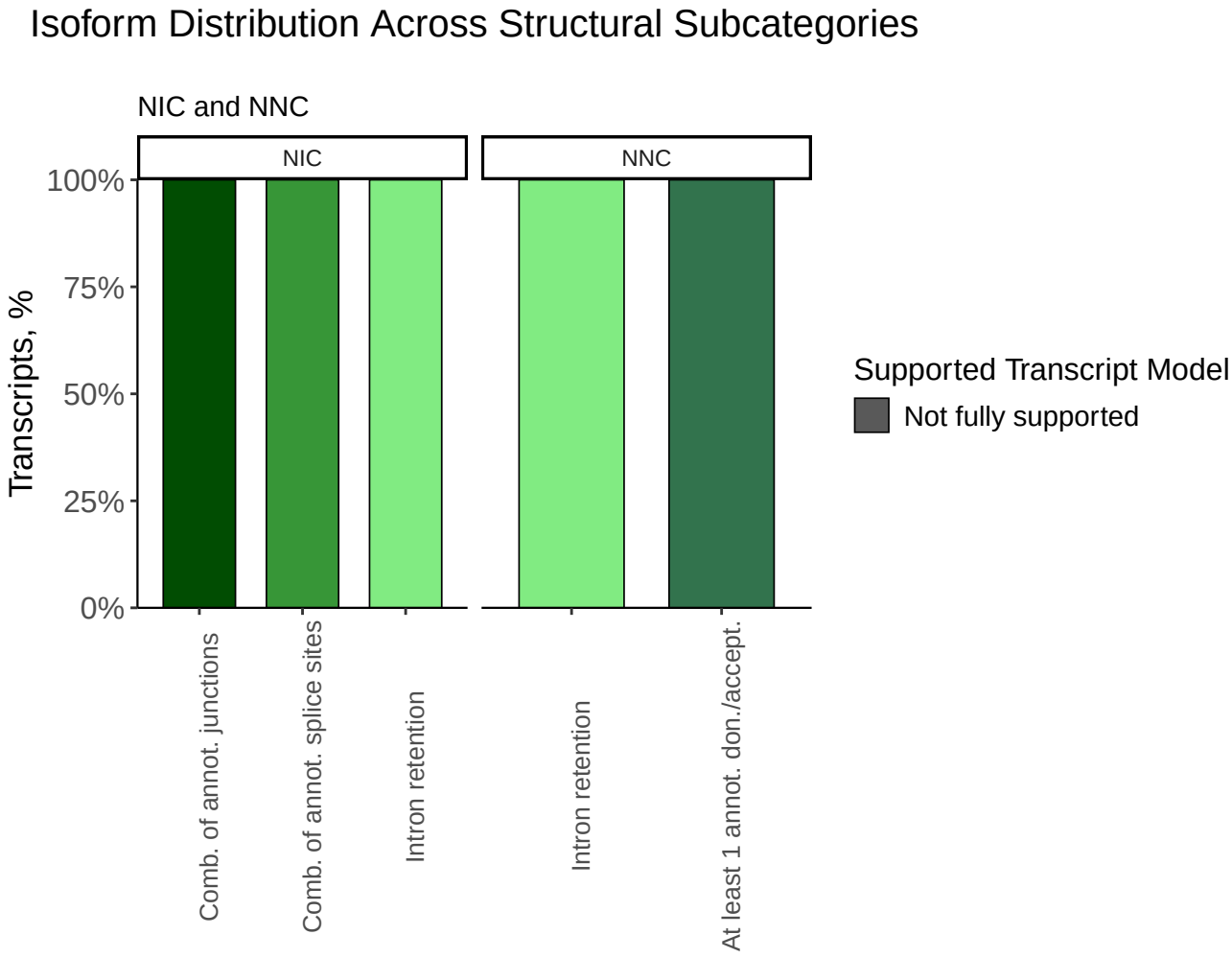

Isoform Distribution Across Structural Categories
Transcripts, count
15000
10000
5000
0
FSM
ISM
NIC
NNC
Supported Transcript Model
Fully supported
Not fully supported

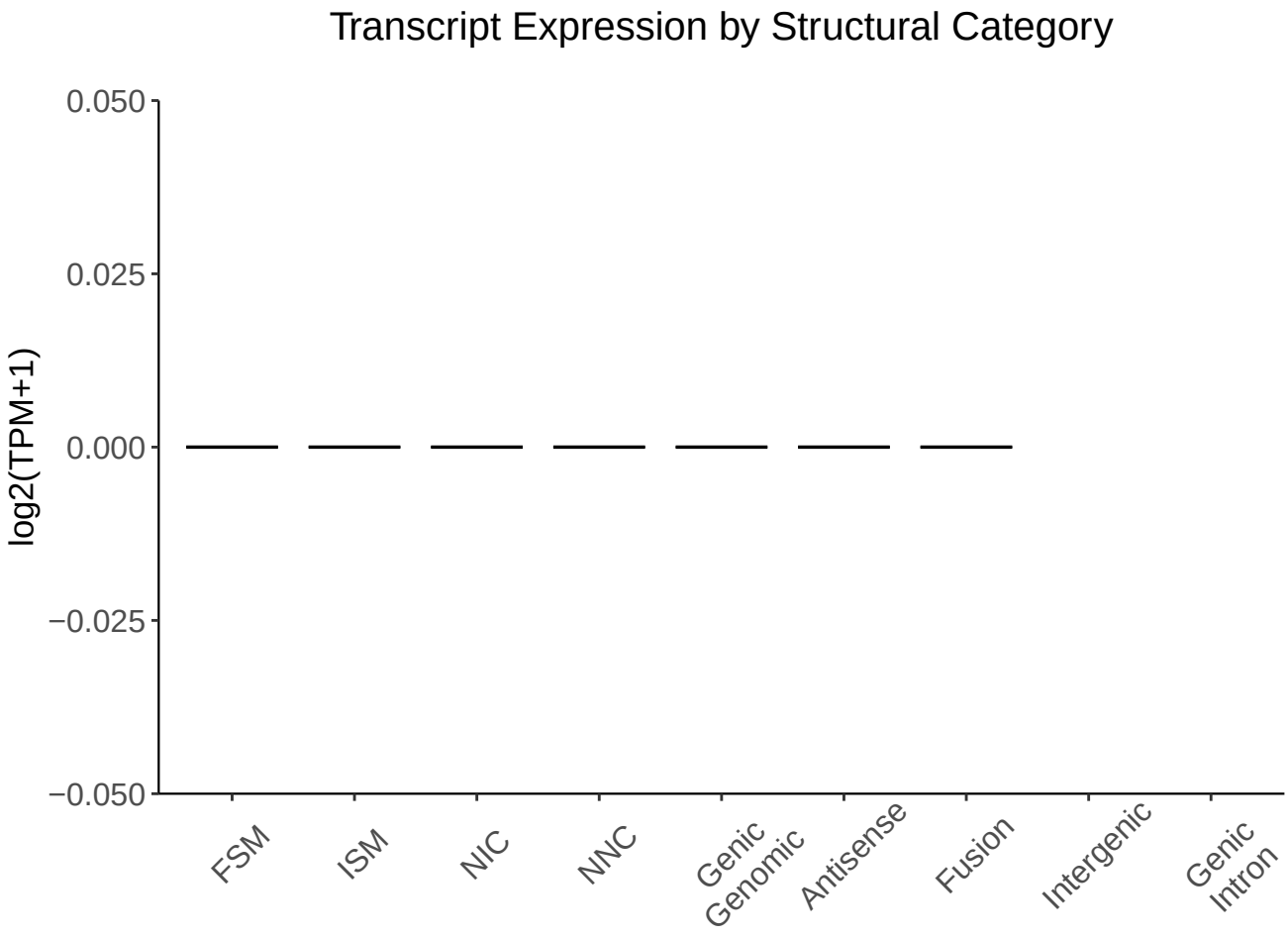

Isoform Distribution Across Structural Categories
Transcripts, %
100%
75%
50%
25%
0%
FSM
ISM
NIC
NNC
Supported Transcript Model
Fully supported
Not fully supported


Isoform Distribution Across Structural Subcategories
FSM and ISM
FSM
ISM
Transcripts, count
15000
10000
5000
0
Reference match
Mono−exon
3' fragment
Internal fragment
5' fragment
Intron retention
Supported Transcript Model
Fully supported
Not fully supported

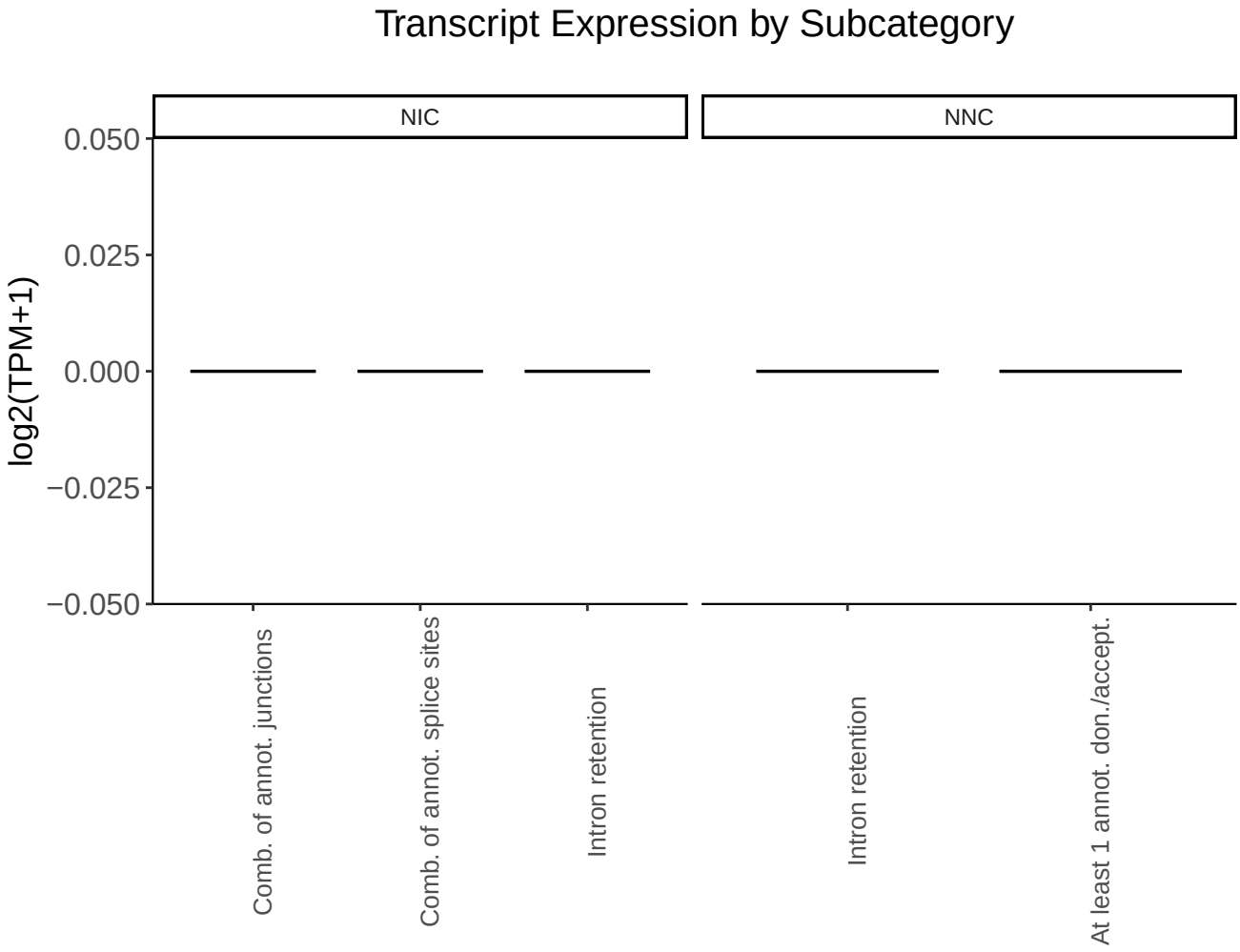

Isoform Distribution Across Structural Subcategories
FSM and ISM
FSM
ISM
100%
75%
50%
25%
0%
Transcripts, %
Reference match
Mono−exon
3' fragment
Internal fragment
5' fragment
Intron retention
Supported Transcript Model
Fully supported
Not fully supported

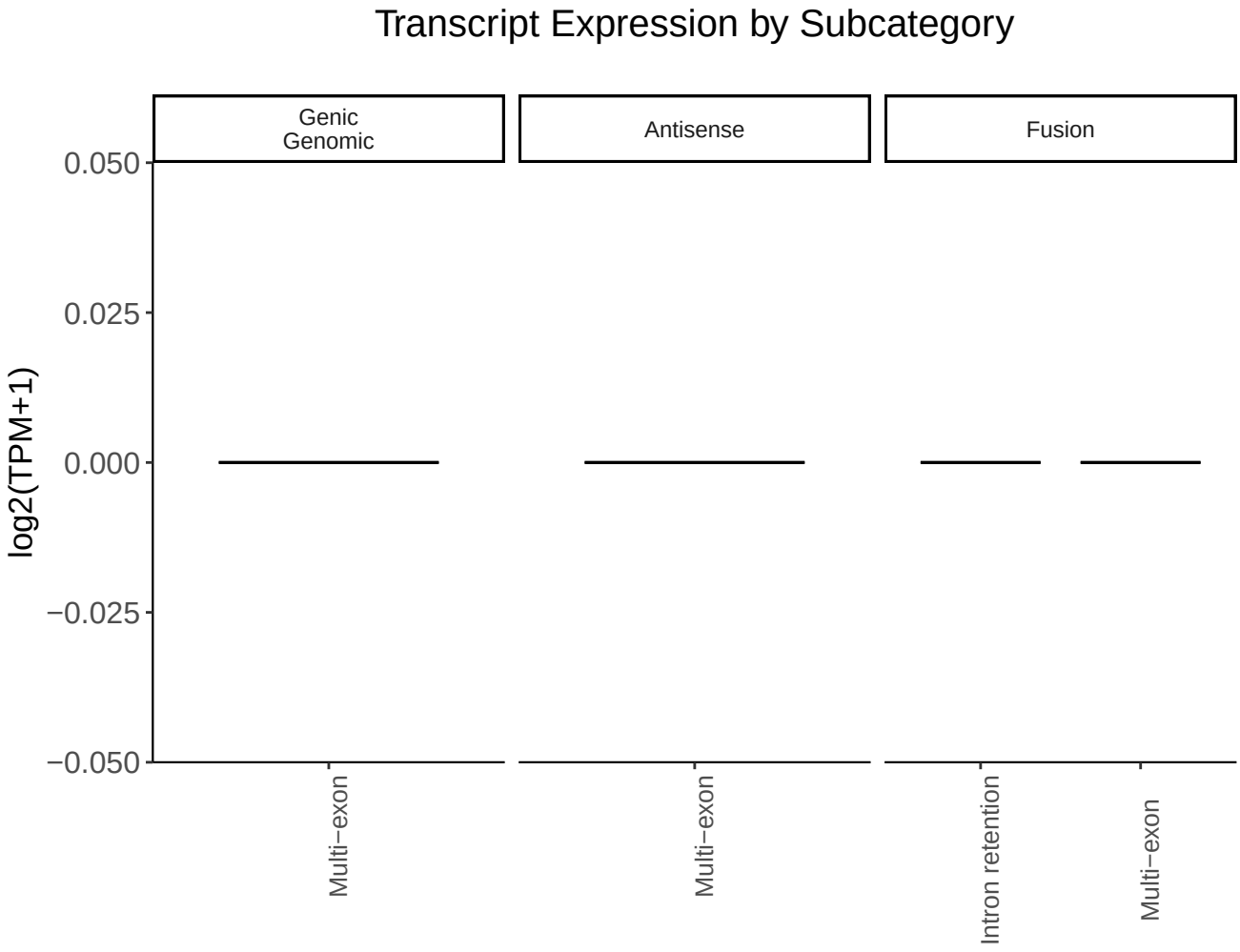

Isoform Distribution Across Structural Subcategories
NIC and NNC
NIC
NNC
Transcripts, count
160
120
80
40
0
Comb. of annot. junctions
Comb. of annot. splice sites
Intron retention
Intron retention
At least 1 annot. don./accept.
Supported Transcript Model
Not fully supported

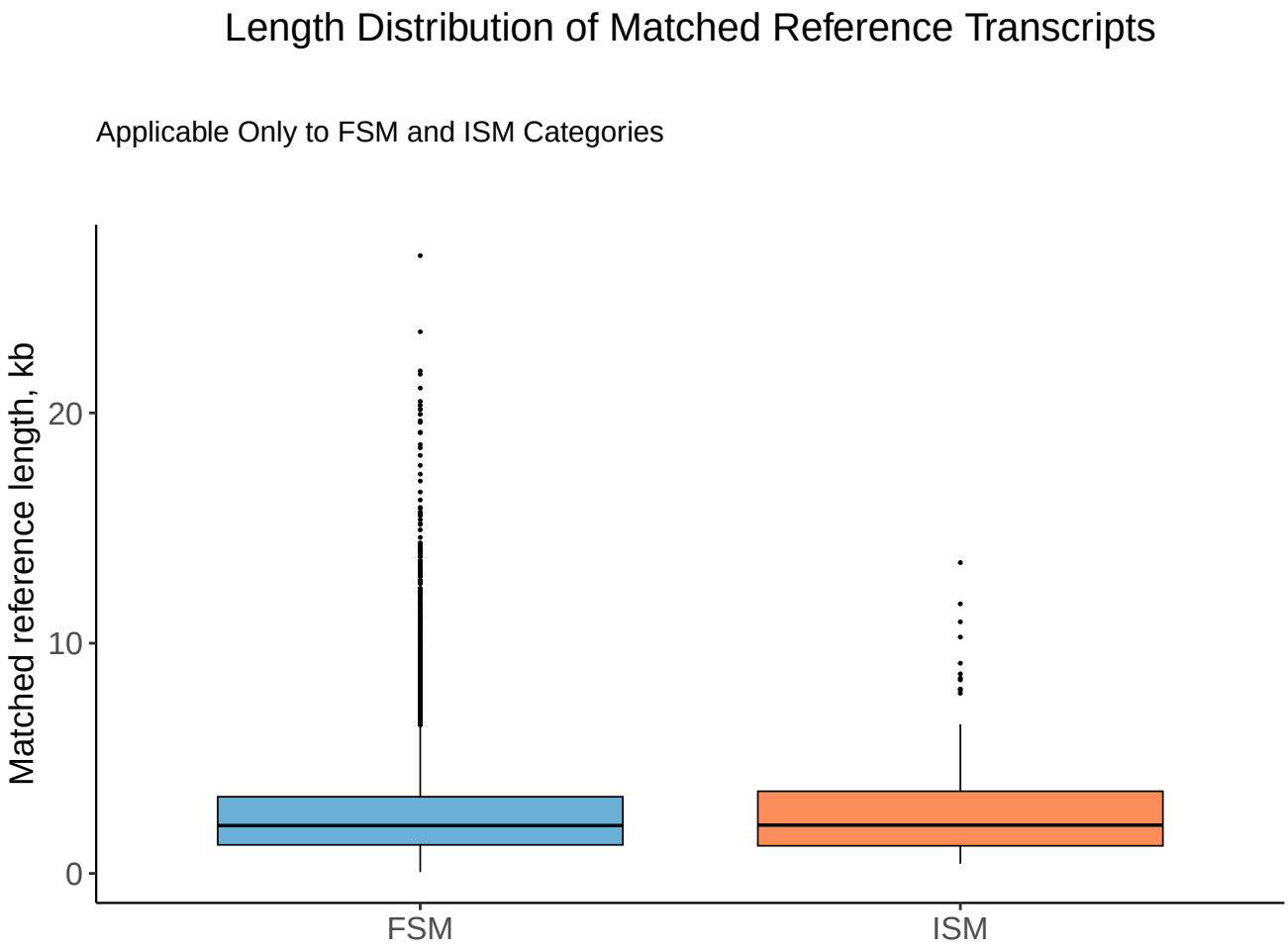

Isoform Distribution Across Structural Subcategories
NIC and NNC
NIC
NNC
Transcripts, %
100%
75%
50%
25%
0%
Comb. of annot. junctions
Comb. of annot. splice sites
Intron retention
Intron retention
At least 1 annot. don./accept.
Supported Transcript Model
Not fully supported

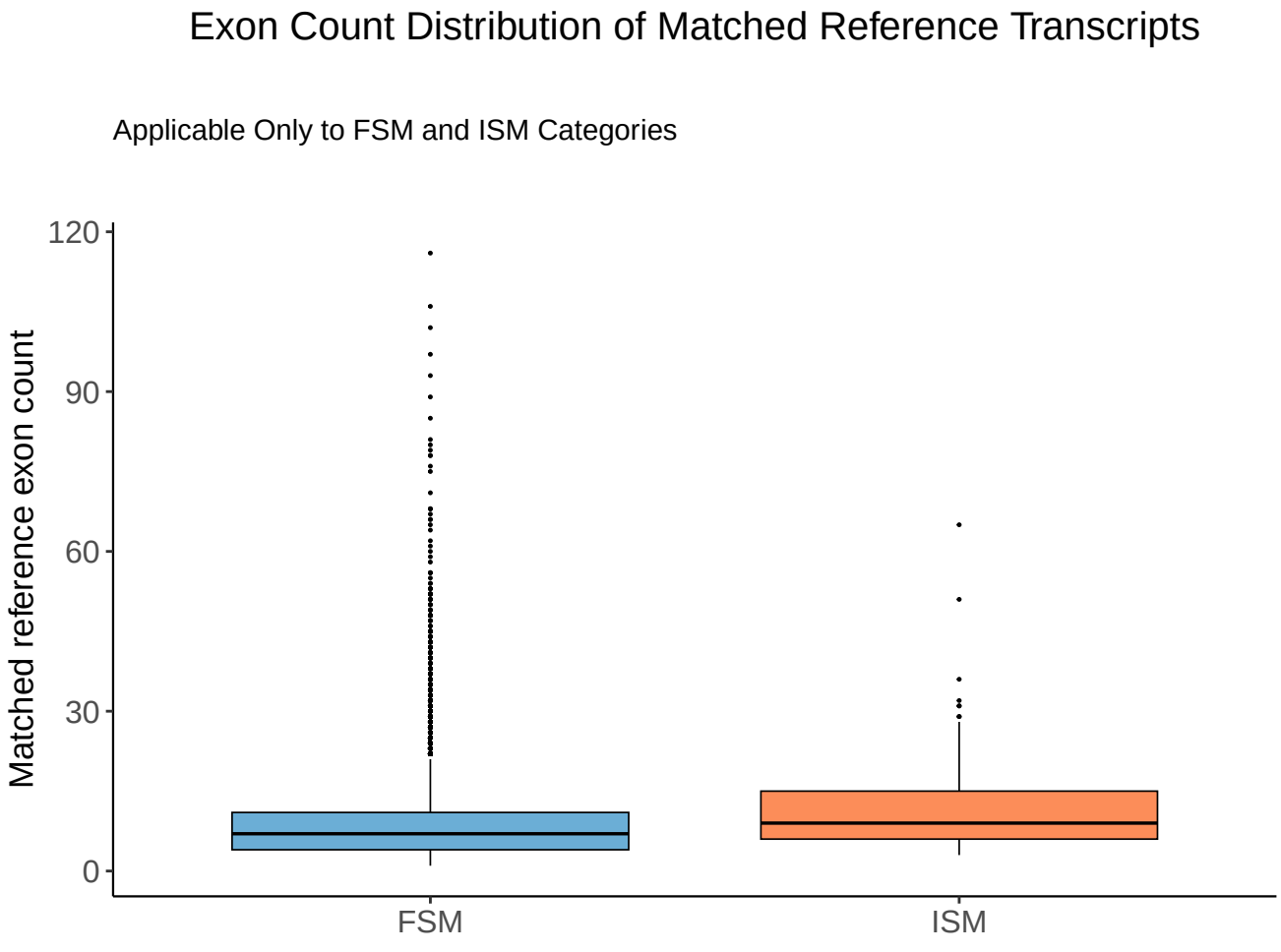

Transcript Expression by Structural Category
log2(TPM+1)
0.050
0.025
0.000
−0.025
−0.050
FSM
ISM
NIC
NNC
Genic Genomic
Antisense
Fusion
Intergenic
Genic Intron

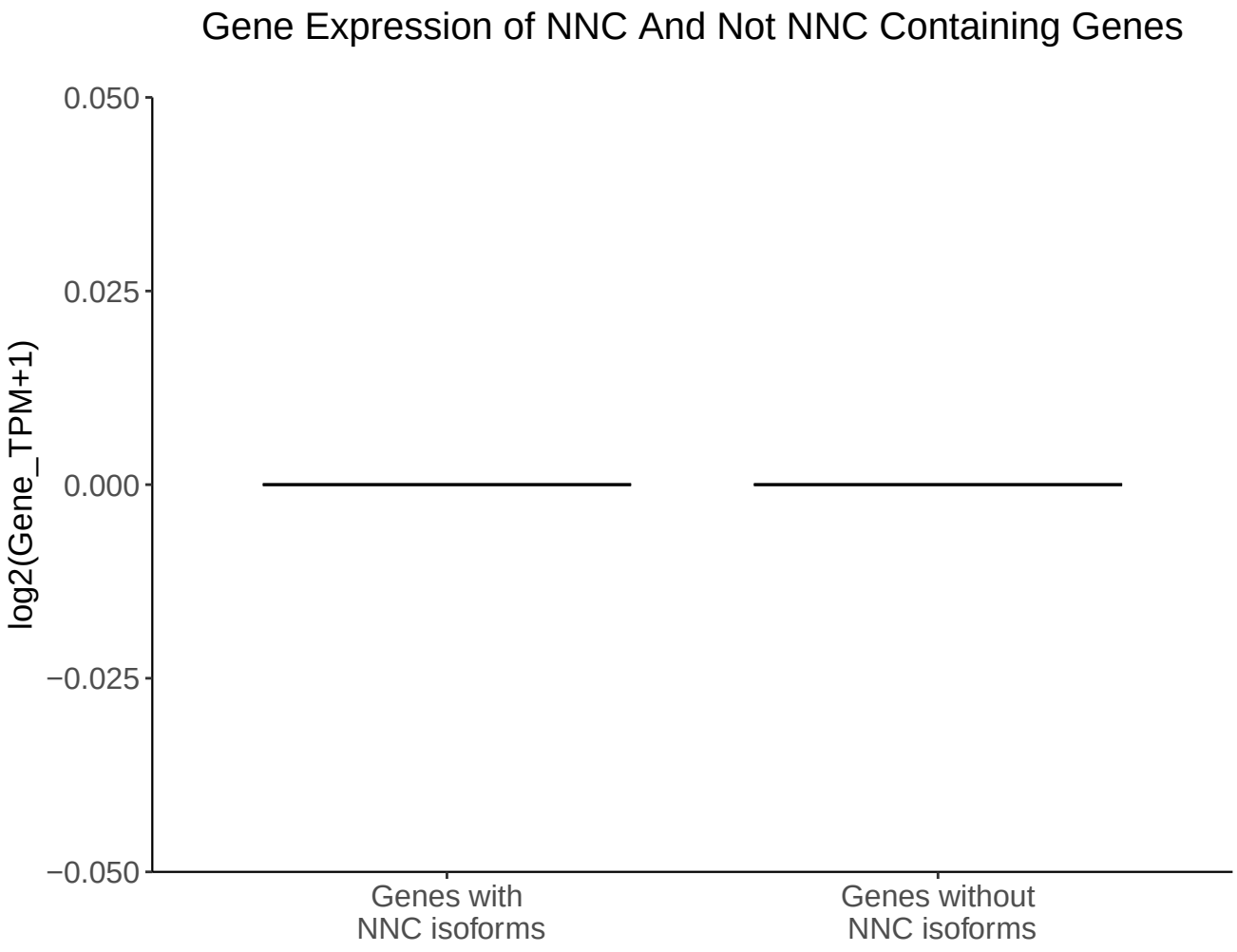
Transcript Expression by Subcategory
FSM
ISM
log2(TPM+1)
0.050
0.025
0.000
−0.025
−0.050
Reference match
Mono−exon
3' fragment
Internal fragment
5' fragment
Intron retention


Transcript Expression by Subcategory
NIC
NNC
log2(TPM+1)
0.050
0.025
0.000
−0.025
−0.050
Comb. of annot. junctions
Comb. of annot. splice sites
Intron retention
Intron retention
At least 1 annot. don./accept.

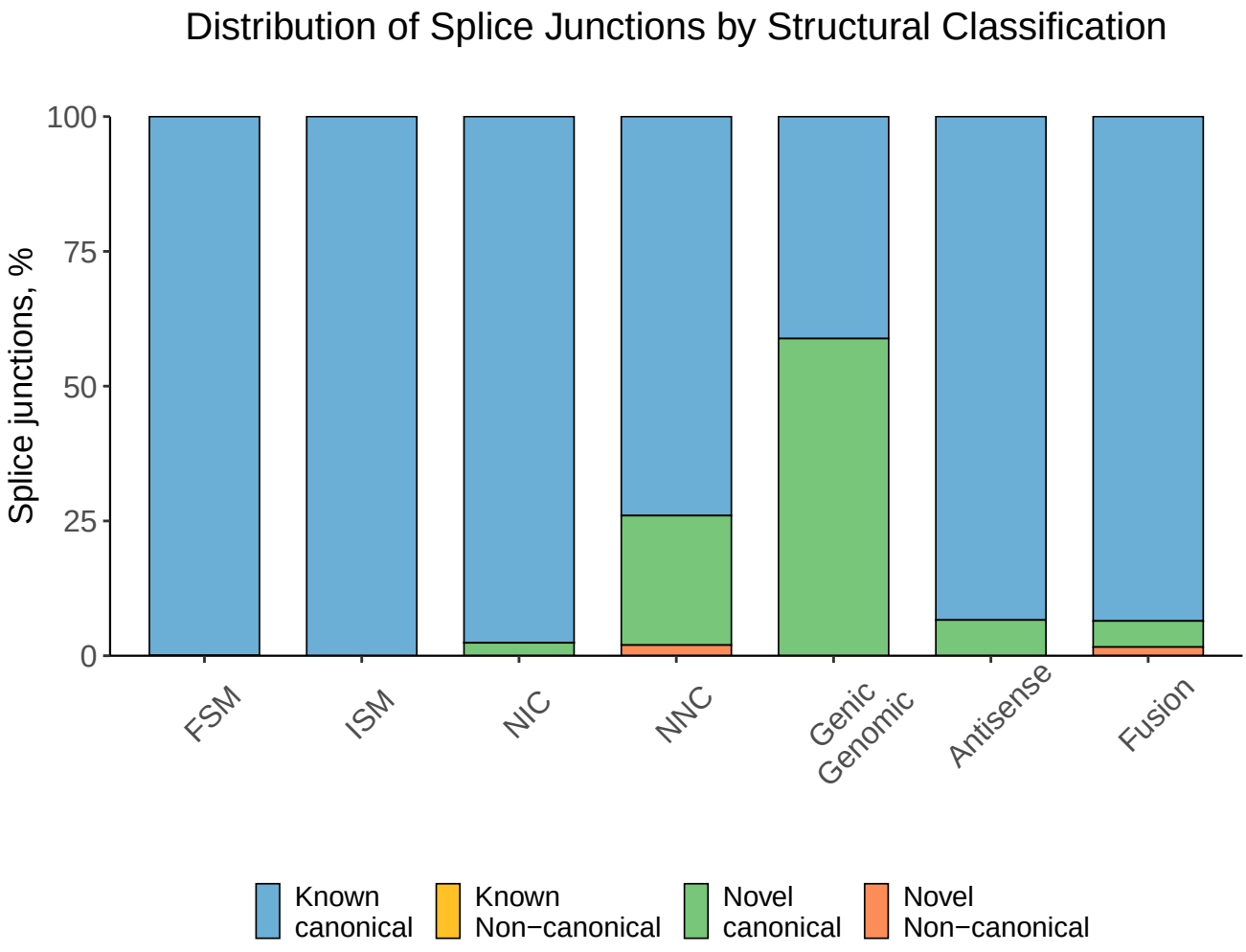

Transcript Expression by Subcategory
Genic Genomic
Antisense
Fusion
log2(TPM+1)
0.050
0.025
0.000
−0.025
−0.050
Multi−exon
Multi−exon
Intron retention
Multi−exon

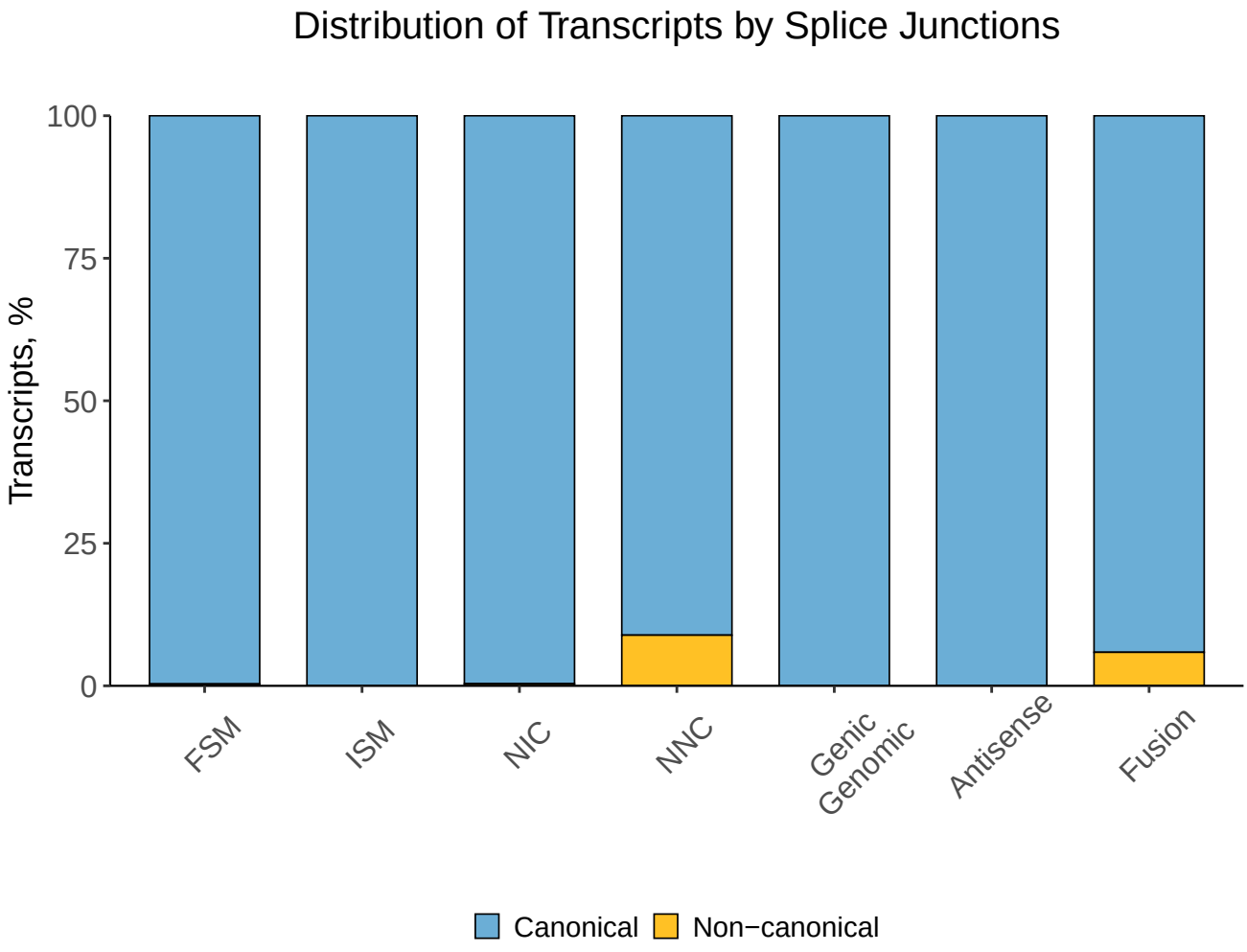

Length Distribution of Matched Reference Transcripts
Applicable Only to FSM and ISM Categories
Matched reference length, kb
0
10
20
FSM
ISM

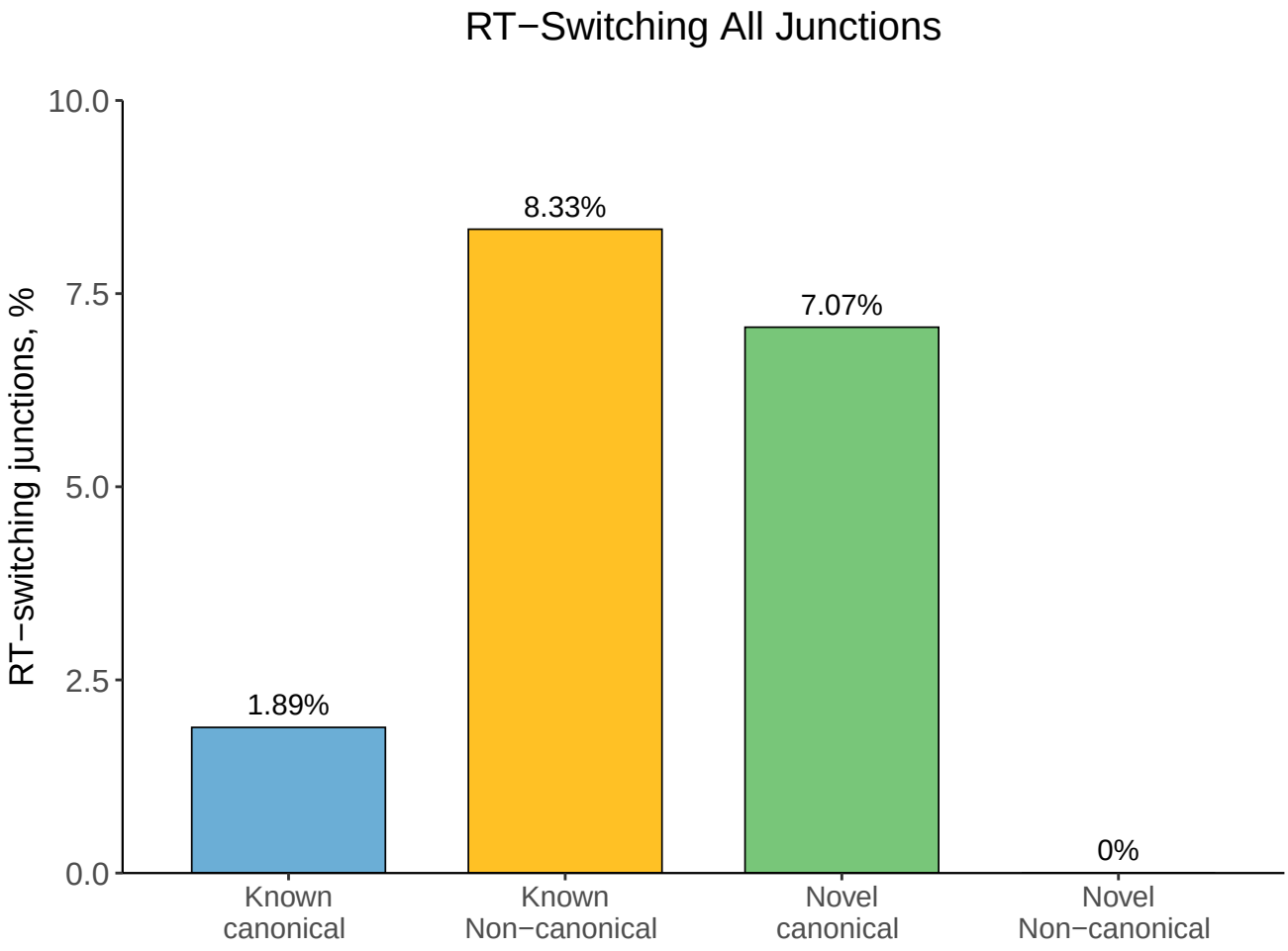

Exon Count Distribution of Matched Reference Transcripts
Applicable Only to FSM and ISM Categories
Matched reference exon count
120
90
60
30
0
FSM
ISM

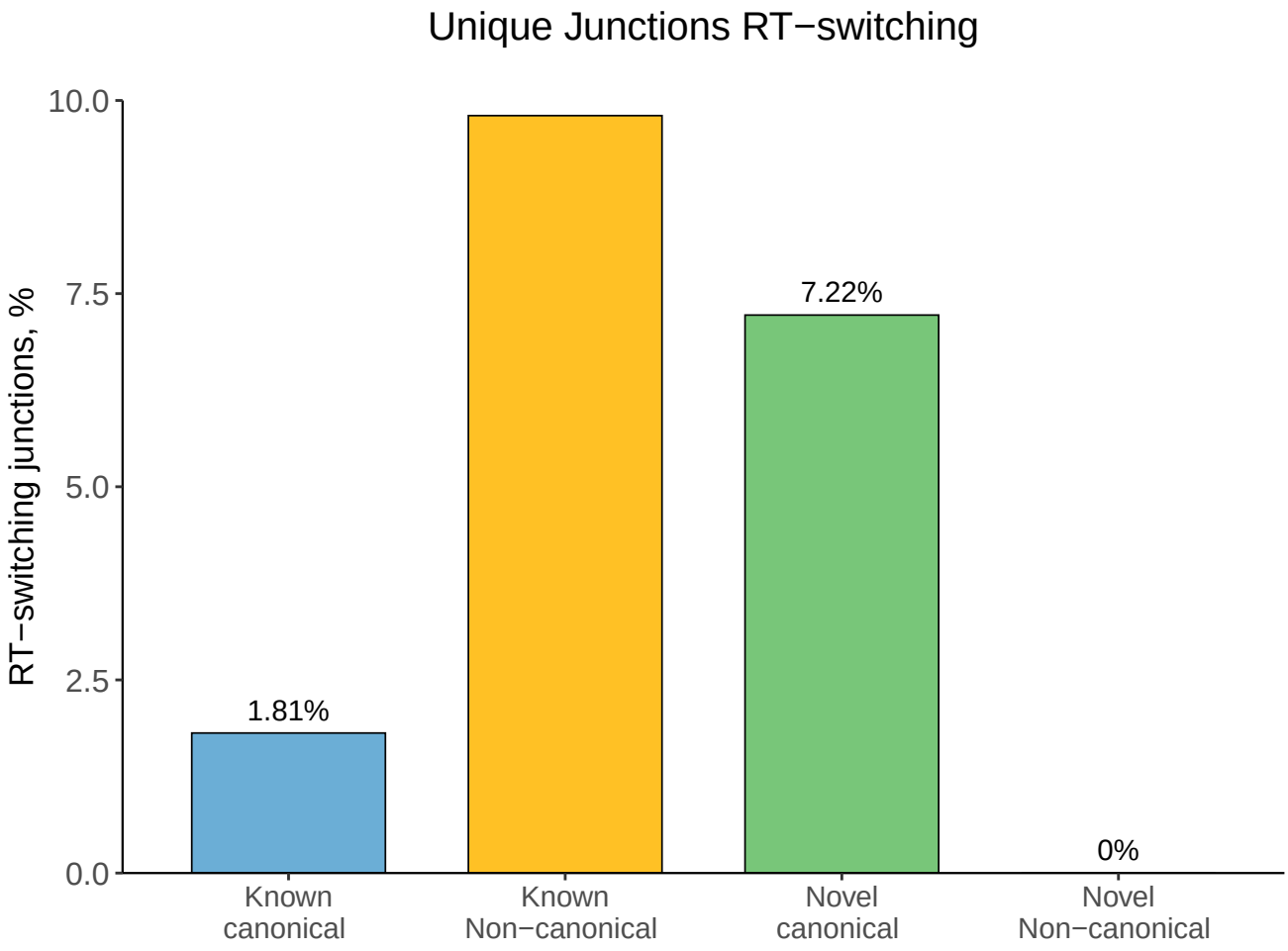

Gene Expression of NNC And Not NNC Containing Genes
0.050
0.025
0.000
−0.025
−0.050
log2(Gene_TPM+1)
Genes with
NNC isoforms
Genes without
NNC isoforms

Splice Junction Characterization

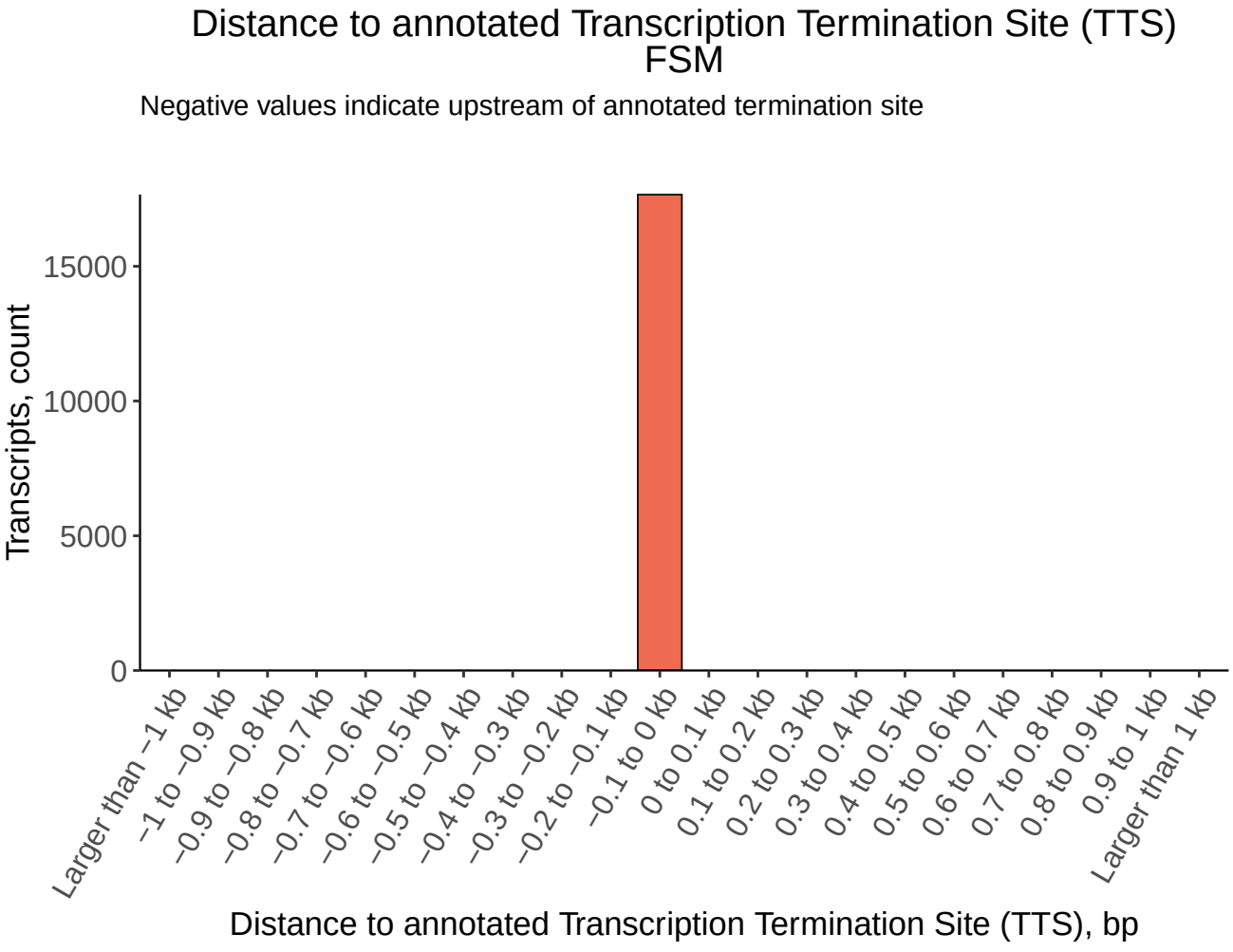

Distribution of Splice Junctions by Structural Classification
Splice junctions, %
100
75
50
25
0
FSM
ISM
NIC
NNC
Genic Genomic
Antisense
Fusion
Known canonical
Known Non−canonical
Novel canonical
Novel Non−canonical

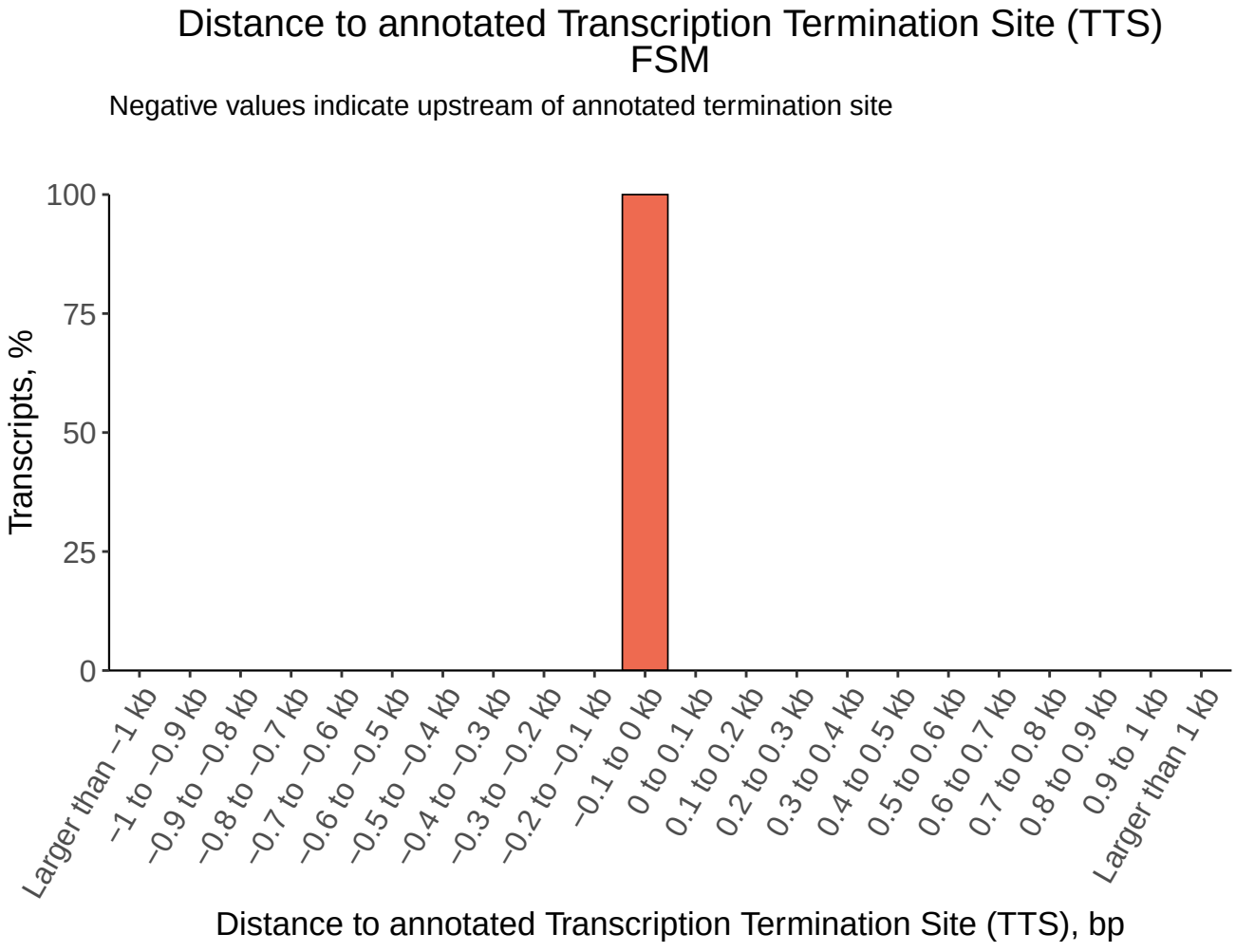

Distribution of Transcripts by Splice Junctions
Transcripts, %
100
75
50
25
0
FSM
ISM
NIC
NNC
Genic Genomic
Antisense
Fusion
Canonical
Non−canonical

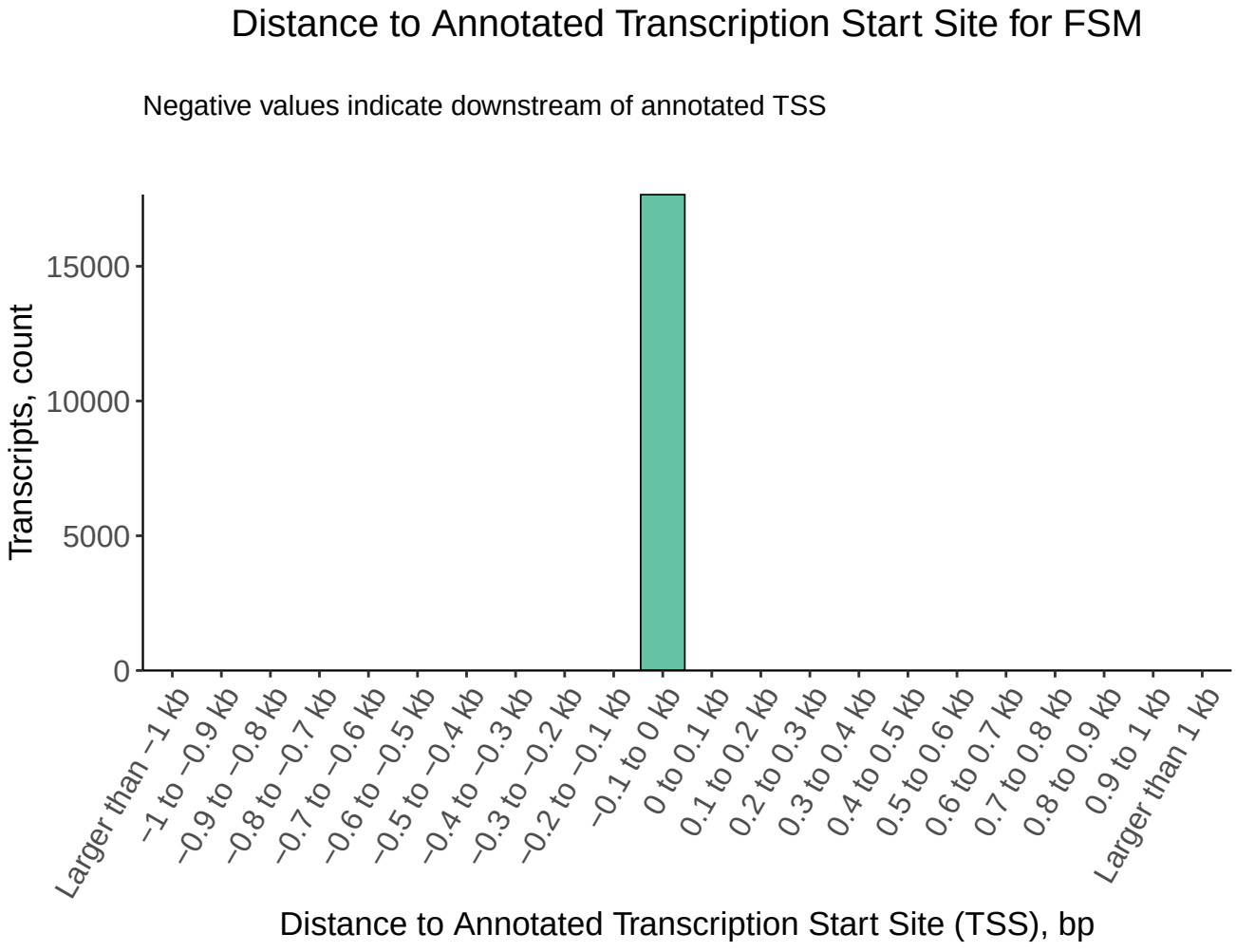

RT−Switching All Junctions
RT−switching junctions, %
10.0
7.5
5.0
2.5
0.0
1.89%
8.33%
7.07%
0%
Known canonical
Known Non−canonical
Novel canonical
Novel Non−canonical

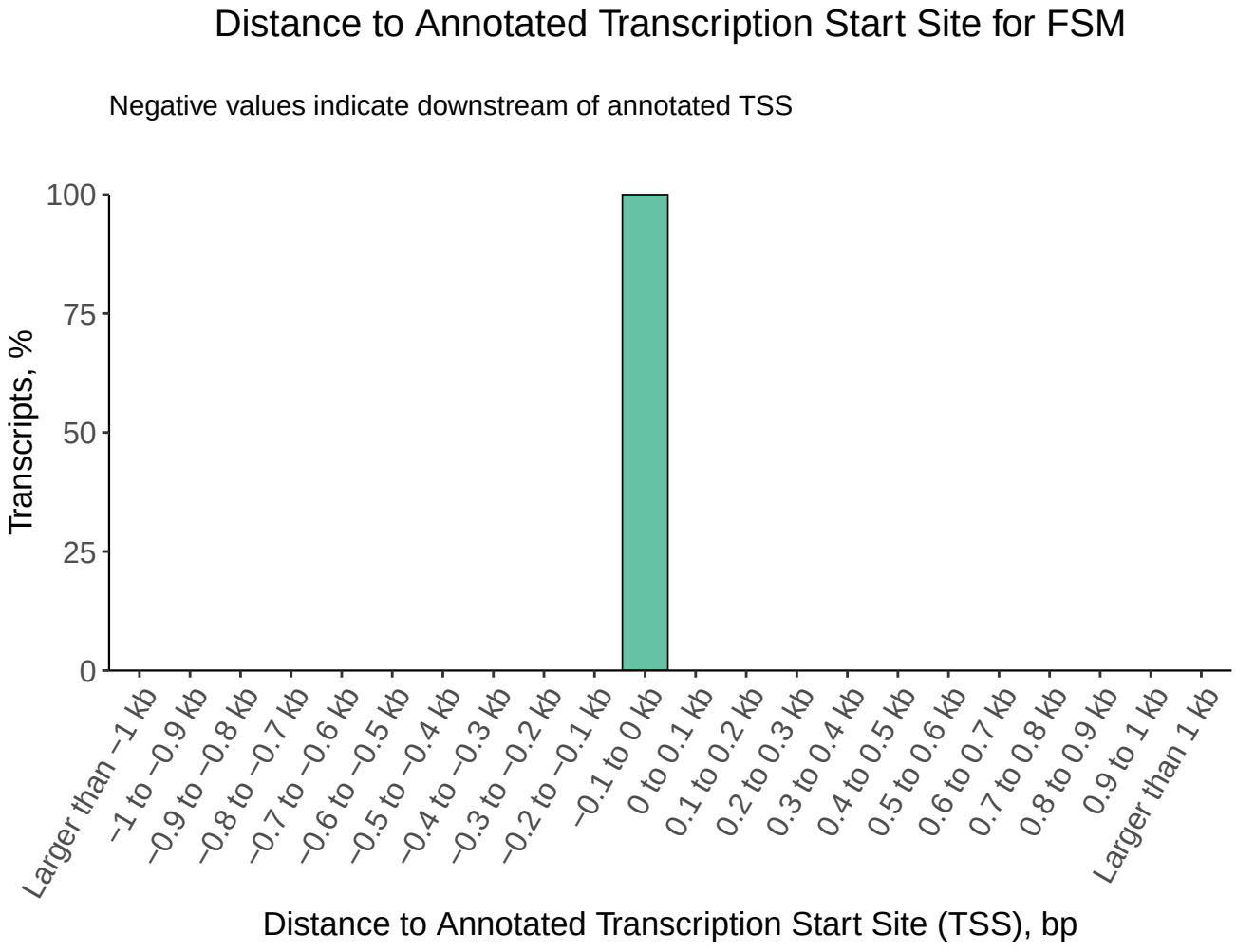

Unique Junctions RT−switching
RT−switching junctions, %
10.0
7.5
5.0
2.5
0.0
1.81%
7.22%
0%
Known canonical
Known Non−canonical
Novel canonical
Novel Non−canonical

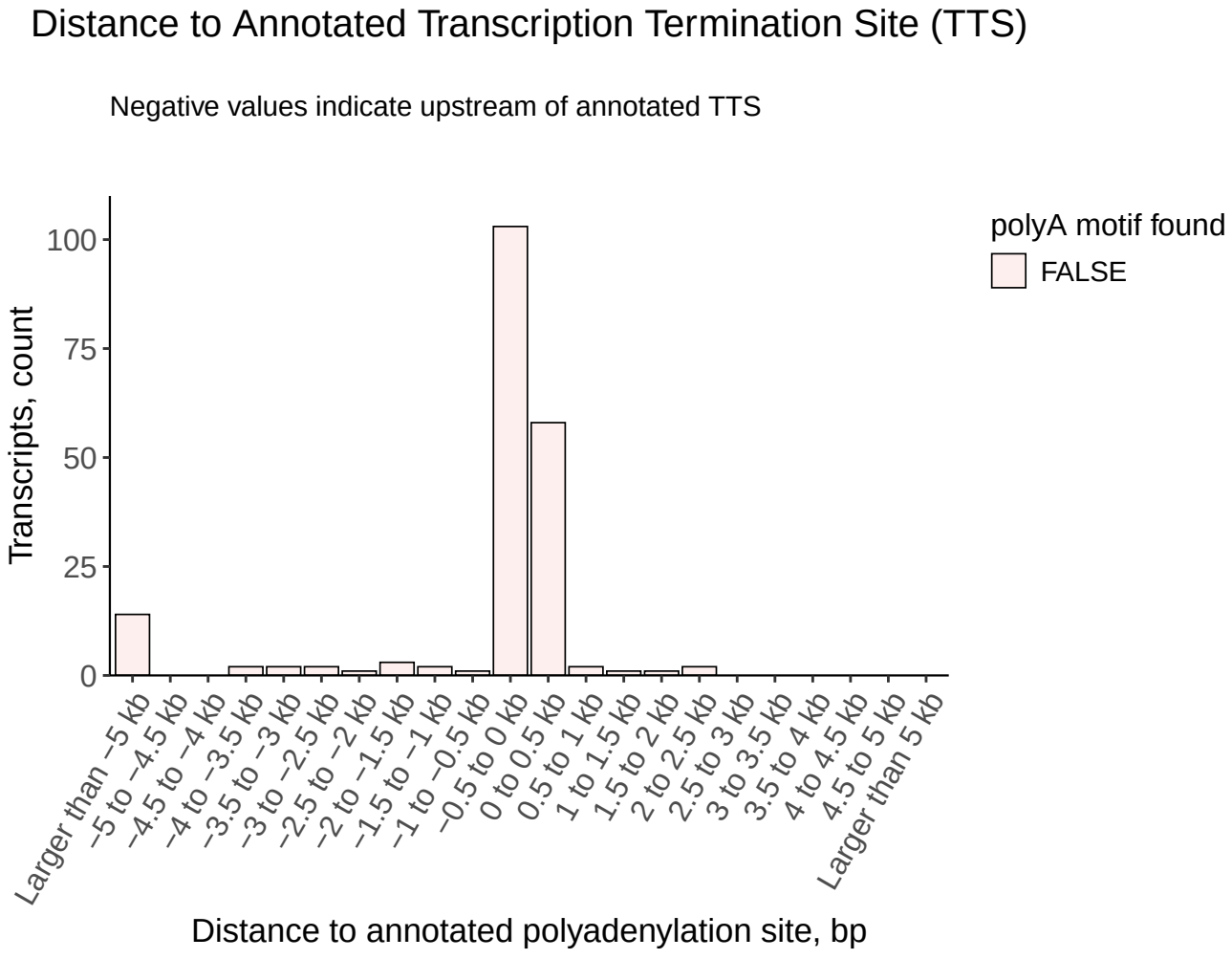
Comparison With Annotated TSS and TTS

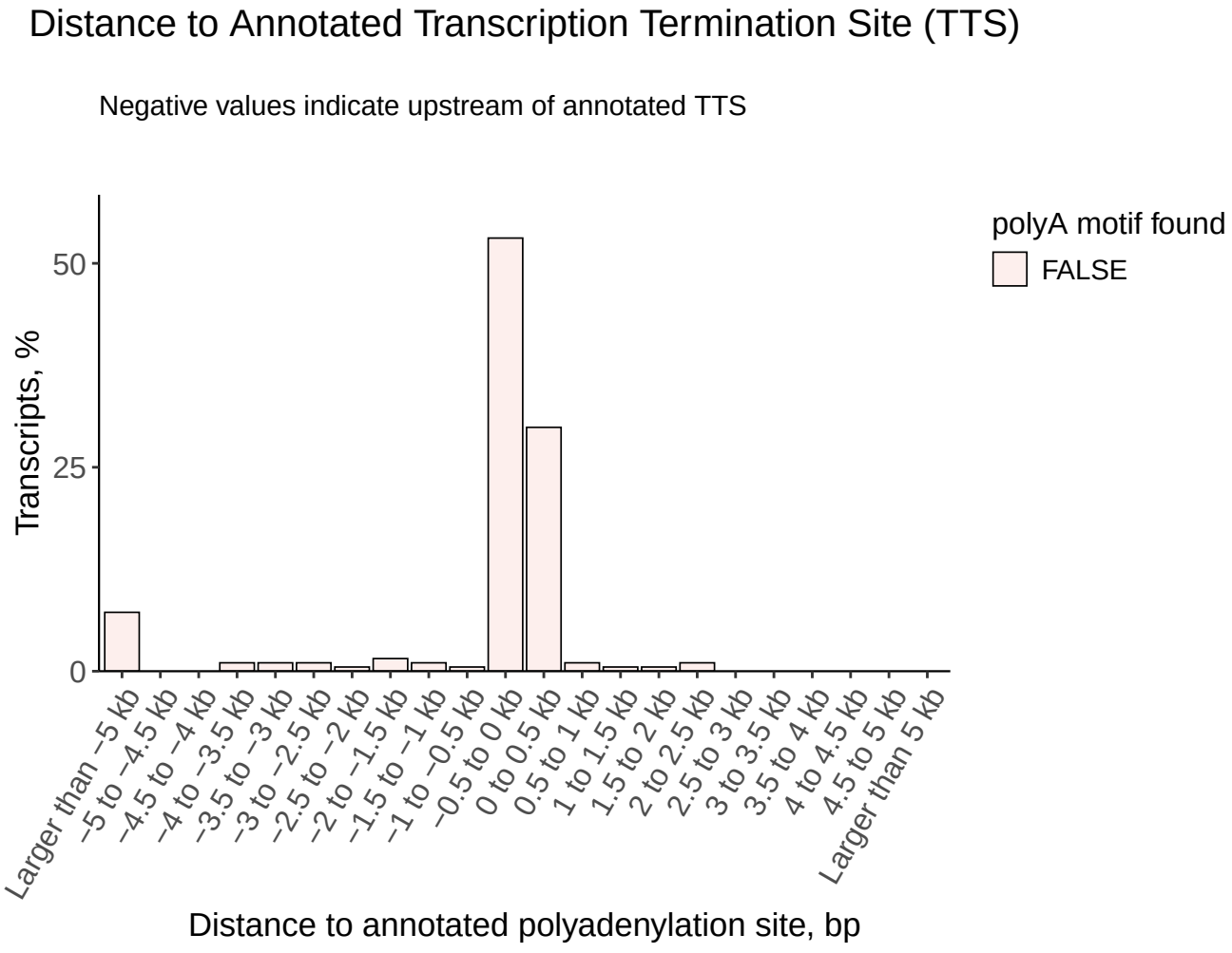

Distance to annotated Transcription Termination Site (TTS)
FSM
Negative values indicate upstream of annotated termination site
Transcripts, count
15000
10000
5000
0
Larger than −1 kb
−1 to −0.9 kb
−0.9 to −0.8 kb
−0.8 to −0.7 kb
−0.7 to −0.6 kb
−0.6 to −0.5 kb
−0.5 to −0.4 kb
−0.4 to −0.3 kb
−0.3 to −0.2 kb
−0.2 to −0.1 kb
−0.1 to 0 kb
0 to 0.1 kb
0.1 to 0.2 kb
0.2 to 0.3 kb
0.3 to 0.4 kb
0.4 to 0.5 kb
0.5 to 0.6 kb
0.6 to 0.7 kb
0.7 to 0.8 kb
0.8 to 0.9 kb
0.9 to 1 kb
Larger than 1 kb
Distance to annotated Transcription Termination Site (TTS), bp


Distance to annotated Transcription Termination Site (TTS)
FSM
Negative values indicate upstream of annotated termination site
Transcripts, %
100
75
50
25
0
Larger than −1 kb
−1 to −0.9 kb
−0.9 to −0.8 kb
−0.8 to −0.7 kb
−0.7 to −0.6 kb
−0.6 to −0.5 kb
−0.5 to −0.4 kb
−0.4 to −0.3 kb
−0.3 to −0.2 kb
−0.2 to −0.1 kb
−0.1 to 0 kb
0 to 0.1 kb
0.1 to 0.2 kb
0.2 to 0.3 kb
0.3 to 0.4 kb
0.4 to 0.5 kb
0.5 to 0.6 kb
0.6 to 0.7 kb
0.7 to 0.8 kb
0.8 to 0.9 kb
0.9 to 1 kb
Larger than 1 kb
Distance to annotated Transcription Termination Site (TTS), bp


Distance to Annotated Transcription Start Site for FSM
Negative values indicate downstream of annotated TSS
Transcripts, count
15000
10000
5000
0
Larger than −1 kb
−1 to −0.9 kb
−0.9 to −0.8 kb
−0.8 to −0.7 kb
−0.7 to −0.6 kb
−0.6 to −0.5 kb
−0.5 to −0.4 kb
−0.4 to −0.3 kb
−0.3 to −0.2 kb
−0.2 to −0.1 kb
−0.1 to 0 kb
0 to 0.1 kb
0.1 to 0.2 kb
0.2 to 0.3 kb
0.3 to 0.4 kb
0.4 to 0.5 kb
0.5 to 0.6 kb
0.6 to 0.7 kb
0.7 to 0.8 kb
0.8 to 0.9 kb
0.9 to 1 kb
Larger than 1 kb
Distance to Annotated Transcription Start Site (TSS), bp

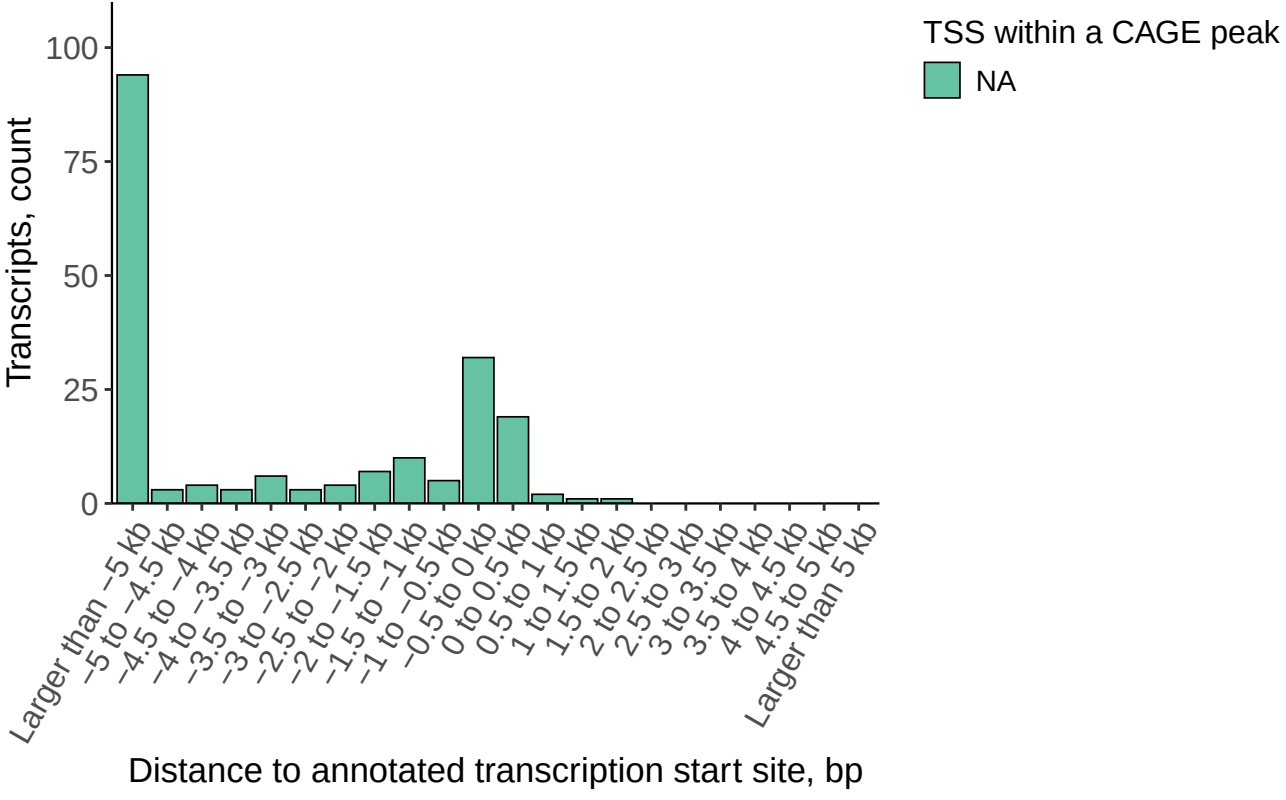

Distance to Annotated Transcription Start Site for FSM
Negative values indicate downstream of annotated TSS
Transcripts, %
100
75
50
25
0
Larger than −1 kb
−1 to −0.9 kb
−0.9 to −0.8 kb
−0.8 to −0.7 kb
−0.7 to −0.6 kb
−0.6 to −0.5 kb
−0.5 to −0.4 kb
−0.4 to −0.3 kb
−0.3 to −0.2 kb
−0.2 to −0.1 kb
−0.1 to 0 kb
0 to 0.1 kb
0.1 to 0.2 kb
0.2 to 0.3 kb
0.3 to 0.4 kb
0.4 to 0.5 kb
0.5 to 0.6 kb
0.6 to 0.7 kb
0.7 to 0.8 kb
0.8 to 0.9 kb
0.9 to 1 kb
Larger than 1 kb
Distance to Annotated Transcription Start Site (TSS), bp

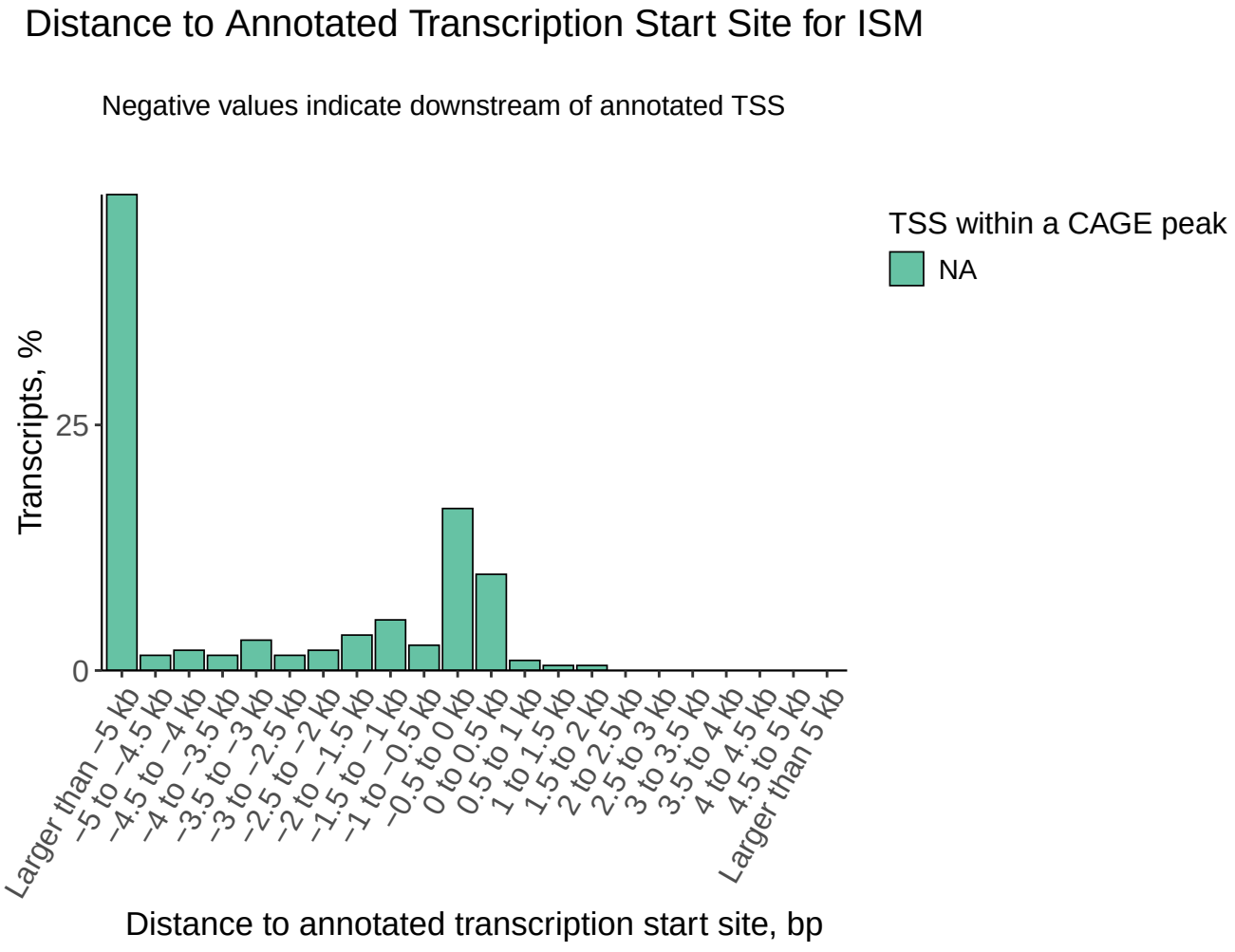

Distance to Annotated Transcription Termination Site (TTS)
Negative values indicate upstream of annotated TTS
Transcripts, count
100
75
50
25
0
Larger than −5 kb
−5 to −4.5 kb
−4.5 to −4 kb
−4 to −3.5 kb
−3.5 to −3 kb
−3 to −2.5 kb
−2.5 to −2 kb
−2 to −1.5 kb
−1.5 to −1 kb
−1 to −0.5 kb
−0.5 to 0 kb
0 to 0.5 kb
0.5 to 1 kb
1 to 1.5 kb
1.5 to 2 kb
2 to 2.5 kb
2.5 to 3 kb
3 to 3.5 kb
3.5 to 4 kb
4 to 4.5 kb
4.5 to 5 kb
Larger than 5 kb
Distance to annotated polyadenylation site, bp
polyA motif found
FALSE


Distance to Annotated Transcription Termination Site (TTS)
Negative values indicate upstream of annotated TTS
Transcripts, %
50
25
0
Larger than −5 kb
−5 to −4.5 kb
−4.5 to −4 kb
−4 to −3.5 kb
−3.5 to −3 kb
−3 to −2.5 kb
−2.5 to −2 kb
−2 to −1.5 kb
−1.5 to −1 kb
−1 to −0.5 kb
−0.5 to 0 kb
0 to 0.5 kb
0.5 to 1 kb
1 to 1.5 kb
1.5 to 2 kb
2 to 2.5 kb
2.5 to 3 kb
3 to 3.5 kb
3.5 to 4 kb
4 to 4.5 kb
4.5 to 5 kb
Larger than 5 kb
Distance to annotated polyadenylation site, bp
polyA motif found
FALSE

Distance to Annotated Transcription Start Site for ISM

Negative values indicate downstream of annotated TSS

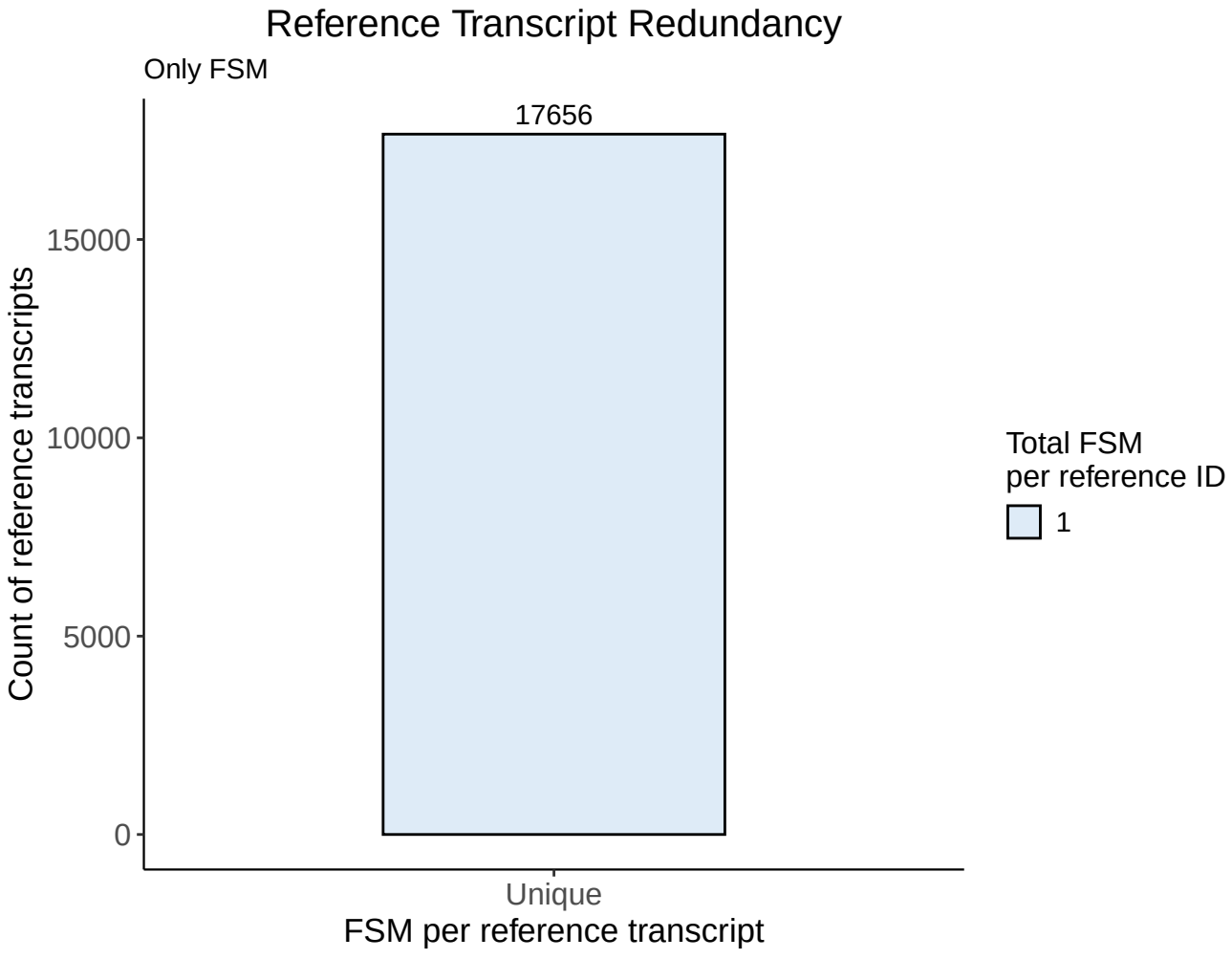


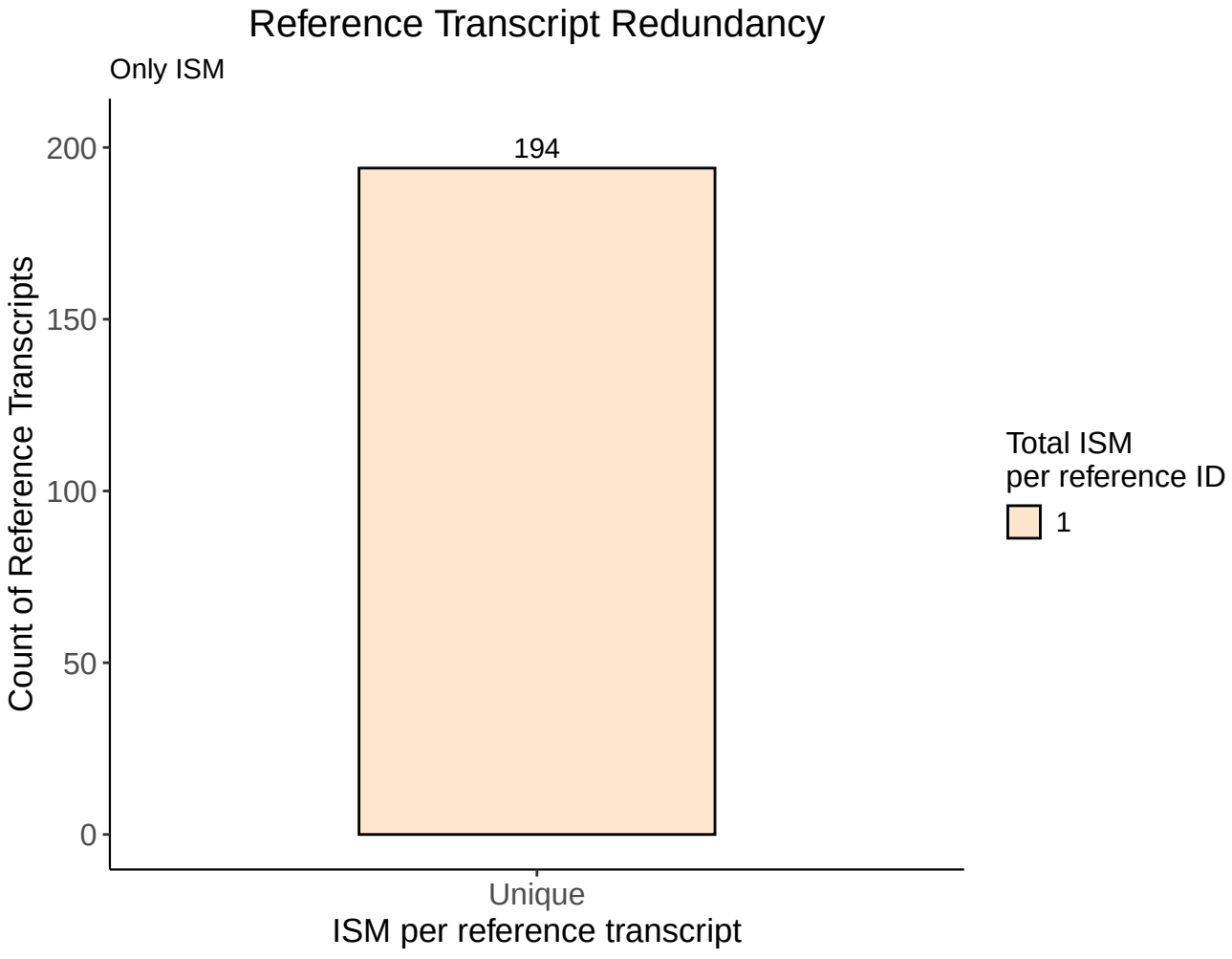

Distance to Annotated Transcription Start Site for ISM
Negative values indicate downstream of annotated TSS
TSS within a CAGE peak
NA
Transcripts, %
25
0
Larger than −5 kb
−5 to −4.5 kb
−4.5 to −4 kb
−4 to −3.5 kb
−3.5 to −3 kb
−3 to −2.5 kb
−2.5 to −2 kb
−2 to −1.5 kb
−1.5 to −1 kb
−1 to −0.5 kb
−0.5 to 0 kb
0 to 0.5 kb
0.5 to 1 kb
1 to 1.5 kb
1.5 to 2 kb
2 to 2.5 kb
2.5 to 3 kb
3 to 3.5 kb
3.5 to 4 kb
4 to 4.5 kb
4.5 to 5 kb
Larger than 5 kb
Distance to annotated transcription start site, bp

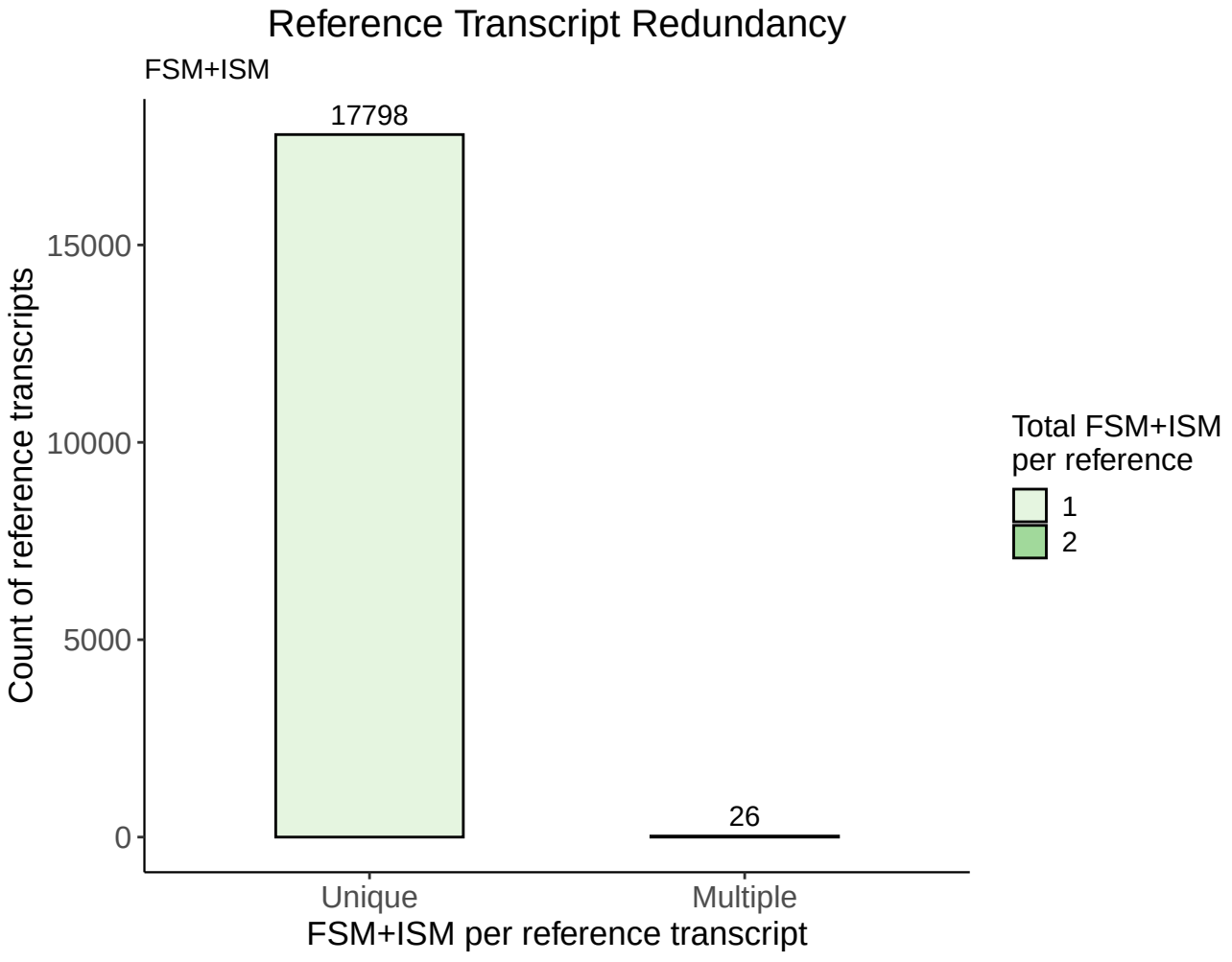
Comparison With Annotated TSS and TTS by Subcategories

PolyA Distance Analysis

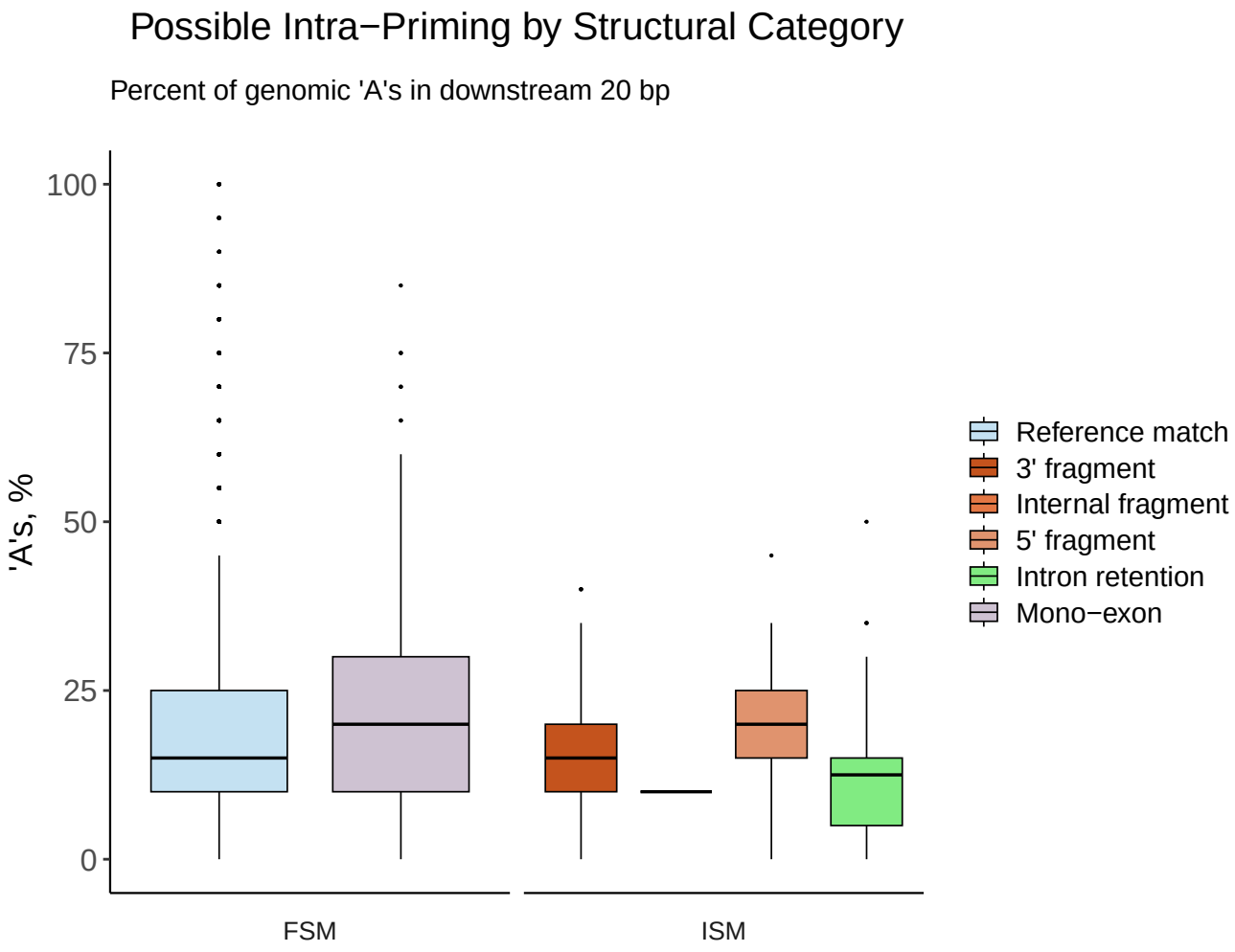
Redundancy Analysis

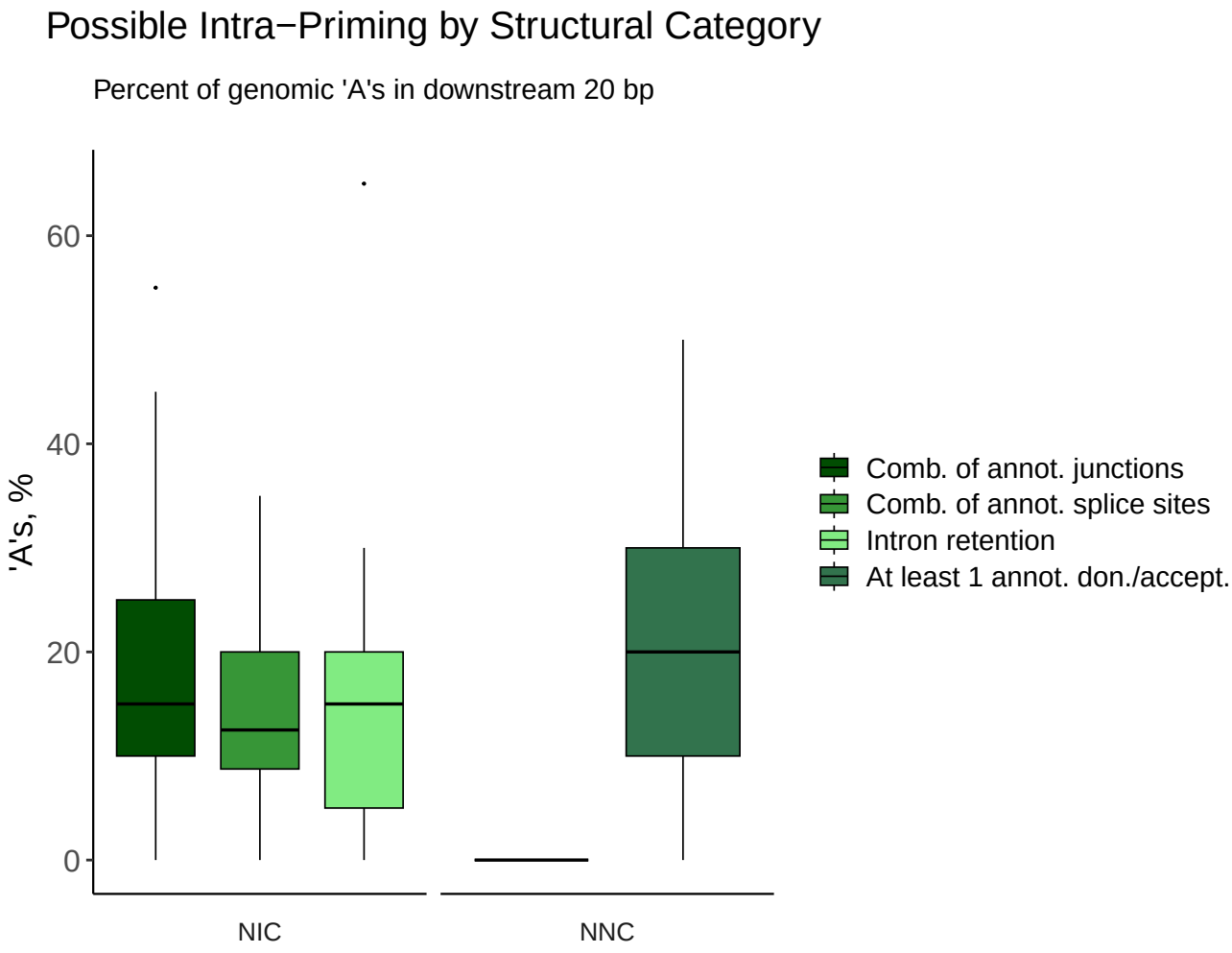

Reference Transcript Redundancy
Only FSM
17656
15000
10000
5000
0
Count of reference transcripts
Unique
FSM per reference transcript
Total FSM
per reference ID
1

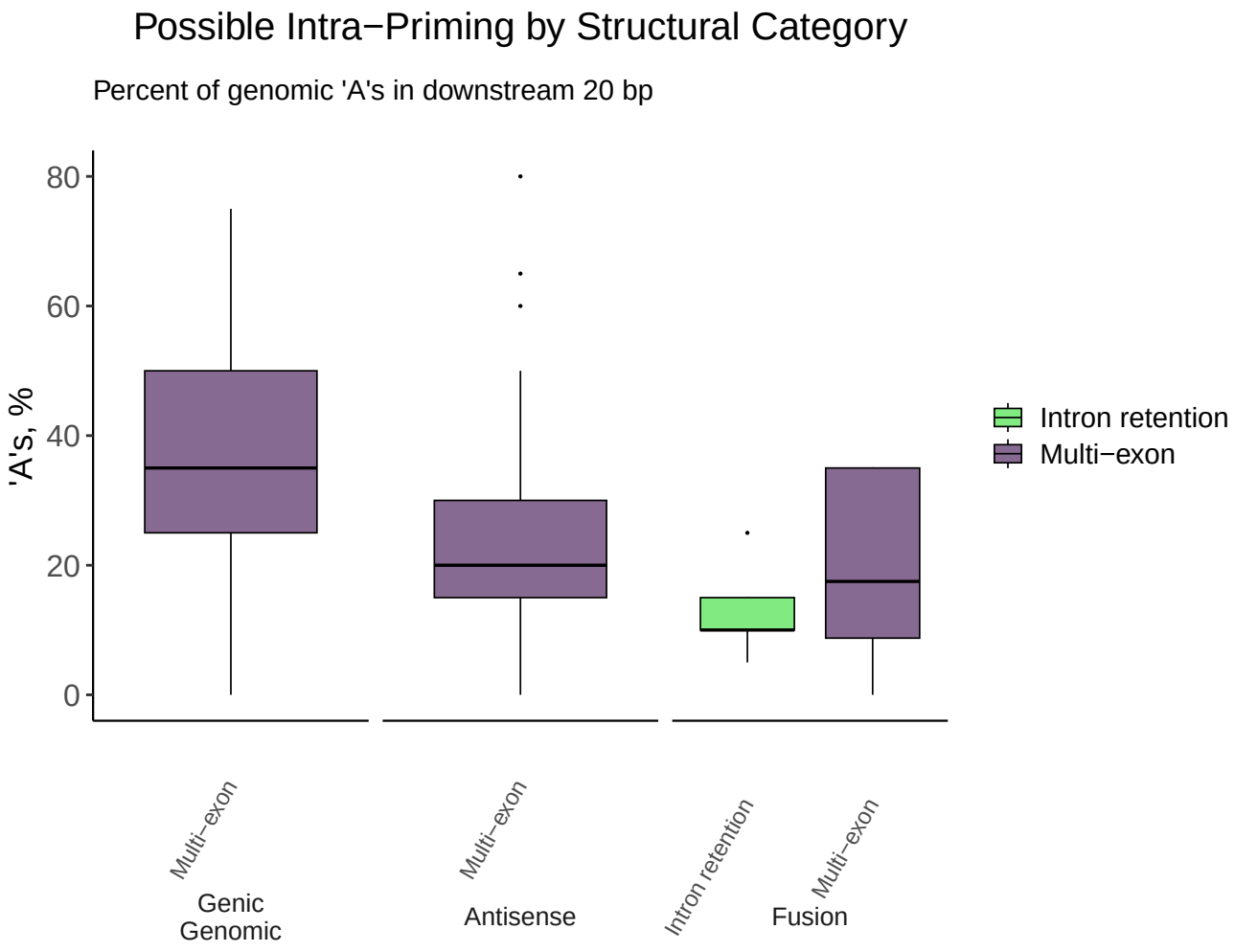

Reference Transcript Redundancy
Only ISM
194
200
150
100
50
0
Count of Reference Transcripts
Unique
ISM per reference transcript
Total ISM
per reference ID
1

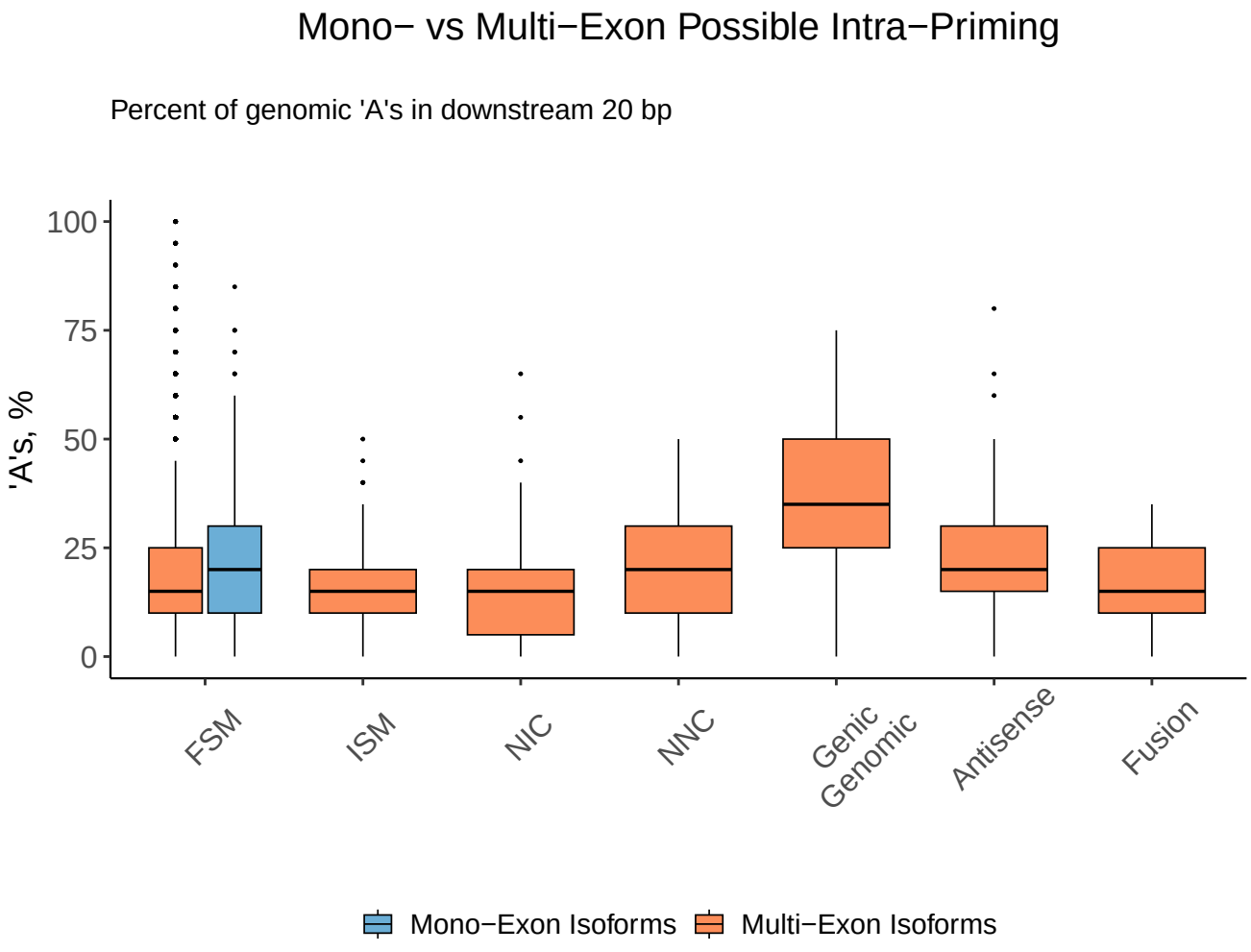

Reference Transcript Redundancy
FSM+ISM
17798
26
15000
10000
5000
0
Count of reference transcripts
Unique
Multiple
FSM+ISM per reference transcript
Total FSM+ISM per reference
1
2

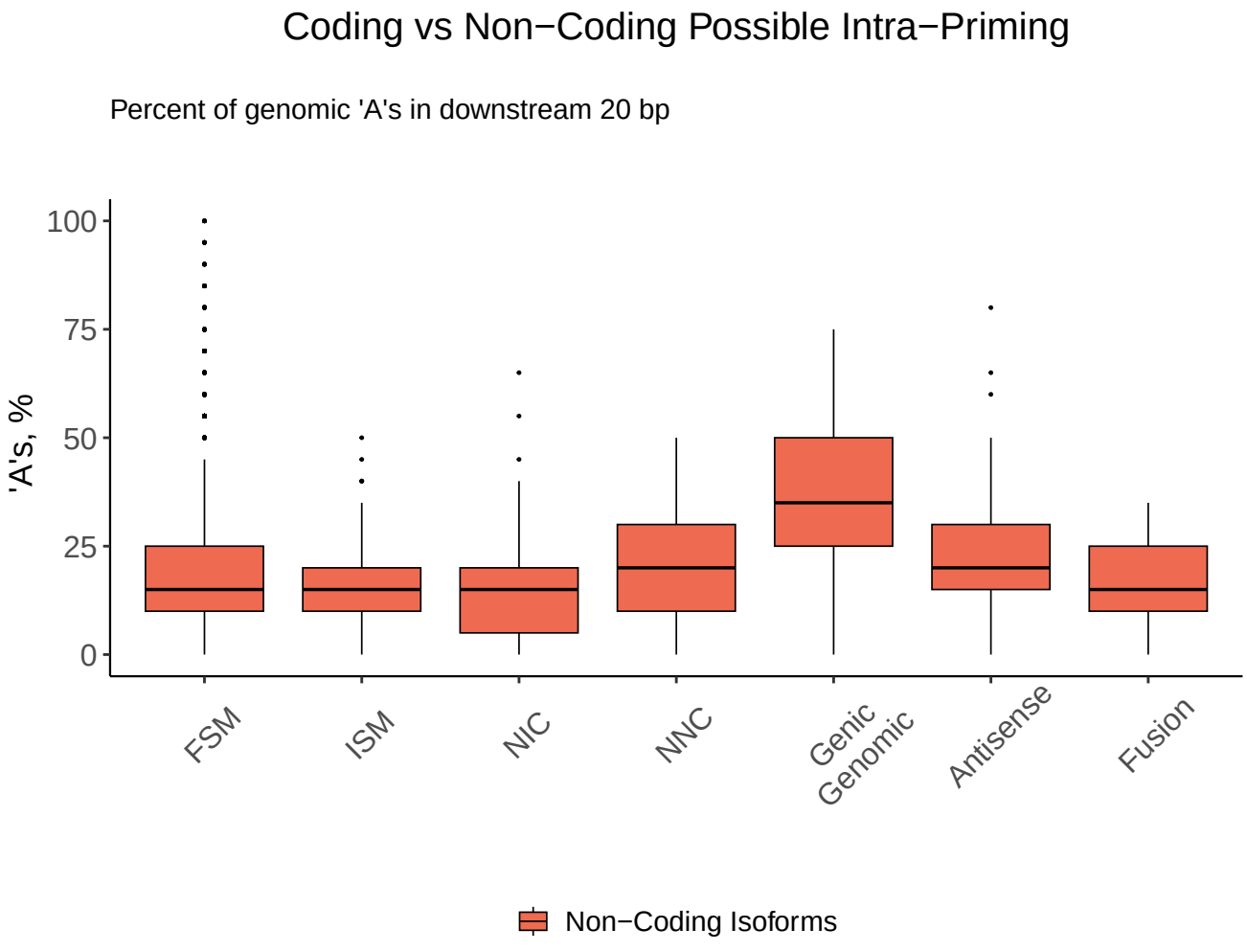
Intra−Priming Quality Check


Possible Intra−Priming by Structural Category
Percent of genomic 'A's in downstream 20 bp
'A's, %
100
75
50
25
0
FSM
ISM
Reference match
3' fragment
Internal fragment
5' fragment
Intron retention
Mono−exon

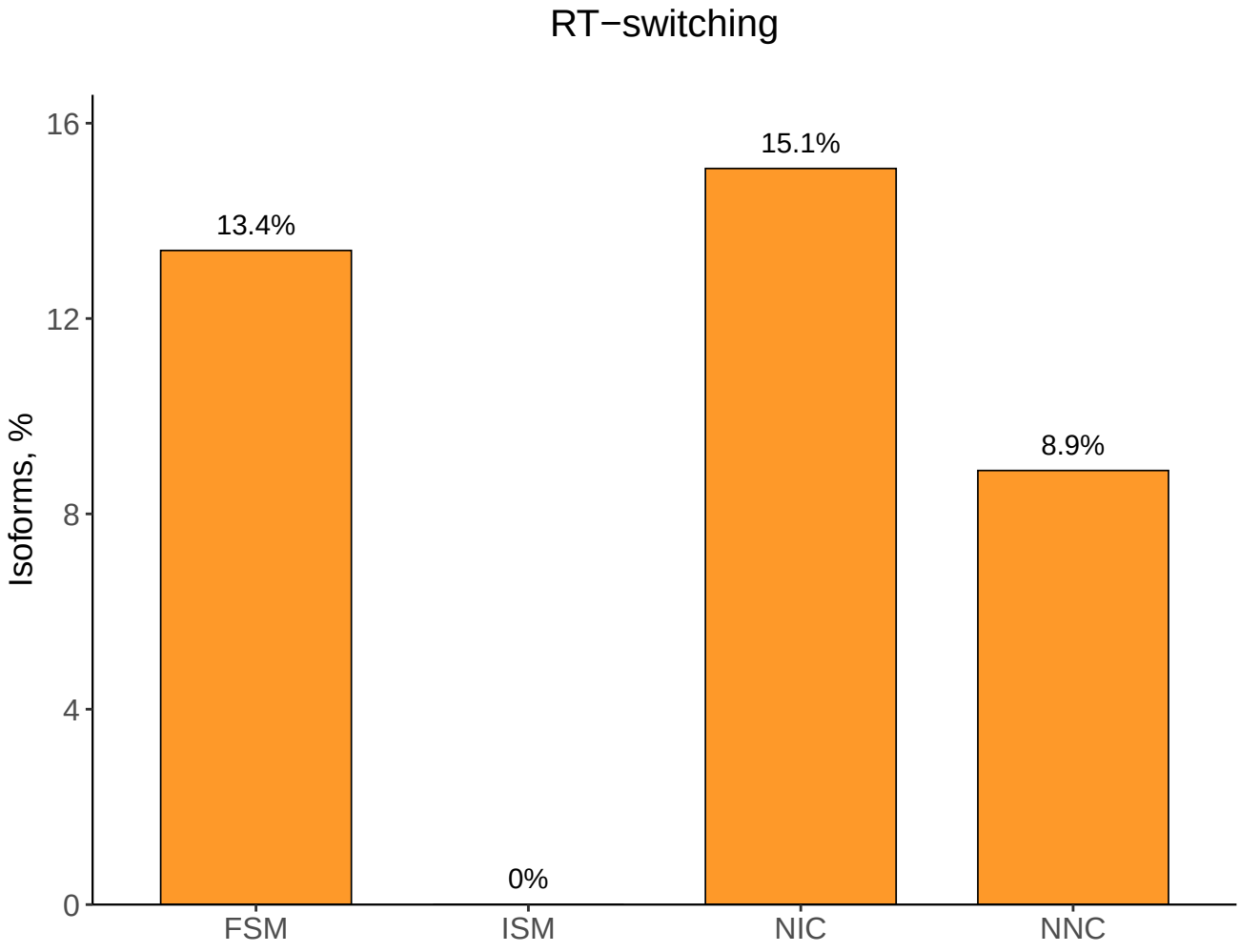

Possible Intra−Priming by Structural Category
Percent of genomic 'A's in downstream 20 bp
'A's, %
60
40
20
0
NIC
NNC
Comb. of annot. junctions
Comb. of annot. splice sites
Intron retention
At least 1 annot. don./accept.

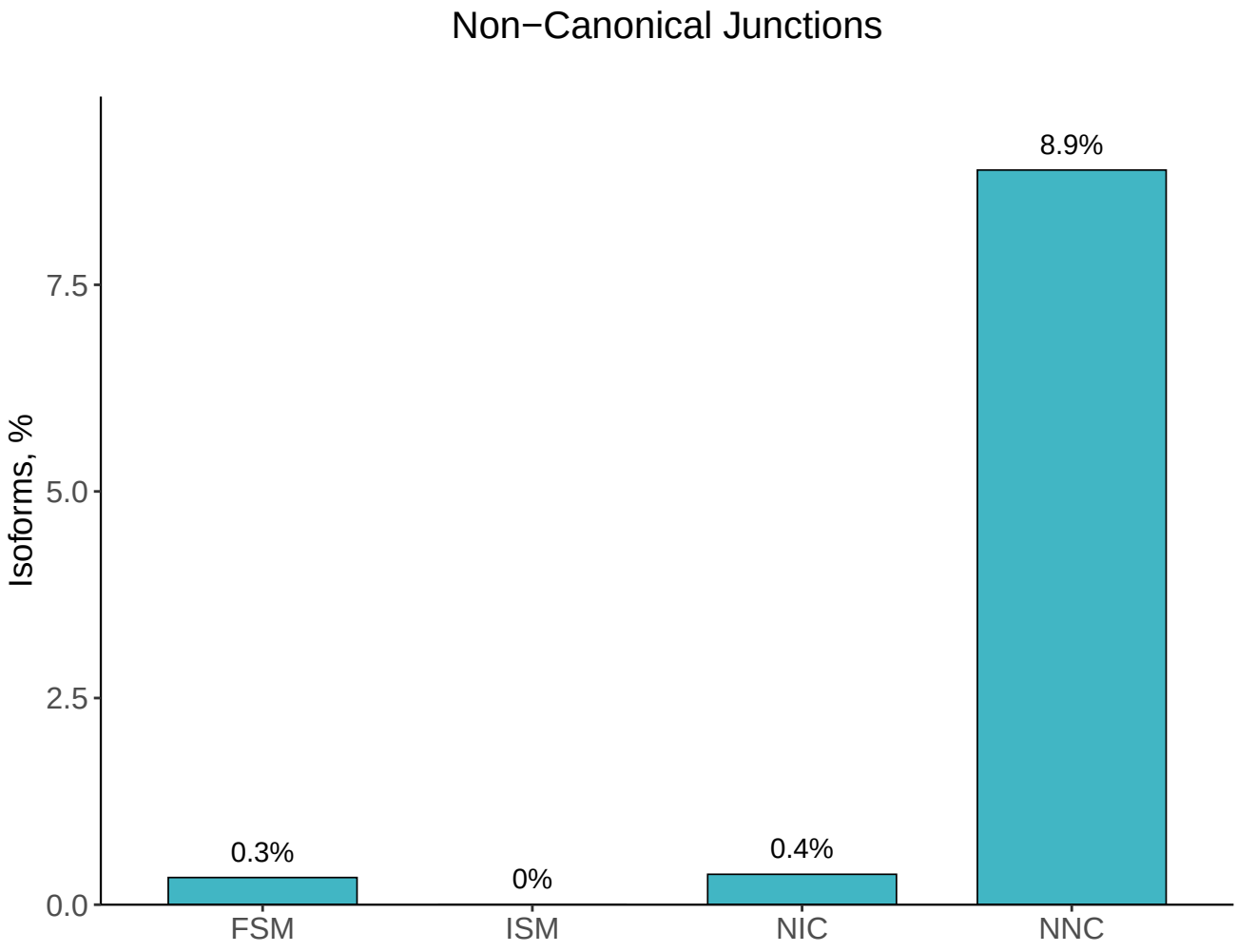

Possible Intra−Priming by Structural Category
Percent of genomic 'A's in downstream 20 bp
'A's, %
80
60
40
20
0
Multi−exon
Genic Genomic
Multi−exon
Antisense
Intron retention
Multi−exon
Fusion
Intron retention
Multi−exon

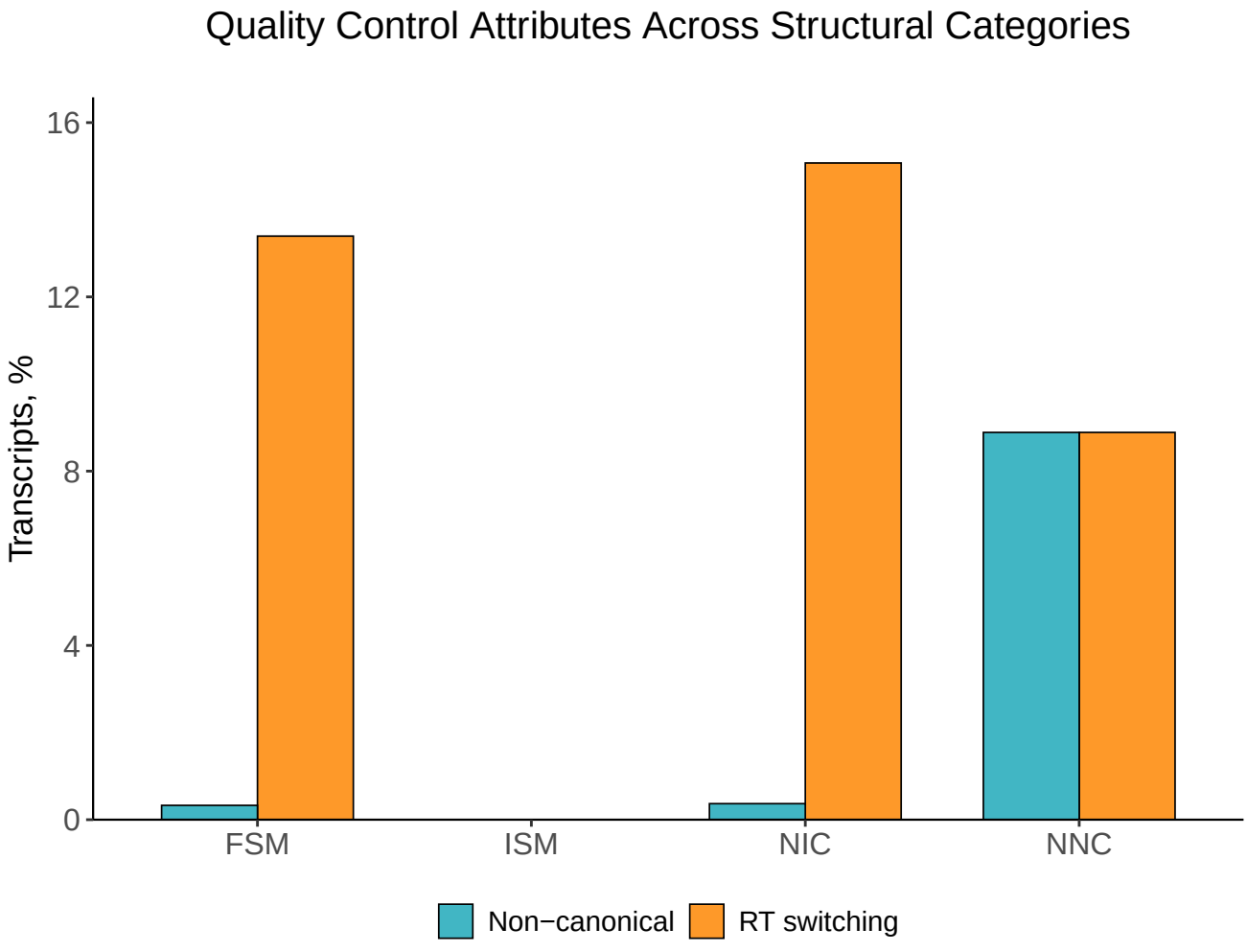

Mono− vs Multi−Exon Possible Intra−Priming
Percent of genomic 'A's in downstream 20 bp
'A's, %
100
75
50
25
0
FSM
ISM
NIC
NNC
Genic Genomic
Antisense
Fusion
Mono−Exon Isoforms
Multi−Exon Isoforms


Coding vs Non−Coding Possible Intra−Priming
Percent of genomic 'A's in downstream 20 bp
'A's, %
100
75
50
25
0
FSM
ISM
NIC
NNC
Genic Genomic
Antisense
Fusion
Non−Coding Isoforms

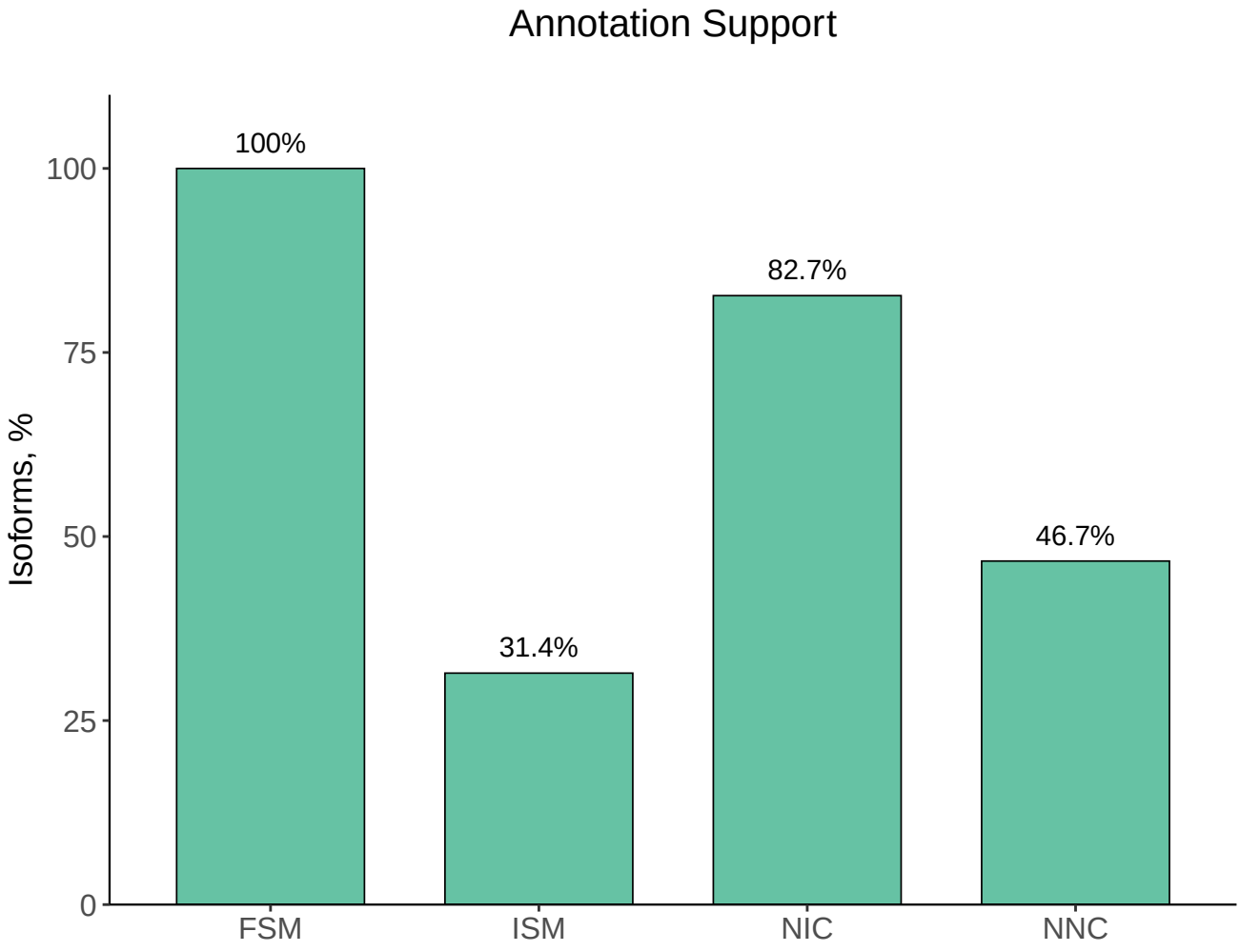
Features of Bad Quality

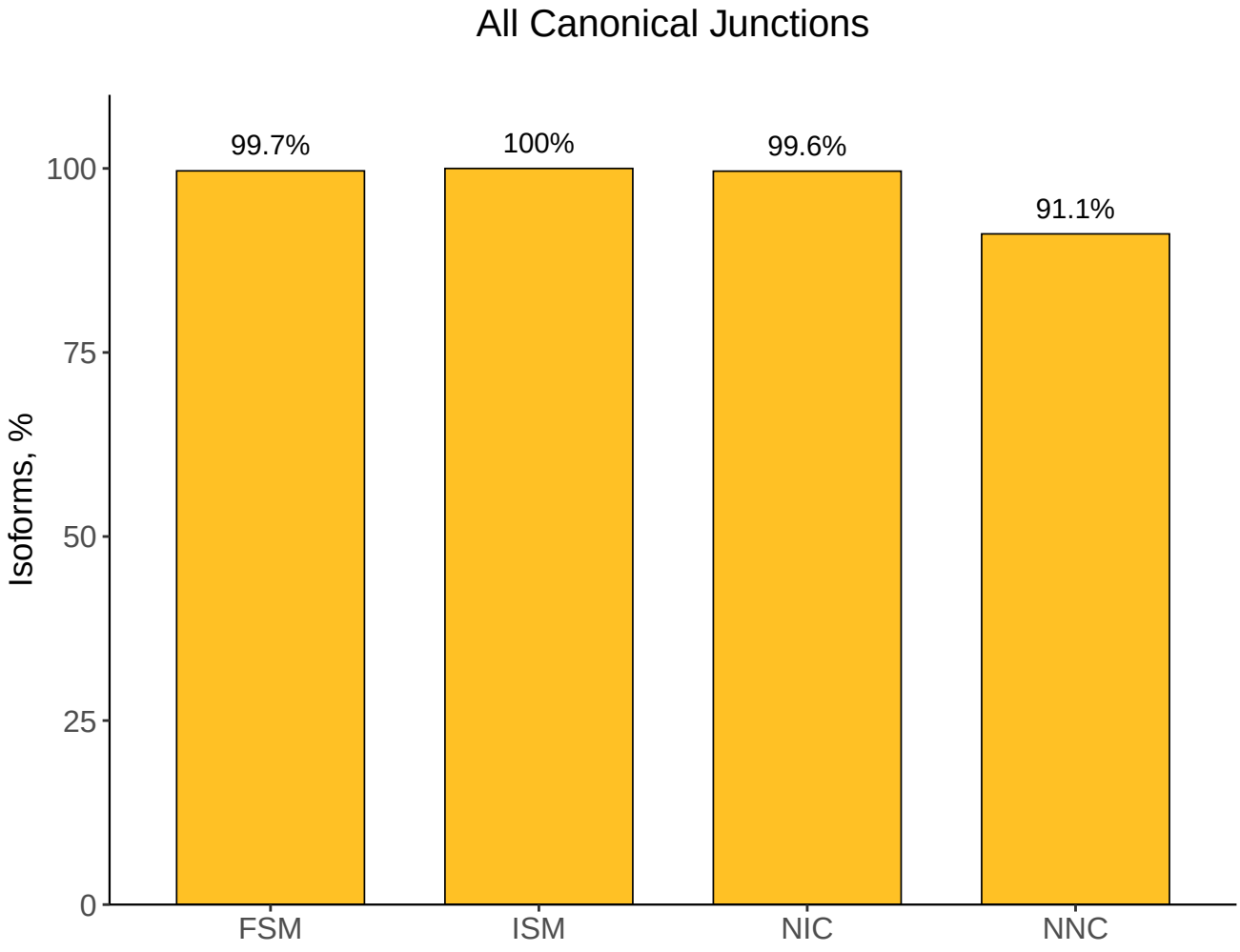

RT−switching
Isoforms, %
16
12
8
4
0
13.4%
0%
15.1%
8.9%
FSM
ISM
NIC
NNC

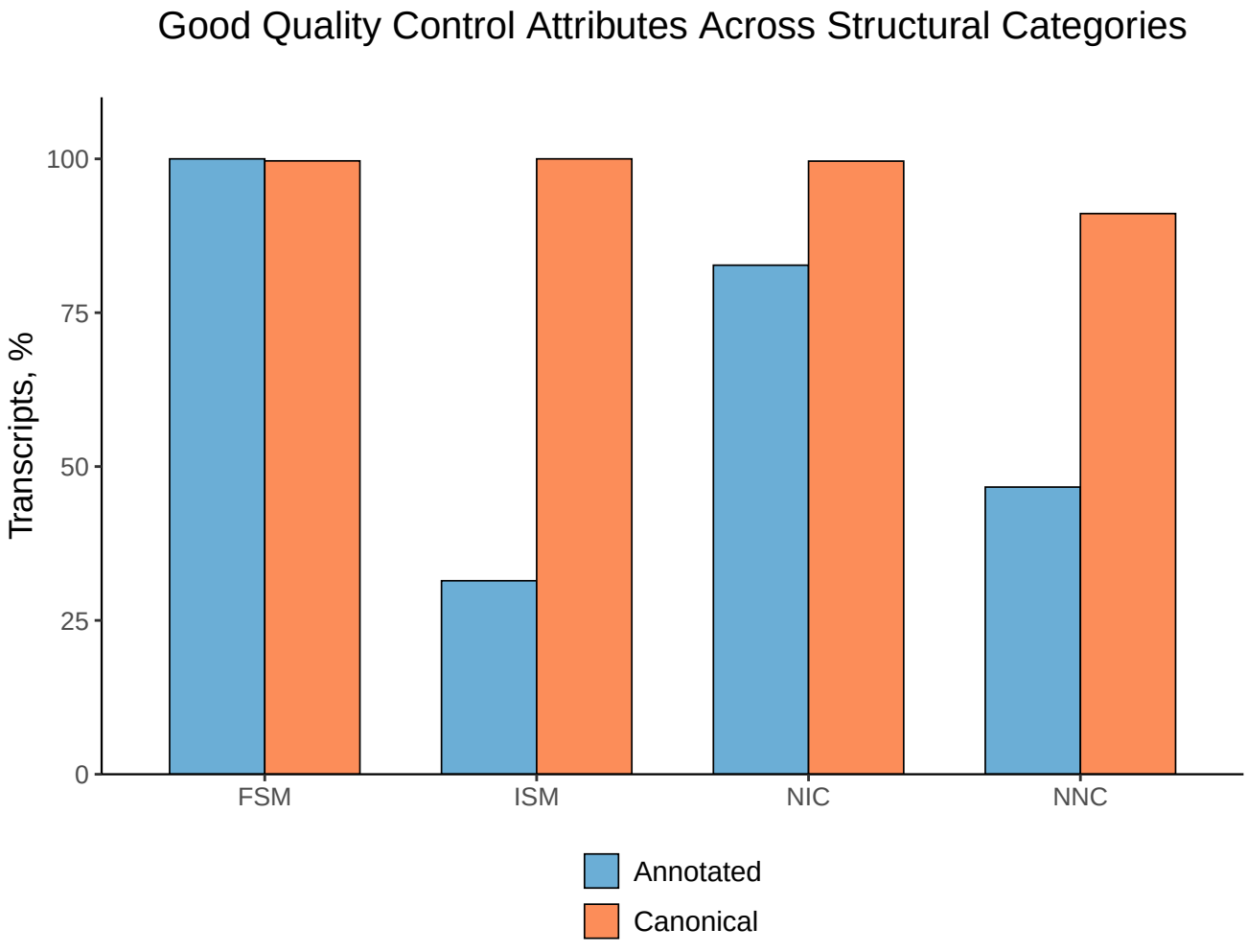

Non−Canonical Junctions
Isoforms, %
7.5
5.0
2.5
0.0
0.3%
0%
0.4%
8.9%
FSM
ISM
NIC
NNC


Quality Control Attributes Across Structural Categories
Transcripts, %
16
12
8
4
0
FSM
ISM
NIC
NNC
Non−canonical
RT switching

Features of Good Quality


Annotation Support
Isoforms, %
100
75
50
25
0
100%
31.4%
82.7%
46.7%
FSM
ISM
NIC
NNC


All Canonical Junctions
Isoforms, %
100
75
50
25
0
99.7%
100%
99.6%
91.1%
FSM
ISM
NIC
NNC


Good Quality Control Attributes Across Structural Categories
Transcripts, %
100
75
50
25
0
FSM
ISM
NIC
NNC
Annotated
Canonical

TransDecoder2 CDS prediction statistics